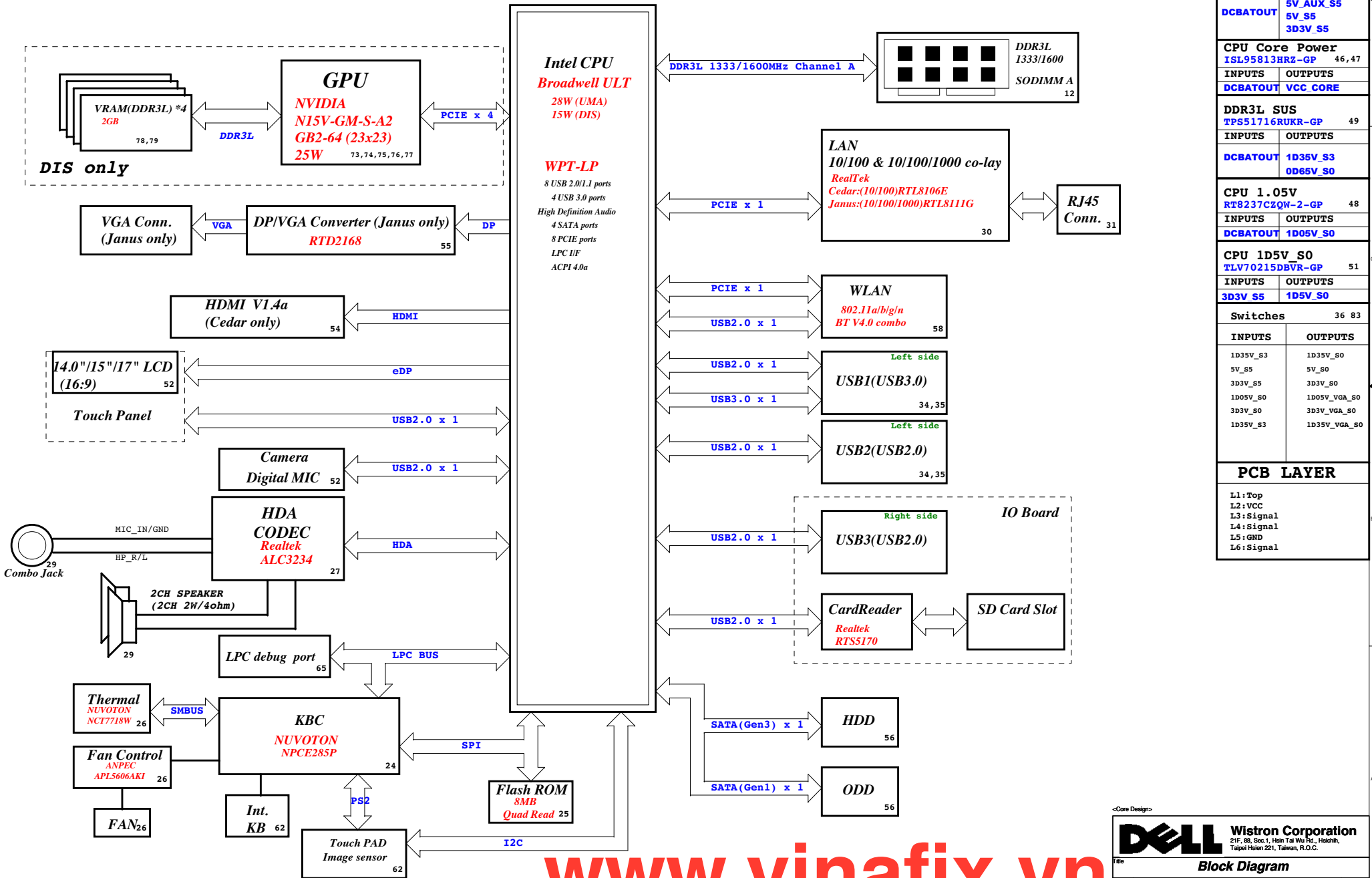


Project code:4PD00I010001
PCB P/N: 13269-1
Revision: X02

Cedar/Janus Block Diagram



www.vinafix.vn

(Blanking)

<Core Design>

		Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title (Reserved)			
Size A4	Document Number Janus HSW 40/50/70		Rev A00
Date: Friday, February 07, 2014		Sheet 3 of	104

SSID = CPU

Layout Note:

Impedance control: 50 ohm

[24,42,44,46] H_PROCHOT# <<>>

[36] H_THERMTRIP_EN <<<

Layout Note: Close to CPU

[12] DDR_PG_CTRL <<<

Layout Note:

Design Guideline:

SM_RCOMP keep routing length less than 500 mils.

71.HASWE.G0U

Layout Note:

Place close to DIMM

<Core Design>



Wistron Corporation

21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

CPU (THERMAL/MISC/PM)

Size
A4

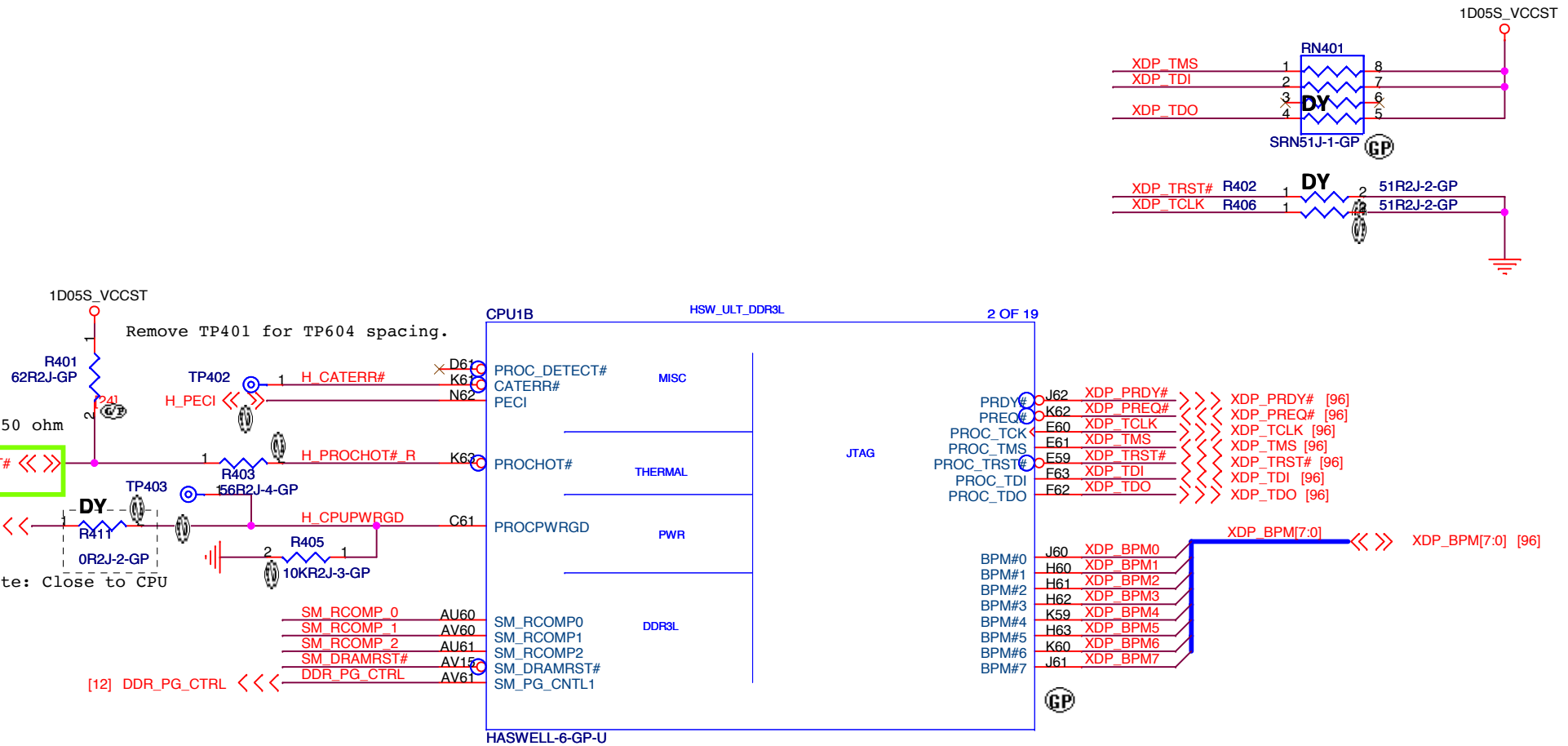
Document Number

Janus HSW 40/50/70

Rev
A00

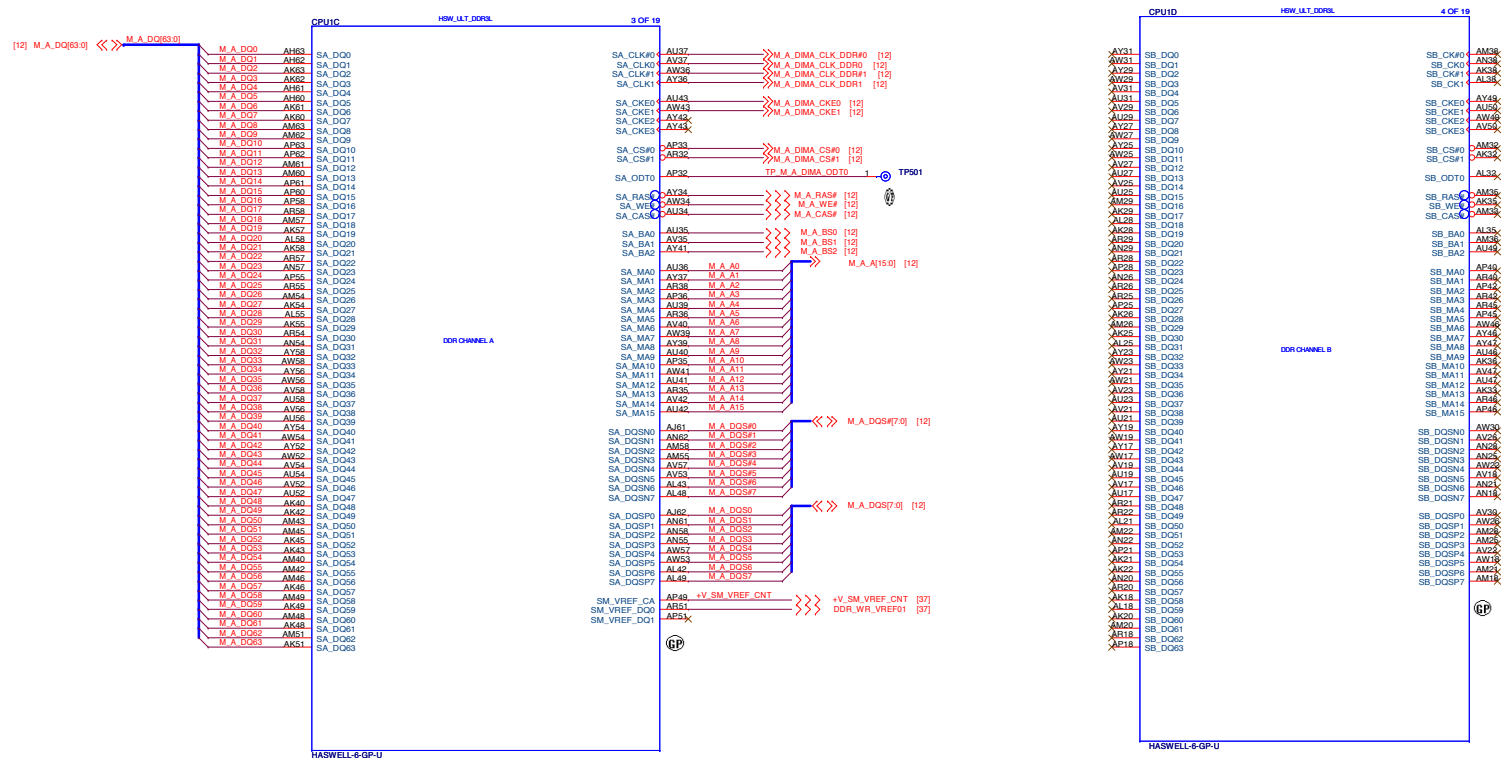
Date: Friday, February 07, 2014

Sheet 4 of 104



SSID = CPU

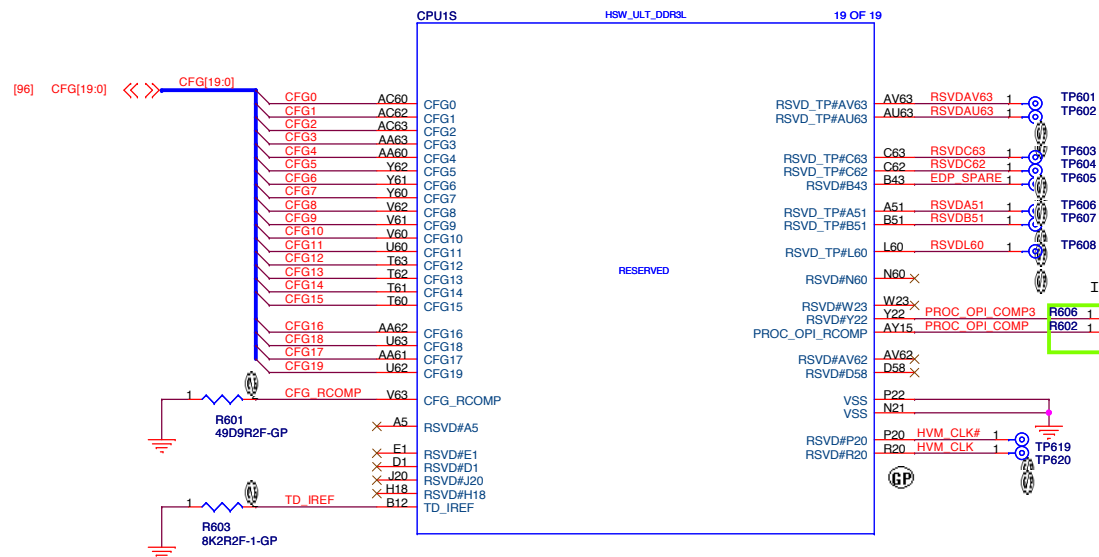
DDR3L ball type: Non-Interleaved Type



Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title			
CPU (DDR)			
Size A2	Document Number	Janus HSW 40/50/70	Rev A00
Date: Friday, February 07, 2014	Sheet 5	of	104

SSID = CPU



#514405

7.4

Reserved or Unused Signals

The following are the general types of reserved (RSVD) signals and connection guidelines:

- RSVD - these signals should not be connected
- RSVD_TP - these signals should be routed to a test point
- RSVD_NCTF - these signals are non-critical to function and may be left unconnected

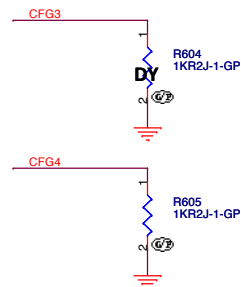
Intel Recommend

Layout Note:

1. Referenced "continuous" VSS plane only.
2. Avoid routing next to clock pins or noisy signals.
3. Trace width: 12~15mil
4. Isolation Spacing: 12mil
5. Max length: 500mil

#514405 PCH strap pin:

Signal Name	Description	Direction / Buffer Type
CFG[19:0]	Configuration Signals: The CFG signals have a default value of '1' if not terminated on the board. Refer to the appropriate Platform Design Guide for pull-down recommendations when a logic low is desired. • CFG[2:0]: Reserved configuration lane. A test point may be placed on the board for these lanes. • CFG[3]: MSR Privacy Bit Feature — 1 = Debug capability is determined by IA32_Debug_Interface_MSR (C80h) bit[0] setting — 0 = IA32_Debug_Interface_MSR (C80h) bit[0] default setting overridden • CFG[4]: eDP enable — 1 = Disabled — 0 = Enabled • CFG[19:5]: Reserved configuration lanes. A test point may be placed on the board for these lanes.	I/O GTL



PHYSICAL_DEBUG_ENABLED (DFX PRIVACY)	
CFG[3]	0 : ENABLED SET DFX ENABLED BIT IN DEBUG INTERFACE MSR 1 : DISABLED

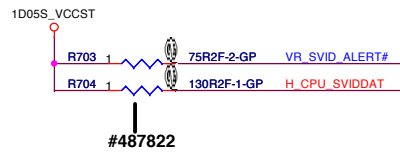
DISPLAY PORT PRESENCE STRAP	
CFG[4]	0 : ENABLED AN EXTERNAL DISPLAY PORT DEVICE IS CONNECTED TO THE EMBEDDED DISPLAY PORT 1 : DISABLED NO PHYSICAL DISPLAY PORT ATTACHED TO EMBEDDED DISPLAY PORT

<Core Design>



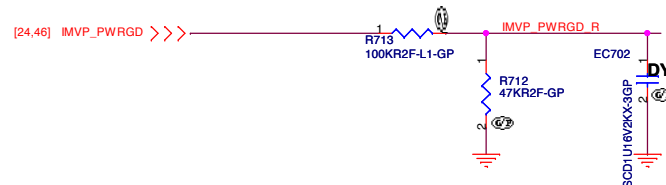
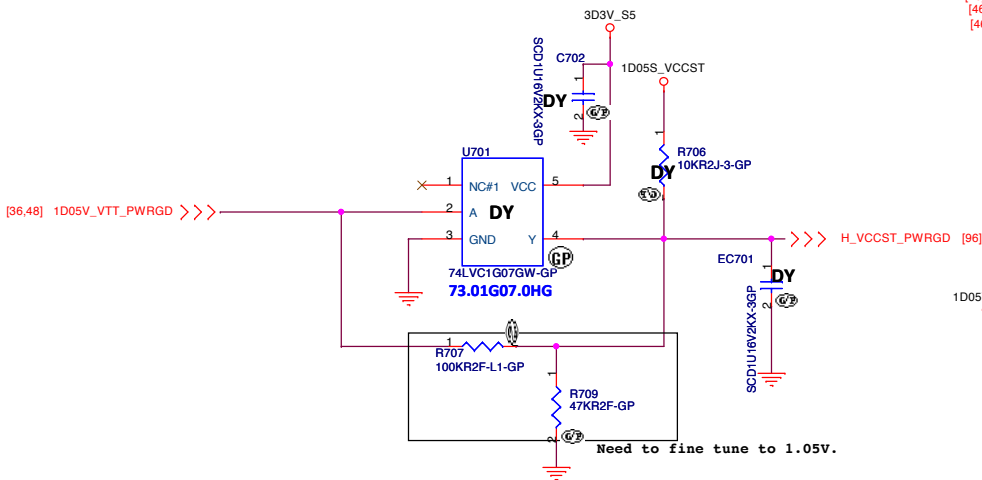
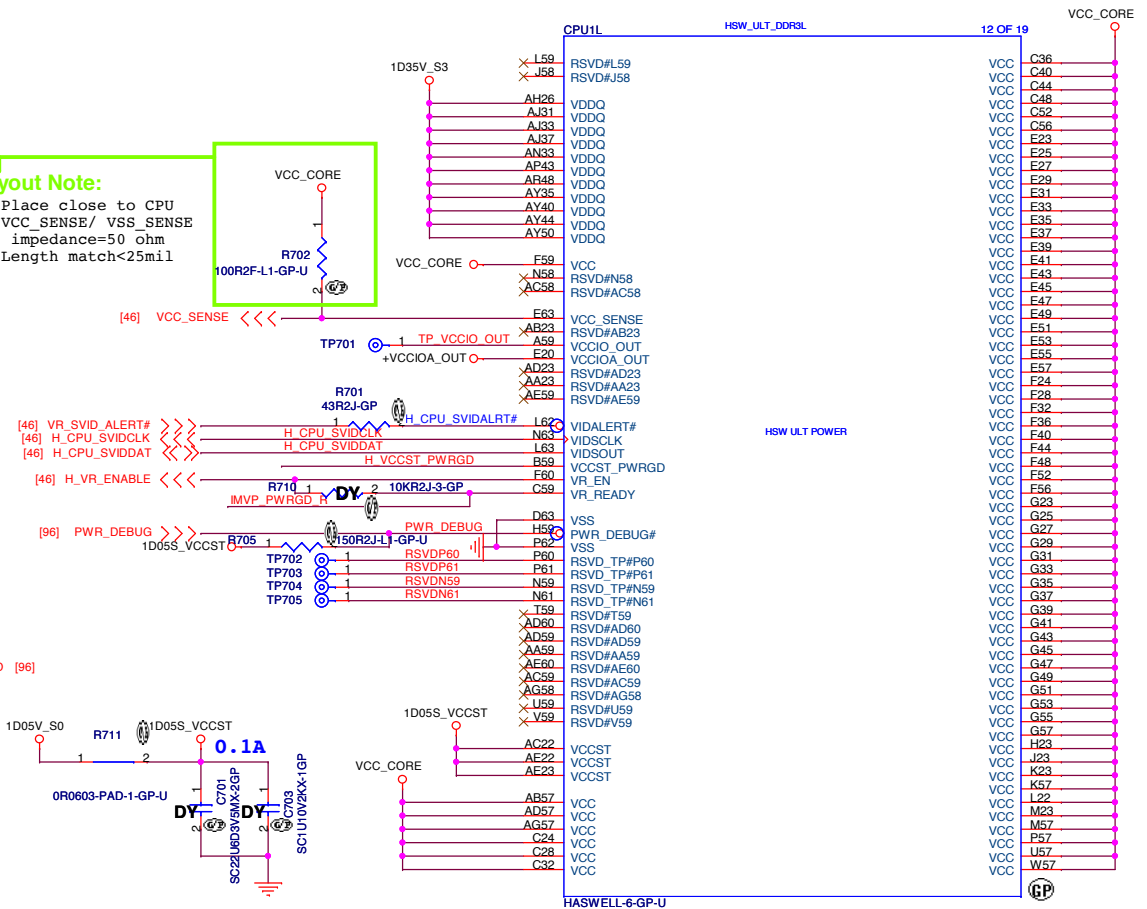
Title			CPU (CFG)	
Size	Document Number	Janus HSW 40/50/70		Rev
A3				A00
Date:	Friday, February 07, 2014	Sheet	6	of 104

SSID = CPU



Layout Note:

1. Place close to CPU
2. VCC_SENSE/ VSS_SENSE impedance=50 ohm
3. Length match<25mil



<Core Design>



Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

CPU (VCC CORE)Size
A3

Document Number

Janus HSW 40/50/70

Rev
400

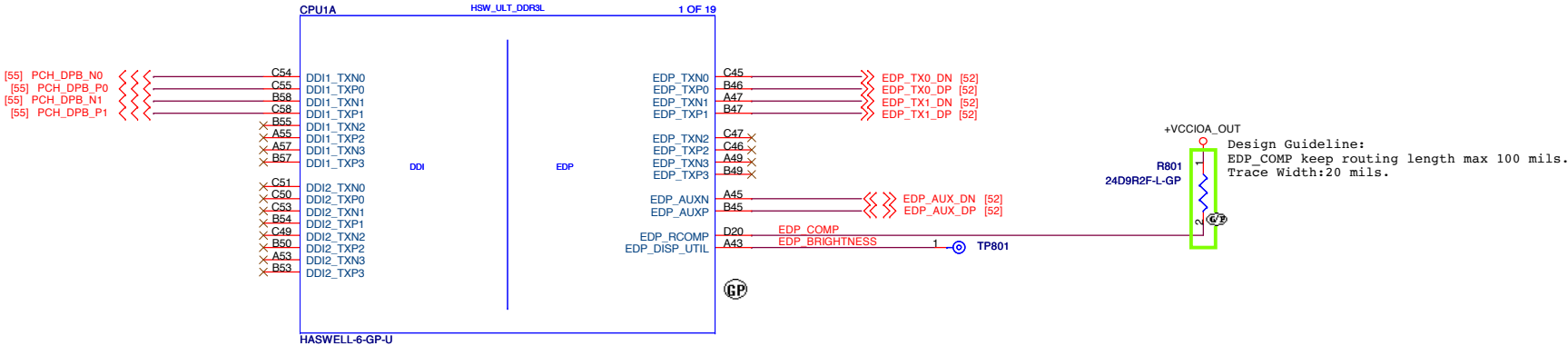
Date: Friday, February 07, 2014

Sheet 7 of 104

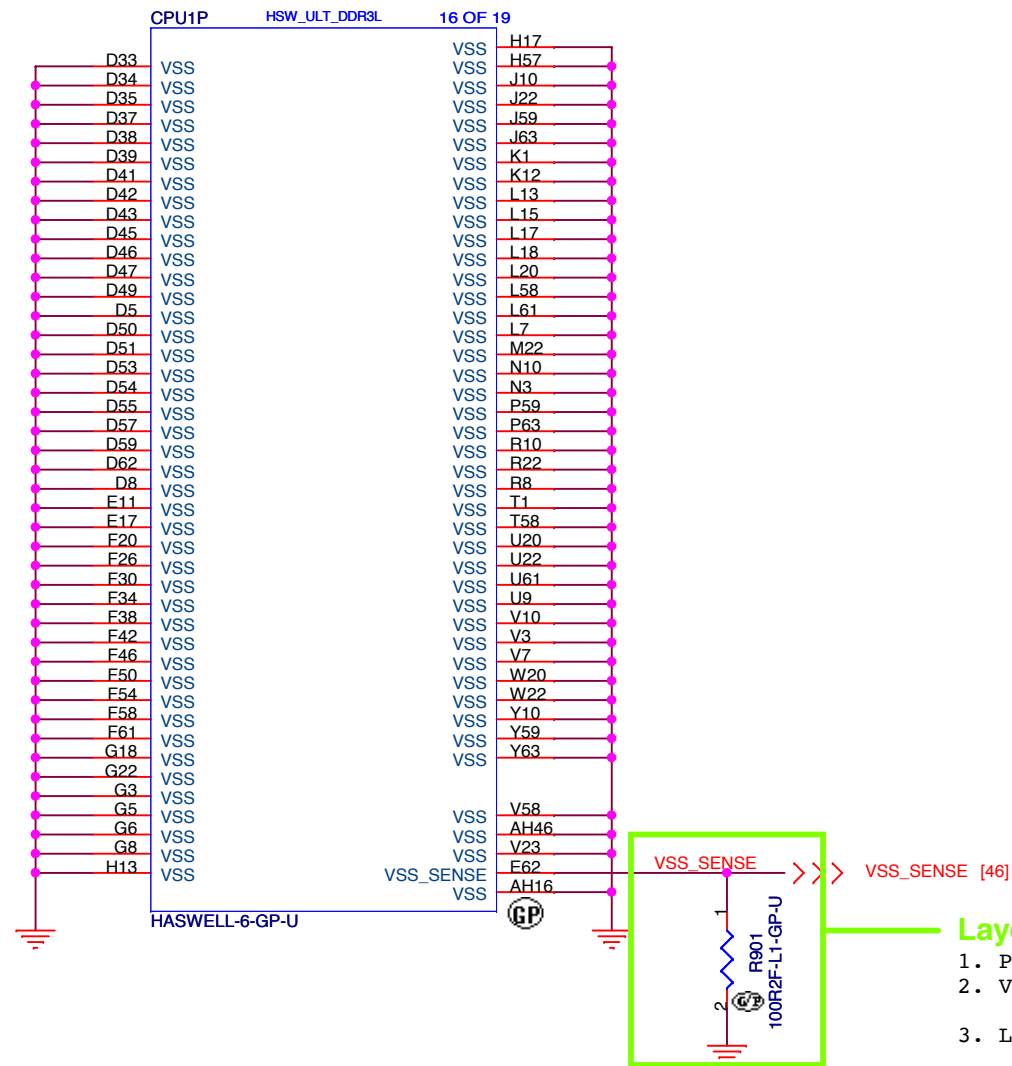
SSID = CPU

www.vinafix.vn

DP to VGA Converter



SSID = CPU



Layout Note:

1. Place close to CPU
2. VCC_SENSE/ VSS_SENSE impedance=50 ohm
3. Length match<25mil

<Core Design>



Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

CPU (VSS)

Size
A4

Document Number

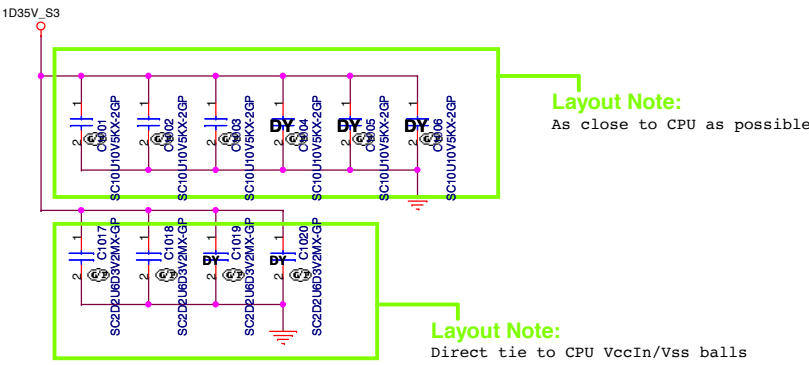
Janus HSW 40/50/70

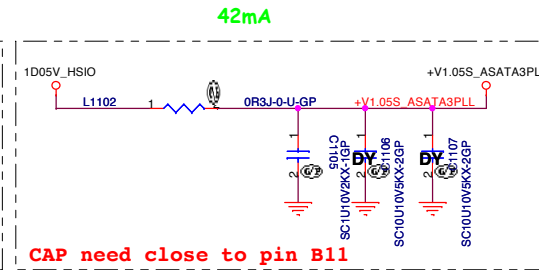
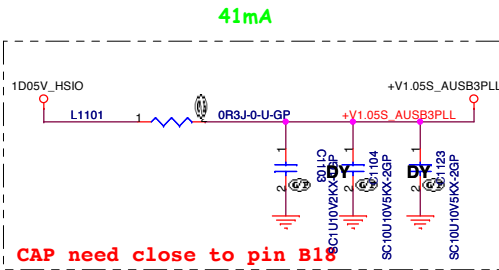
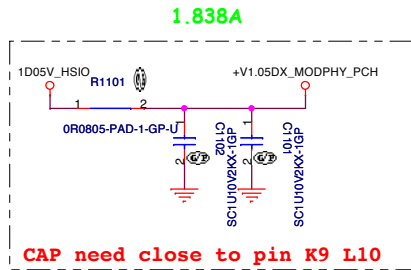
Rev
A00

Date: Friday, February 07, 2014

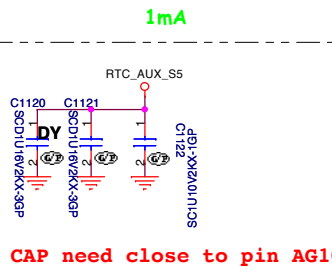
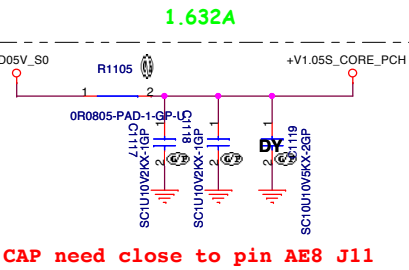
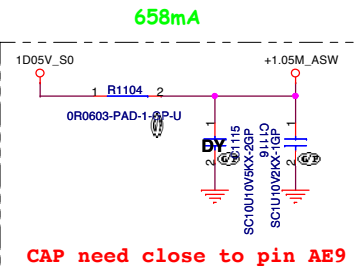
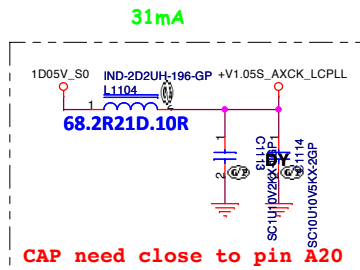
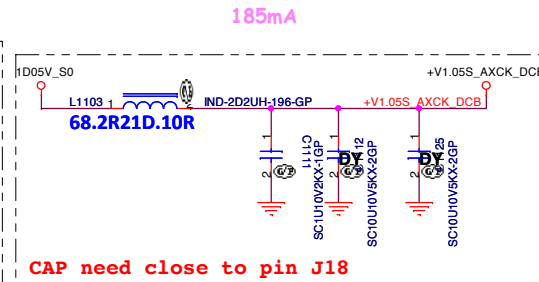
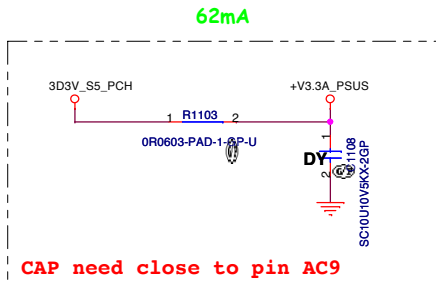
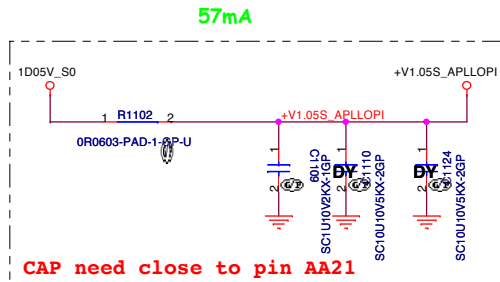
Sheet 9 of 104

SSID = CPU



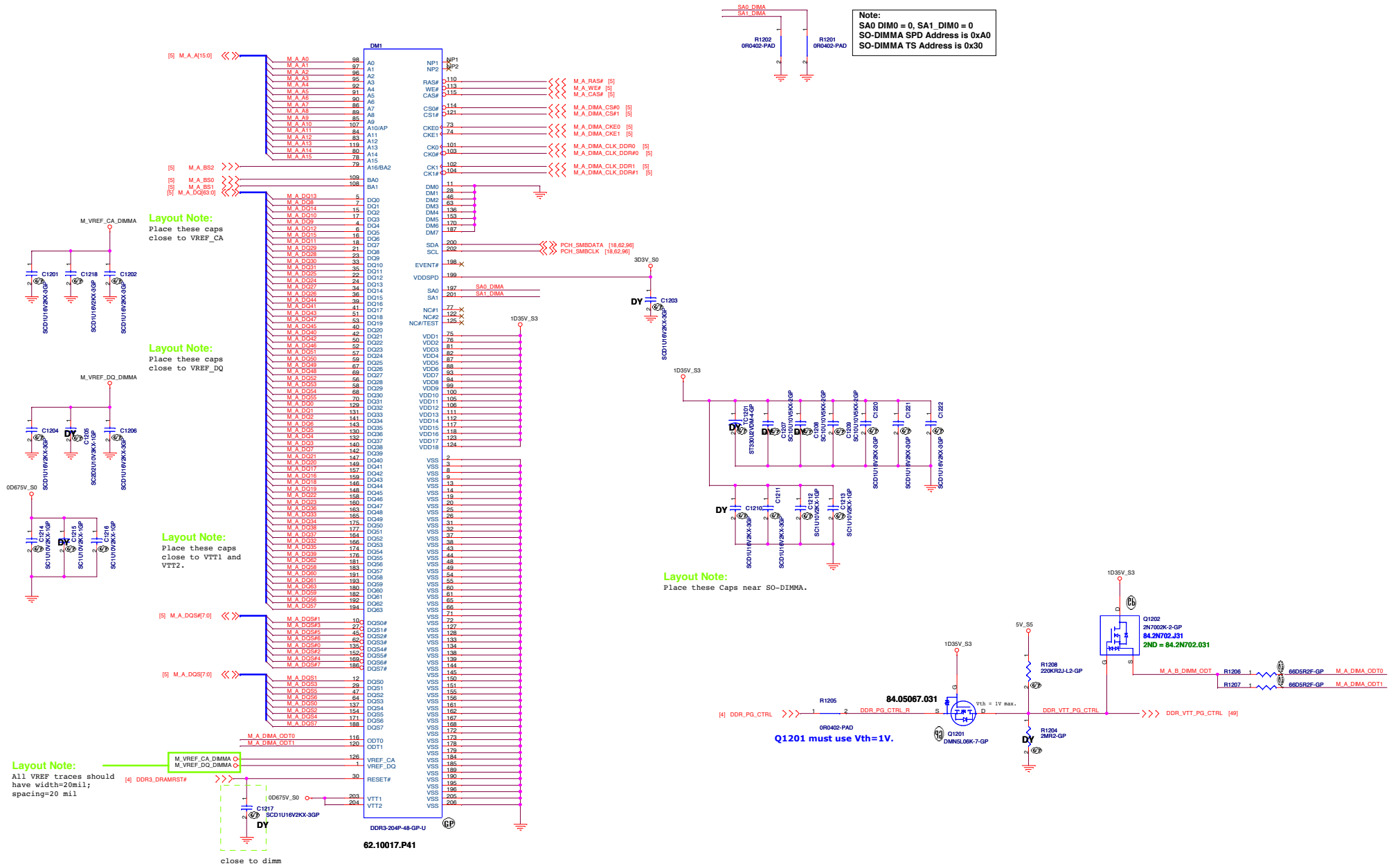


MAX: 1.92A



<Core Design>

SSID = MEMORY



(Blanking)

<Core Design>



Wistron Corporation
21F, 88, Sec. 1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

(Reserved)DDR3-SODIMM2

Size

A3

Document Number

Janus HSW 40/50/70

Rev

A00

Date: Friday, February 07, 2014

Sheet 13 of 104

(Blanking)

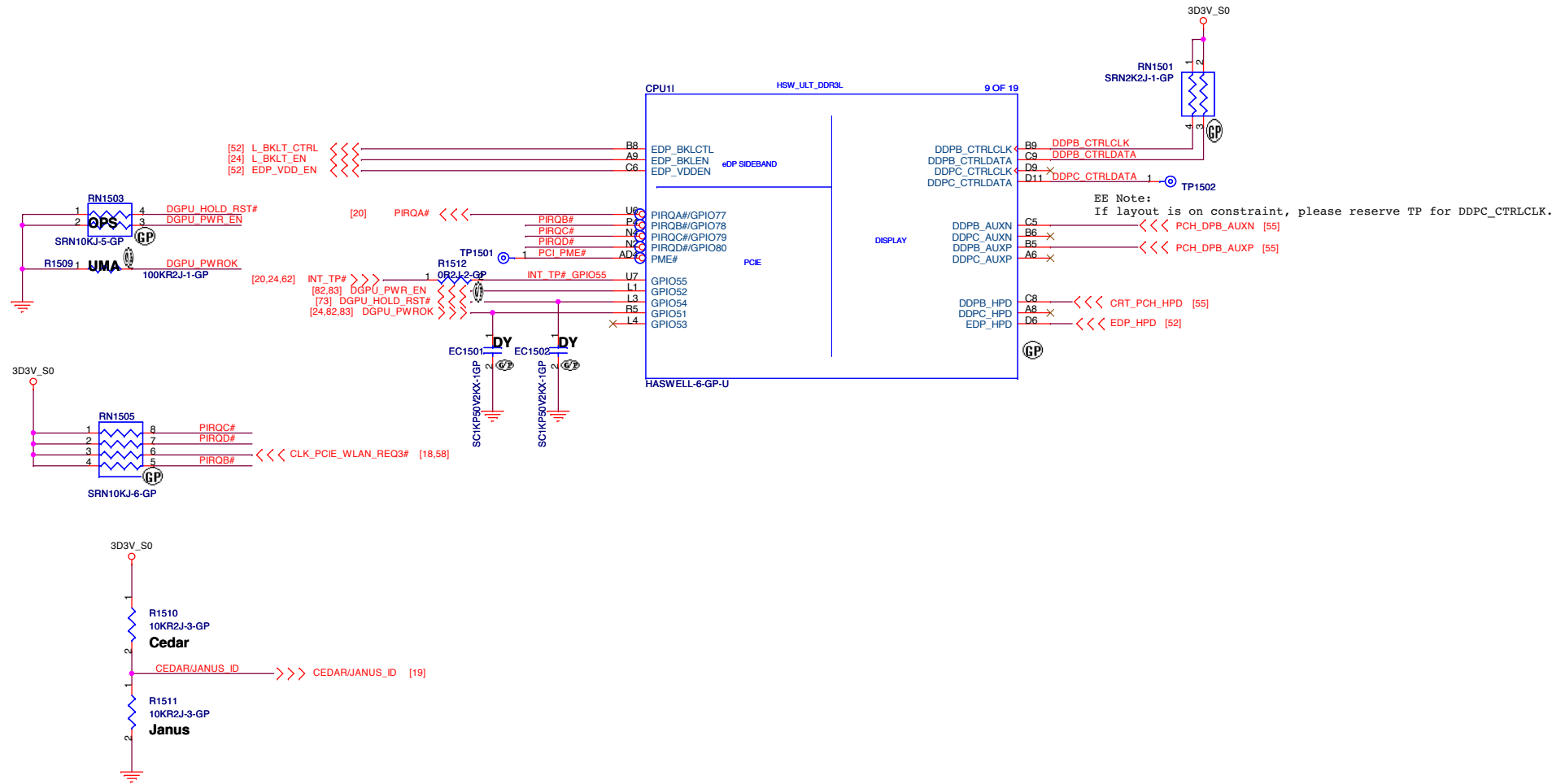
<Core Design>

			Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.		
Title (Reserved)_SODIMM _SODIMM4					
Size A4		Document Number Janus HSW 40/50/70			Rev A00
Date: Friday, February 07, 2014			Sheet 14 of 104		

PCH strap pin:

Port B Detected	
DDPB_CTRLDATA	Low = Disable Port B (default) High = Enable Port B
DDPC_CTRLDATA	Low = Disable Port C (default) High = Enable Port C

The internal pull-down is disabled after PLTRST# deasserts.



<Core Design>



Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

PCH (EDP/GPIO/DDI)

Size
A

Document Number

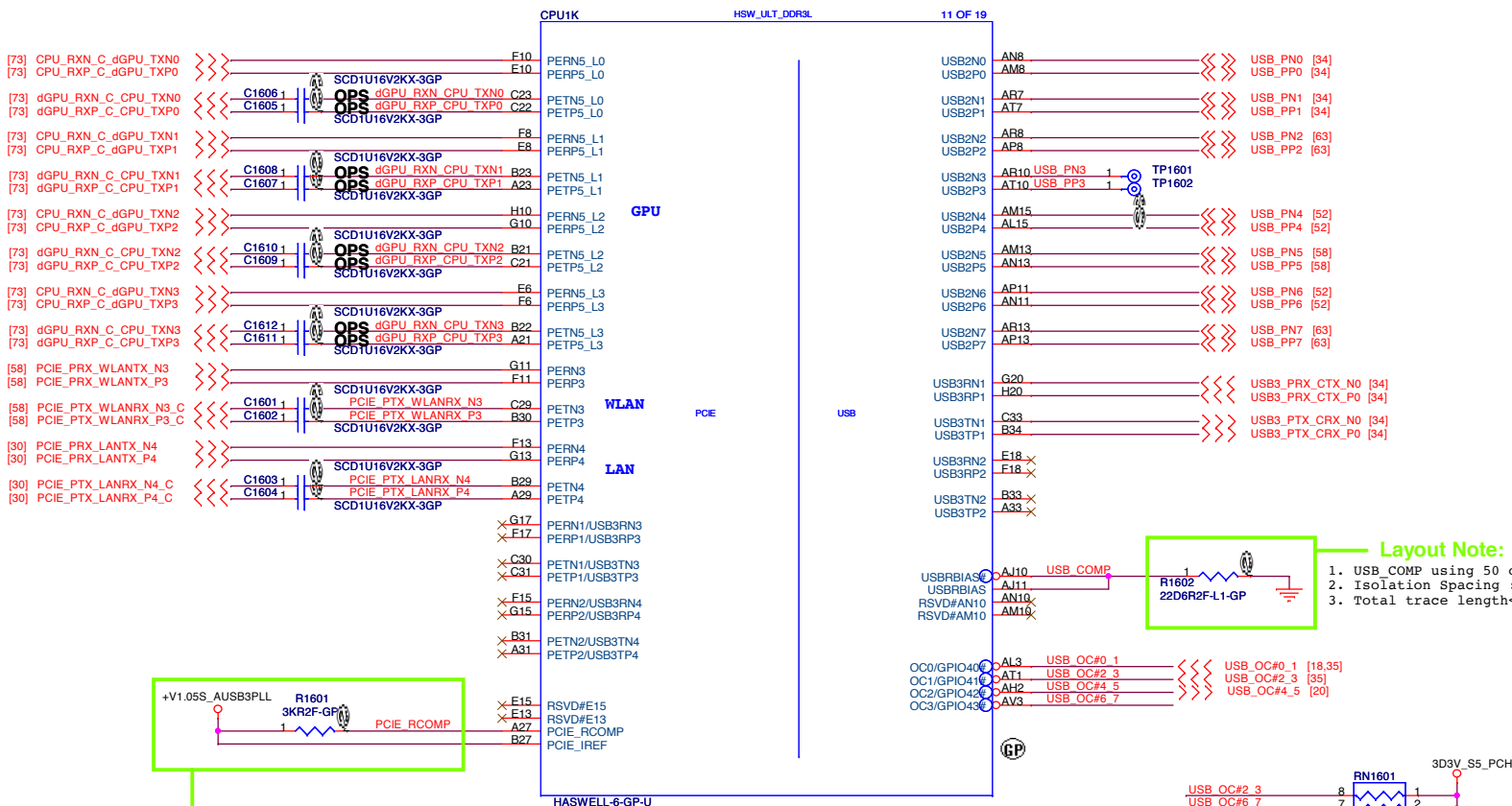
Janus HSW 40/50/70

lev
x

Date: Friday, February 07, 2014

Sheet 15 of 104

SSID = PCH



USB 2.0 Table

Pair	Device
0	USB3.0 port1
1	USB2.0 Port2 (Debug Port)
2	USB2.0 Port3 (IOBD)
3	X
4	CAMERA
5	WLAN
6	Touch Panel
7	Card Reader

Layout Note:

1. USB_COMP using 50 ohm single-ended impedance
2. Isolation Spacing :15mil
3. Total trace length<500mil

Layout Note:

1. PCIE_RCOMP/ PCIE_IREF trace width=12-15mil
2. Isolation Spacing: 12mil
3. Total trace length<500mil

PCIE Table

Port	Device	Share BUS
1	N/A	USB3.0_3
2	N/A	USB3.0_4
3	WLAN	
4	LAN	
5 (L0-L3)	GPU	
6 (L3)	HDD	SATA0
6 (L2)	ODD	SATA1
6 (L0-L1)	N/A	

#515621

Table 1-3. Broadwell U PCH-LP SKUs—Flexible I/O Map

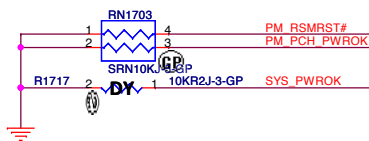
SKU	High Speed I/O Ports													
	Port 1	Port 2	Port 3	Port 4	Port 5	Port 6	Port 7	Port 8	Port 9	Port 10	Port 11	Port 12	Port 13	Port 14
Premium	USB 3.0 Port 1	USB 3.0 Port 2	USB 3.0 Port 3	USB 3.0 Port 4	PCIe* Port 3	PCIe* Port 4	PCIe* Port 5 Lane 0 SSD	PCIe* Port 5 Lane 1 SSD	PCIe* Port 5 Lane 2	PCIe* Port 5 Lane 3	SATA 6Gb/s Port 3	SATA 6Gb/s Port 2	SATA 6Gb/s Port 1	SATA 6Gb/s Port 0
			PCIe* Port 1 SSD	PCIe* Port 2 SSD			GPU	GPU	GPU	GPU	PCIe* Port 6 Lane 0 SSD	PCIe* Port 6 Lane 1 SSD	PCIe* Port 6 Lane 2	PCIe* Port 6 Lane 3
Base	USB 3.0 Port 1	USB 3.0 Port 2	USB 3.0 Port 3	USB 3.0 Port 4	PCIe* Port 3	PCIe* Port 4	PCIe* Port 5 Lane 0 SSD	PCIe* Port 5 Lane 1 SSD	PCIe* Port 5 Lane 2	PCIe* Port 5 Lane 3	PCIe* Port 6 Lane 0 SSD	PCIe* Port 6 Lane 1 SSD	SATA 6Gb/s Port 1	SATA 6Gb/s Port 0
			PCIe* Port 1 SSD	PCIe* Port 2 SSD			GPU	GPU	GPU	GPU				

<Core Design>

DELL Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

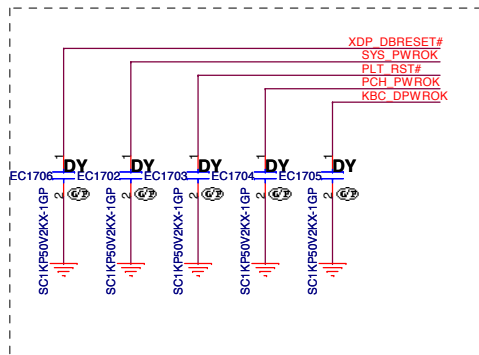
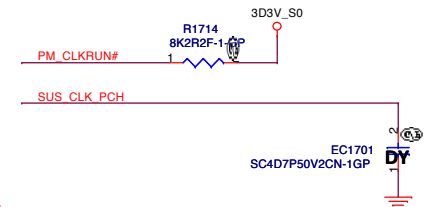
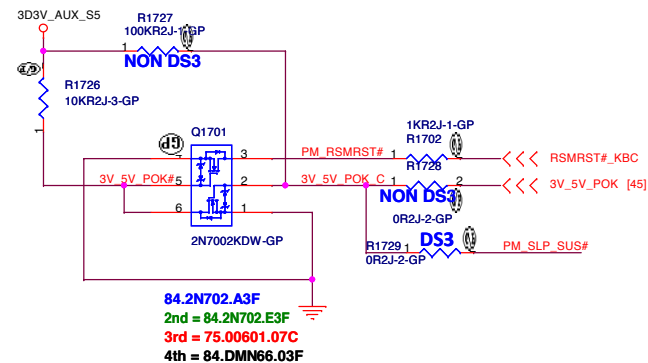
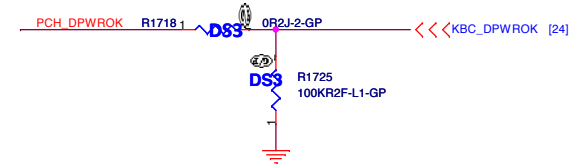
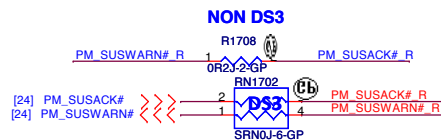
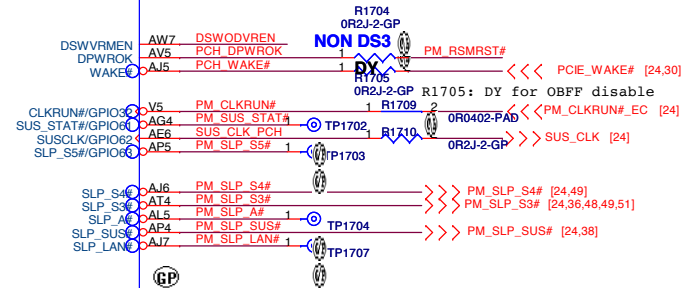
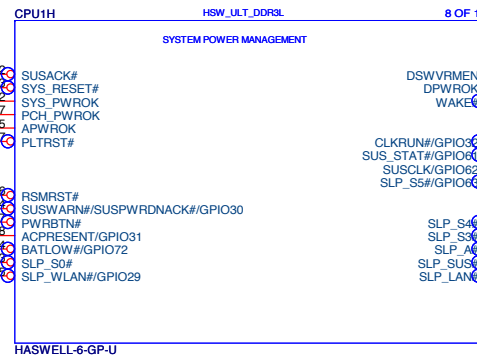
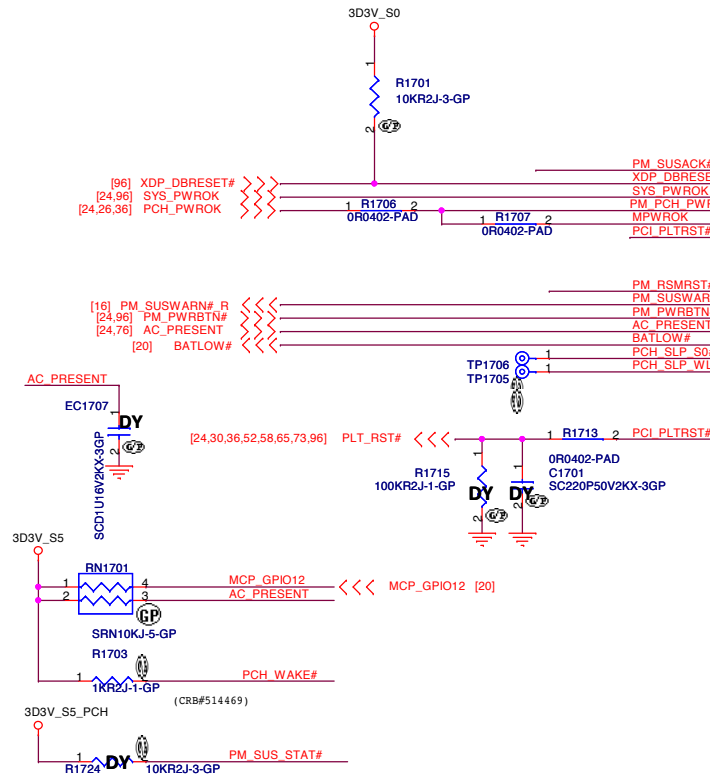
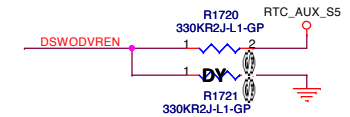
Title			PCH (PCIE/USB)		
Size A3	Document Number	Janus HSW 40/50/70			Rev A00
Date: Friday, February 07, 2014			Sheet 16 of 104		

SSID = PCH

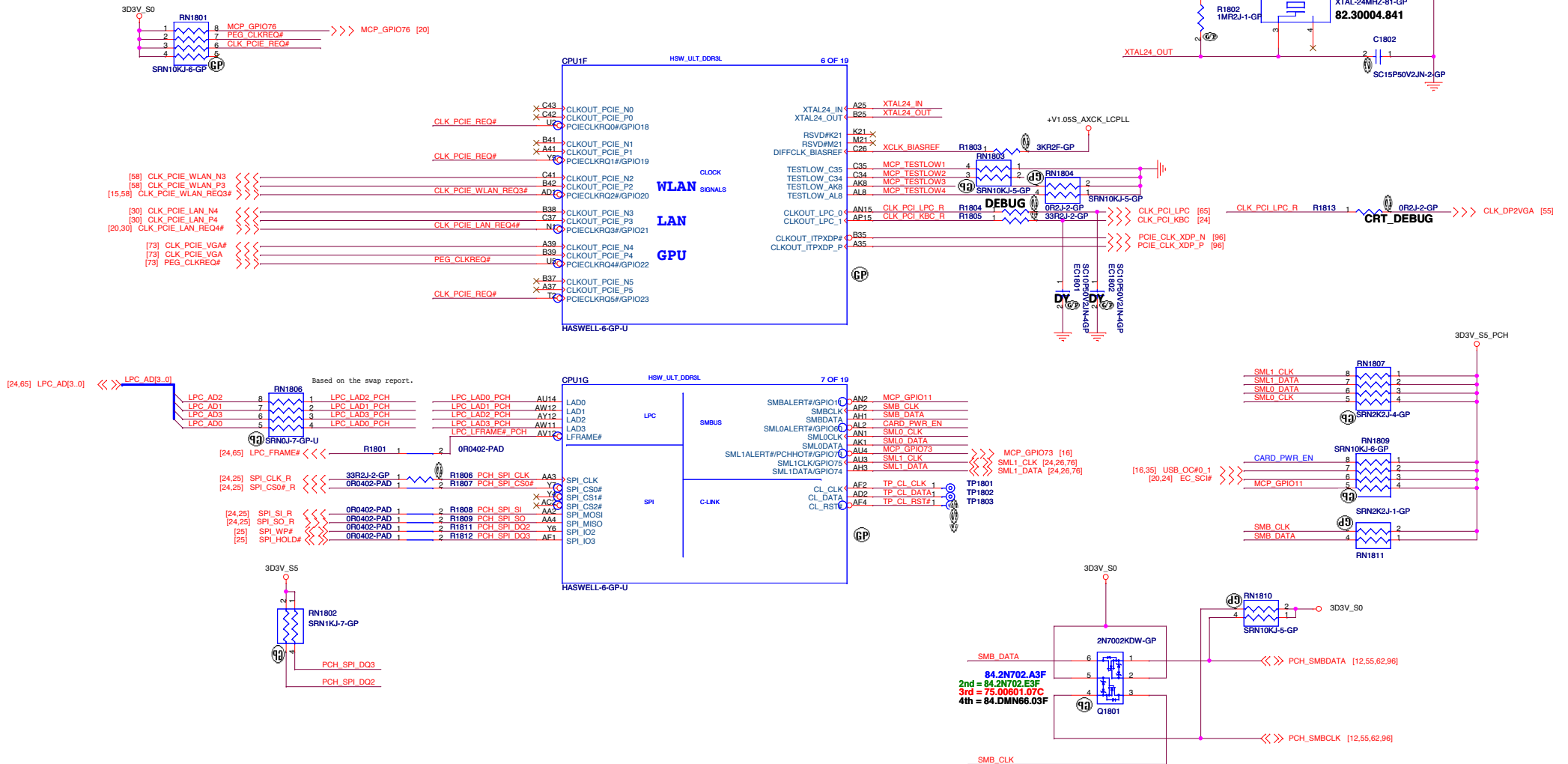


PCH strap pin:

On Die DSW VR Enable	
DSWVRMEN	Low = Disable High = Enable (default)
	* This signal has no integrated pull-up/pull-down.



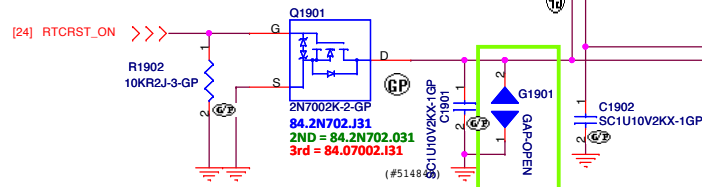
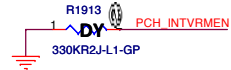
SSID = PCH



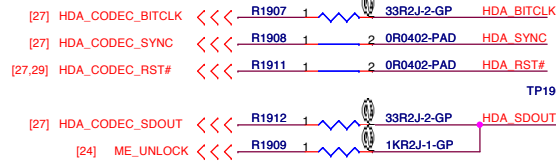
SSID = CPU

PCH strap pin:

Integrated SUS 1V VRM Enable	
INTVRMEN	Low = External VRs High = Internal VRs*



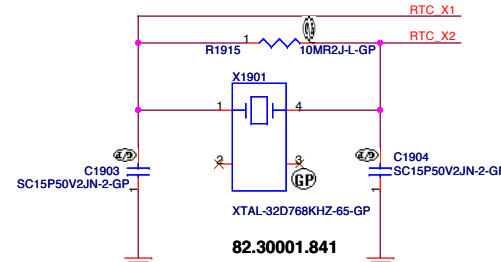
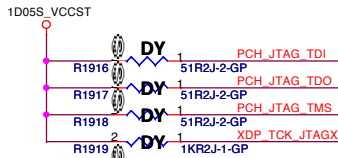
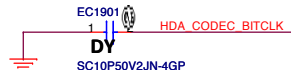
Layout: Place at the open door area.



PCH strap pin:

Flash Descriptor Security Override/ Intel ME Debug Mode	
HDA_SDOUT	Low = Default * High = Enable

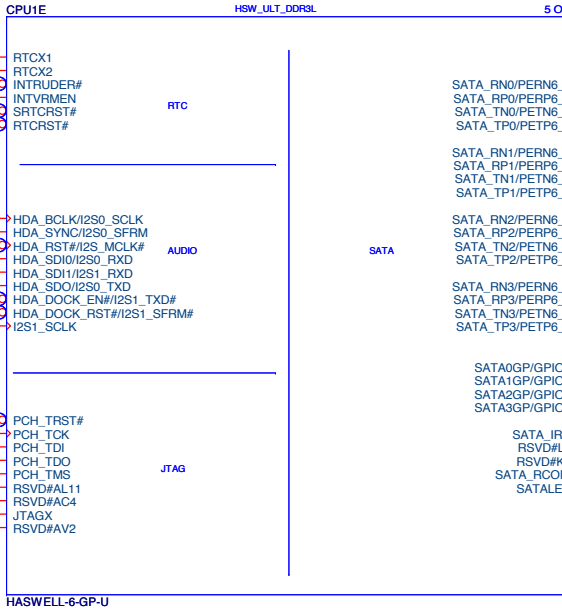
The internal pull-down is disabled after PLTRST# deasserts



RTC X1
RTC X2
SM INTRUDER#
PCH_INTVRMEN
SRTC_RST#
RTC_RST#

HDA_BITCLK
HDA_SYNC
HDA_RST#
HDA_SDIO0
HDA_SDOUT
TP HDA DOCK_ENF#

PCH_JTAG_TRST#
PCH_JTAG_TCK
PCH_JTAG_TDI
PCH_JTAG_TDO
PCH_JTAG_TMS
XDP_TCK_JTAGX



SATA_RN0/PERN6_L3
SATA_RP0/PERP6_L3
SATA_TN0/PETN6_L3
SATA_TP0/PETP6_L3
SATA_RN1/PERN6_L2
SATA_RP1/PERP6_L2
SATA_TN1/PETN6_L2
SATA_TP1/PETP6_L2
SATA_RN2/PERN6_L1
SATA_RP2/PERP6_L1
SATA_TN2/PETN6_L1
SATA_TP2/PETP6_L1
SATA_RN3/PERN6_L0
SATA_RP3/PERP6_L0
SATA_TN3/PETN6_L0
SATA_TP3/PETP6_L0

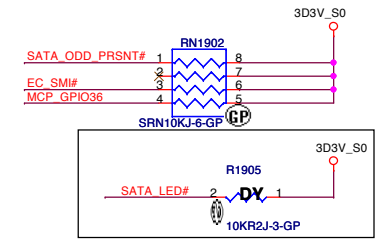
V1 EC SMI#
U1 MCP GPIO36
AC1 CEDAR/JANUS_ID [15]
A12 SATA_IREF
L11
K10
C12
U3
SATA_LED#

HDD1

ODD

Layout Note:

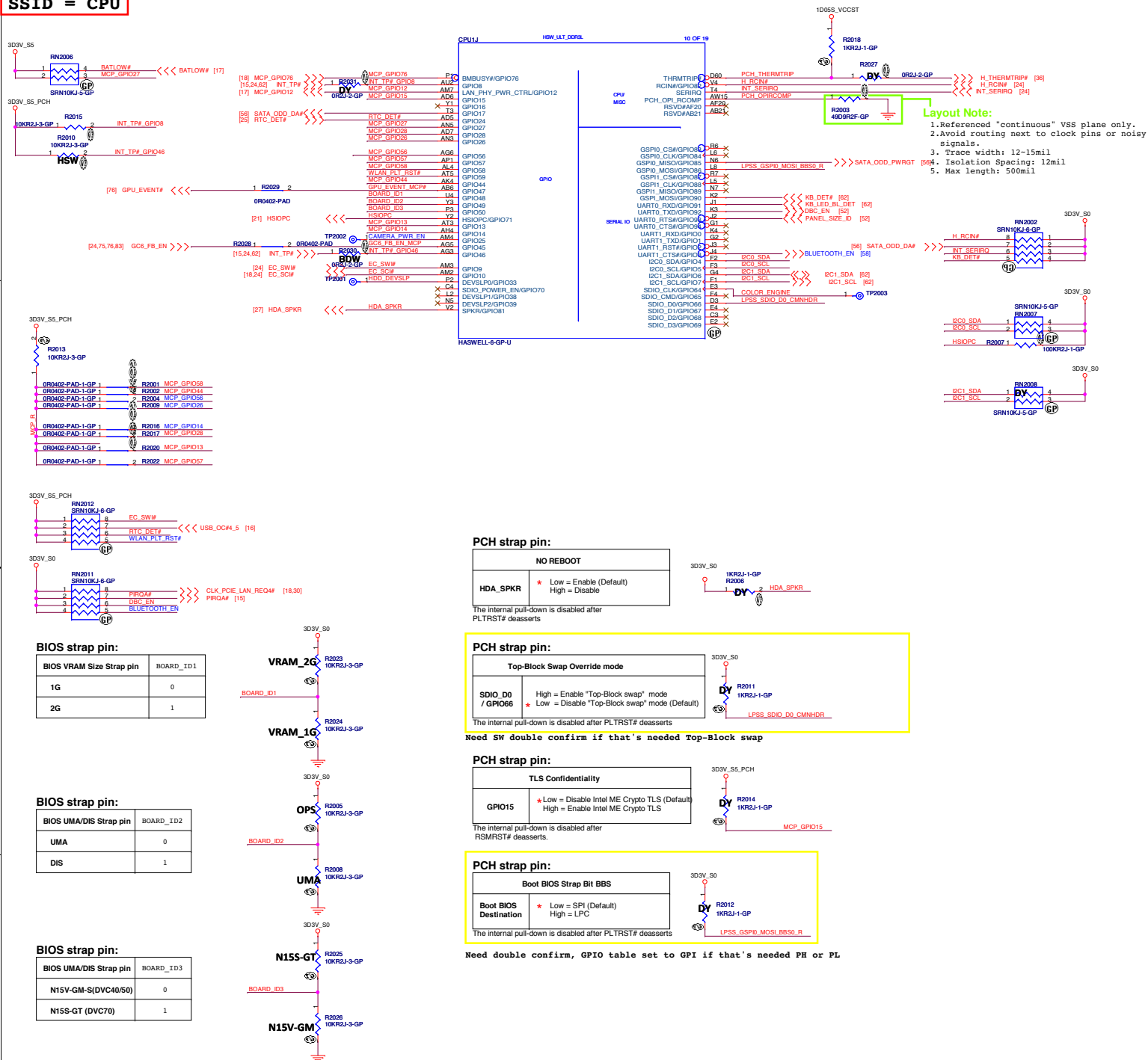
4mil trace at break-out and 3
12-15mil trace with <0.2 ohms
and length total <= 500mils.



Unused SATA[3:0]GP pins must be terminated to either
3.3V rail or GND using 8.2K to 10K on the
motherboard. Either pull-up or pull-down is acceptable.

<Core Design>

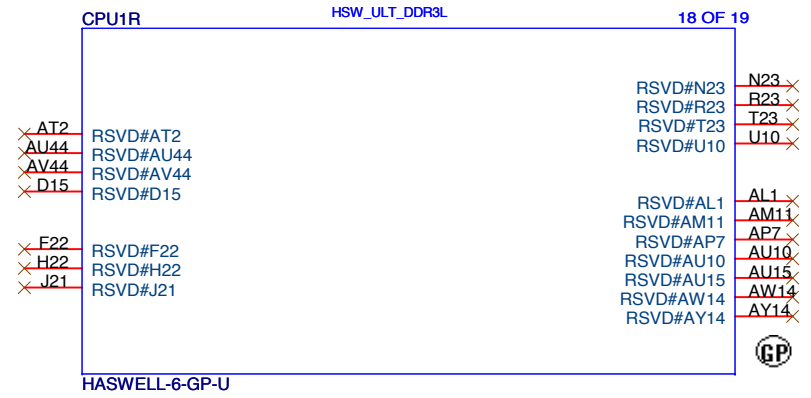
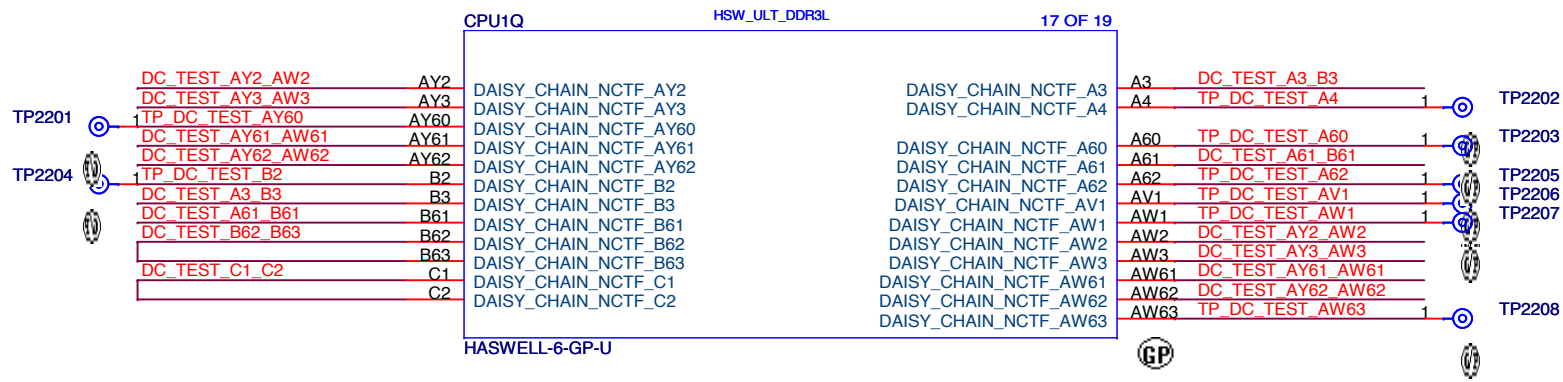
SSTD = CPI



<Core Design>

		Wistron Corporation 21F, 88, Sec. 1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title			
Size A3		CPU (POWER2) Janus HSW 40/50/70	
Document Number		Rev A00	
Date:	Friday, February 07, 2014	Sheet 21	of 104

SSID = PCH



<Core Design>



Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

CPU (RSVD)

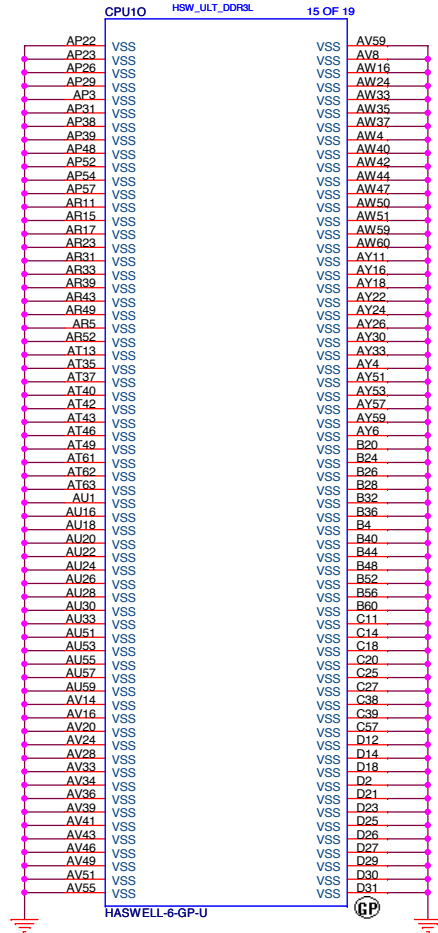
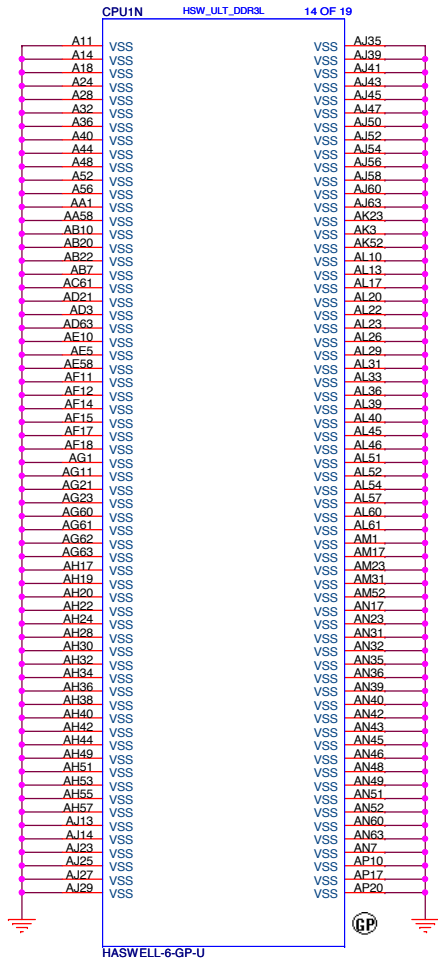
SizeA4

Document Number

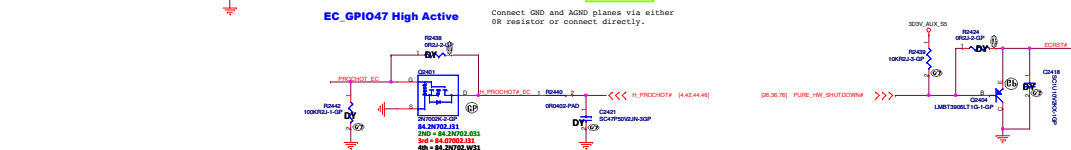
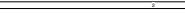
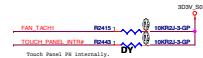
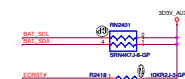
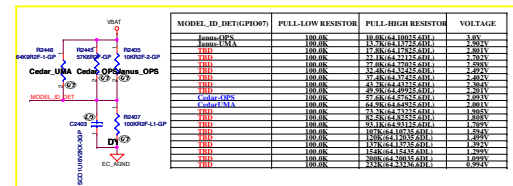
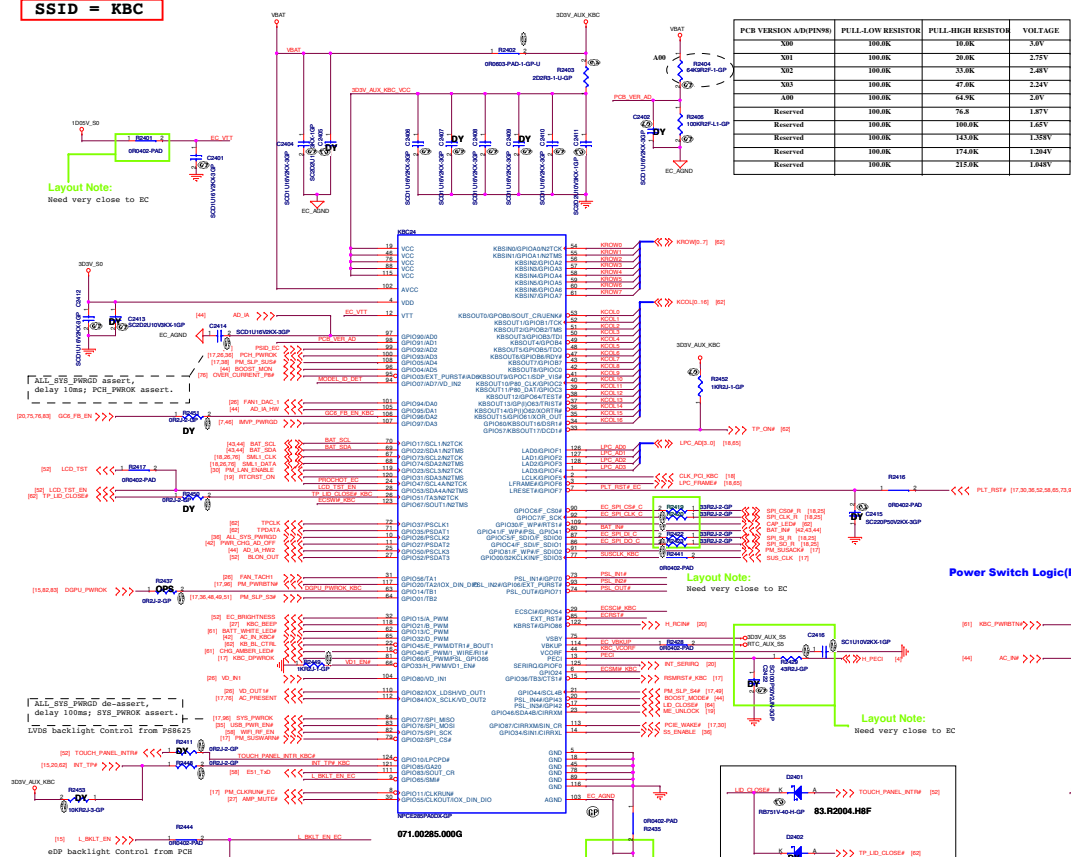
RevA00

Date: Friday, February 07, 2014Sheet 22 of 104

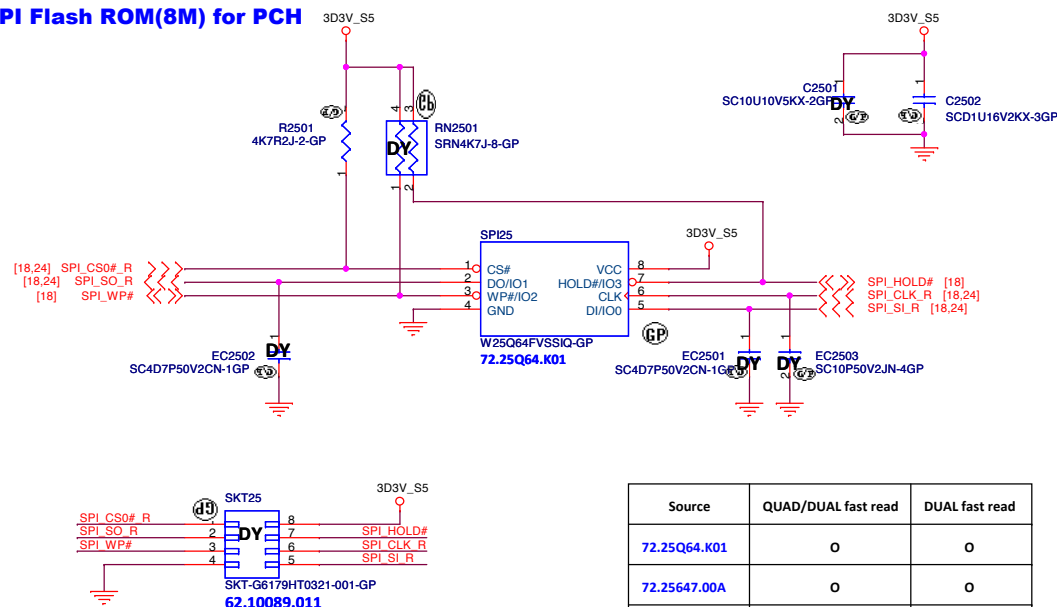
SSID = PCH



SSID = KBC

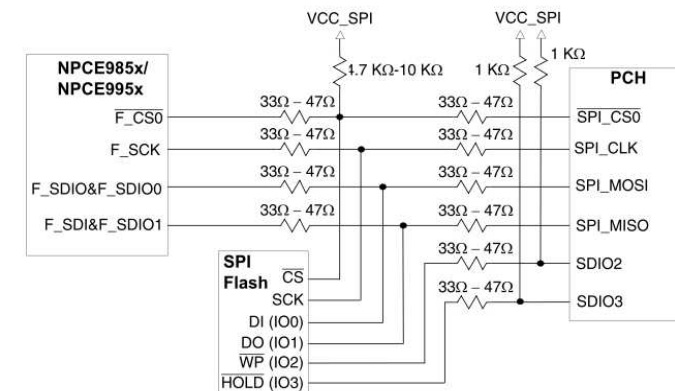


SPI Flash ROM(8M) for PCH

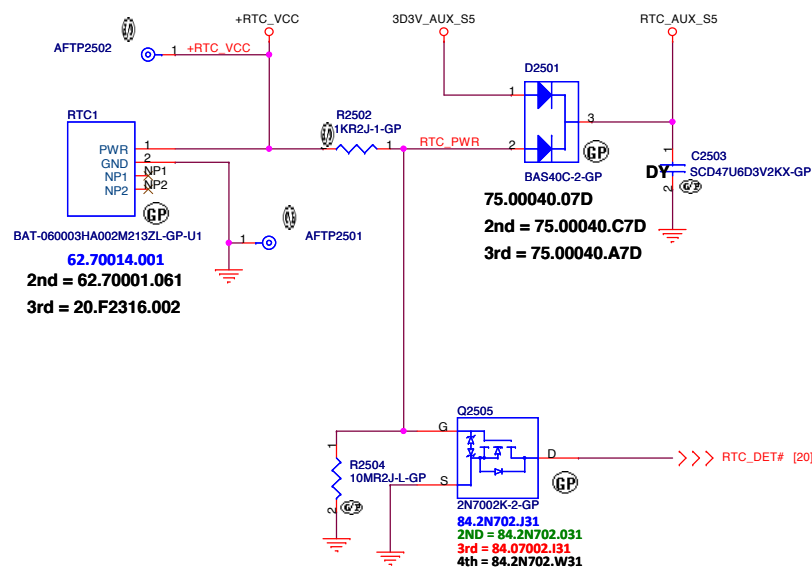


Source	QUAD/DUAL fast read	DUAL fast read
72.25Q64.K01	O	O
72.25647.00A	O	O
072.25B64.0001	O	O

Single SPI shared flash connection (SPI Quad I/O mode)



Refer to "NCPE985x/ NPCE995x board design reference guide"

SSID = RBATT

<Core Design>

DELL

Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Flash/RTC

Size
A3

Document Number	
-----------------	--

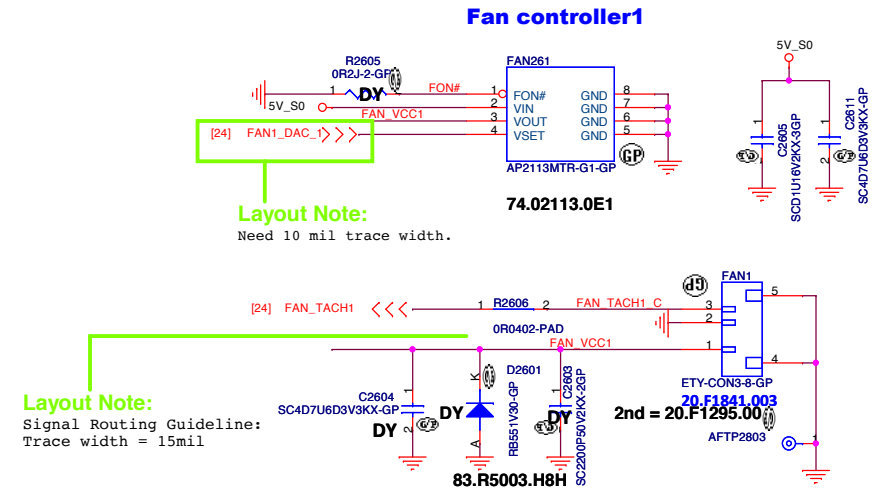
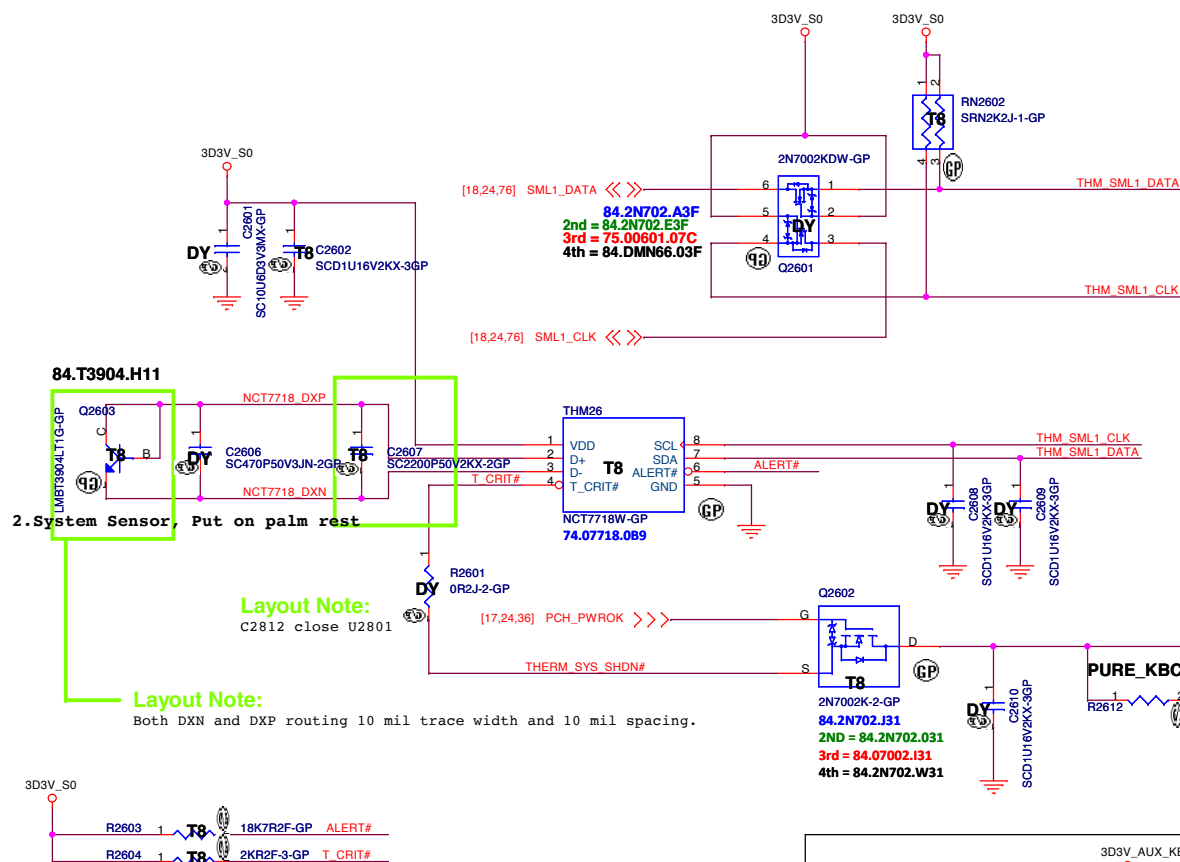
Janus HSW 40/50/70

Rev	A00
-----	------------

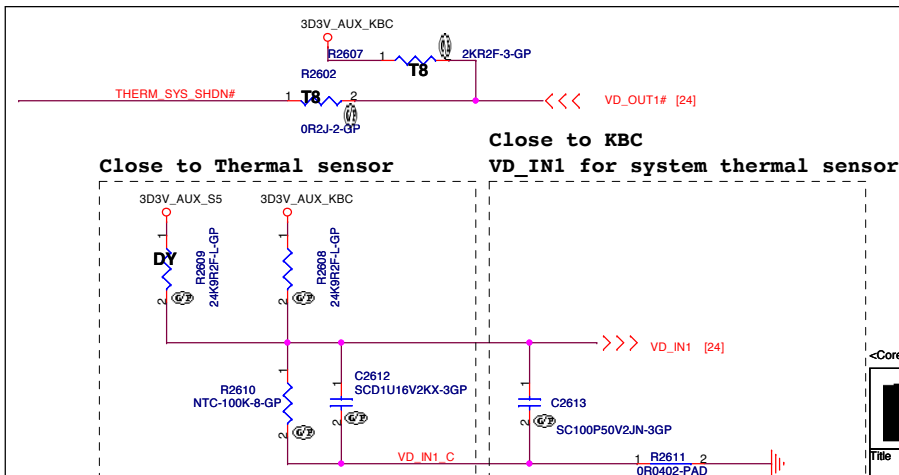
Date: Friday, February 07, 2014

Sheet 25 of 104

SSID = Thermal



TEMPERATURE (°C)		T_CRIT#				
		2KΩ	7.5KΩ	10.5KΩ	14KΩ	18.7KΩ
ALERT#	2KΩ	77	87	97	107	117
	7.5KΩ	79	89	99	109	119
	10.5KΩ	81	91	101	111	121
	14KΩ	83	93	103	113	123
	18.7KΩ	85	95	105	115	125





5

4

3

2

1

D

D

C

C

B

B

A

A

(Blanking)

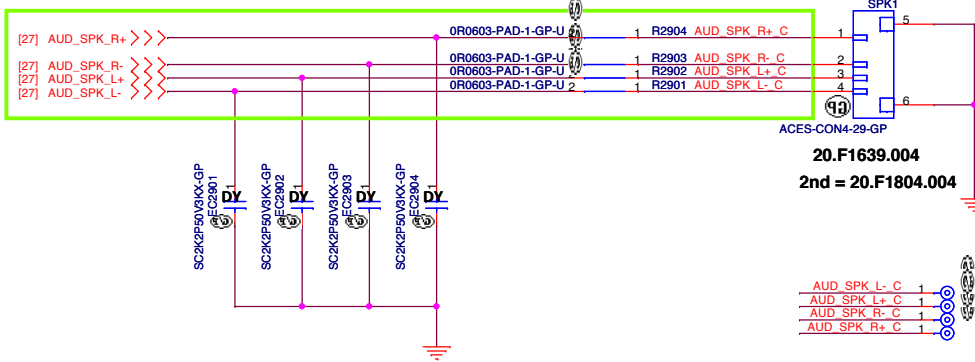
<Core Design>

			Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.		
Title					
Reserved					
Size	Document Number				Rev
A4	Janus HSW 40/50/70				A00
Date: Friday, February 07, 2014			Sheet	28	of 104

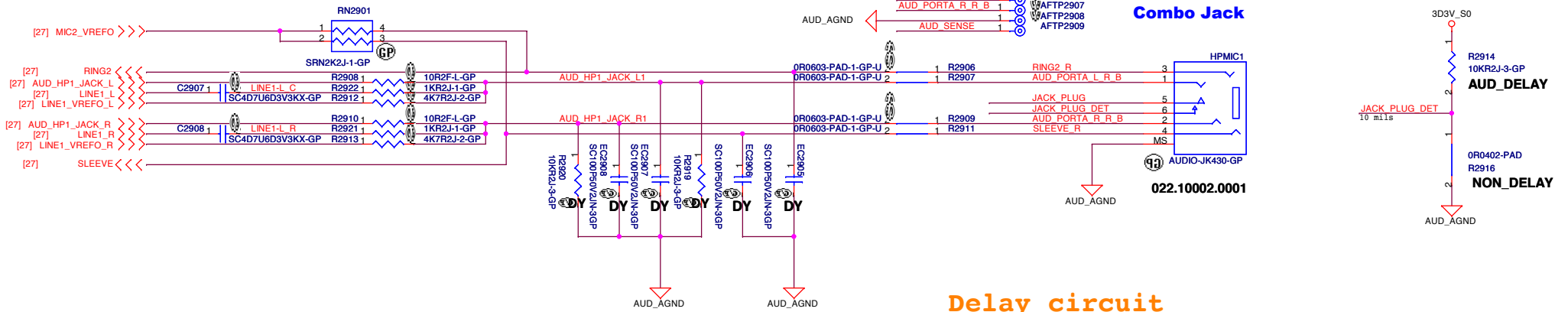
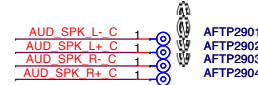
SSID = AUDIO

Layout Note:

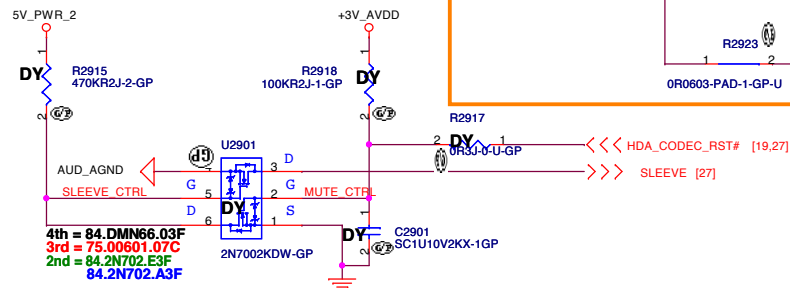
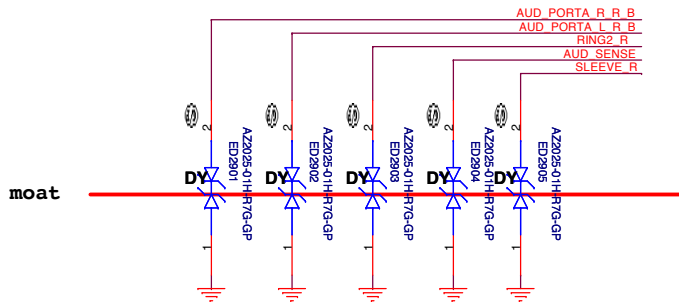
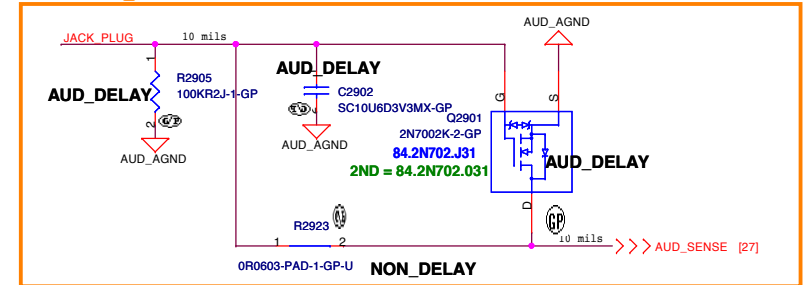
Speaker trace width >40mil @ 2W4ohm speaker power



CONN Pin	Net name
Pin1	SPK_R+
Pin2	SPK_R-
Pin3	SPK_L+
Pin4	SPK_L-



Delay circuit



<Core Design>

DELL **Wistron Corporation**
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Speaker/HPMIC

Size

Document Number

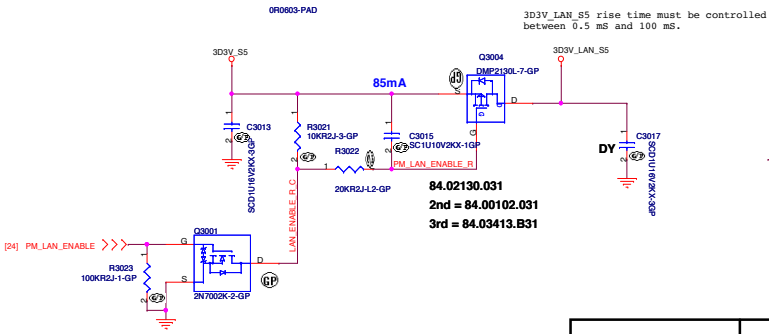
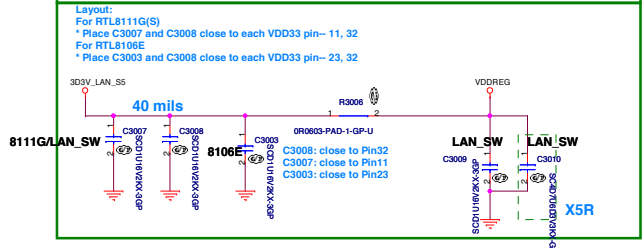
Janus HSW 40/50/70

Rev	
-----	--

Date: Friday, February 07, 2014

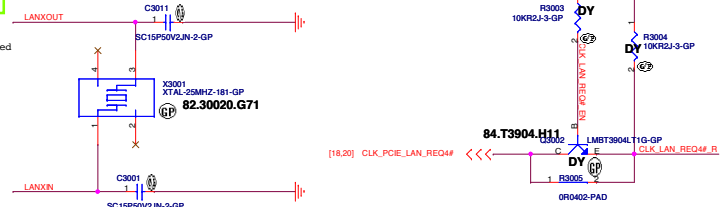
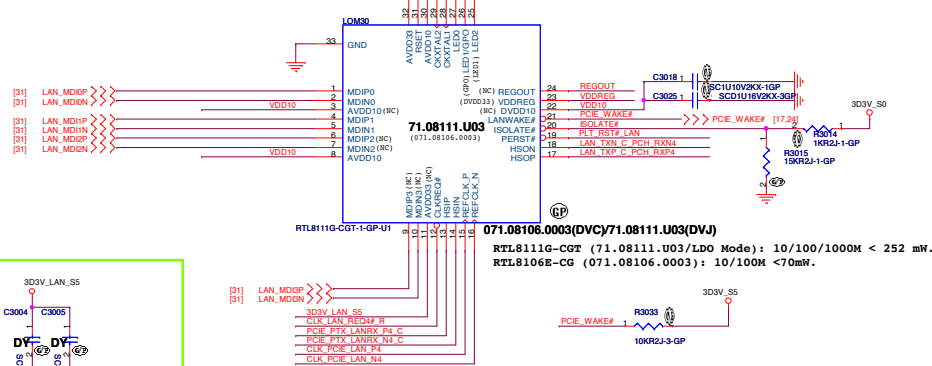
Sheet 29 of 104

104



LAN CHIP (10/100/1000M & 10/100M co-lay)

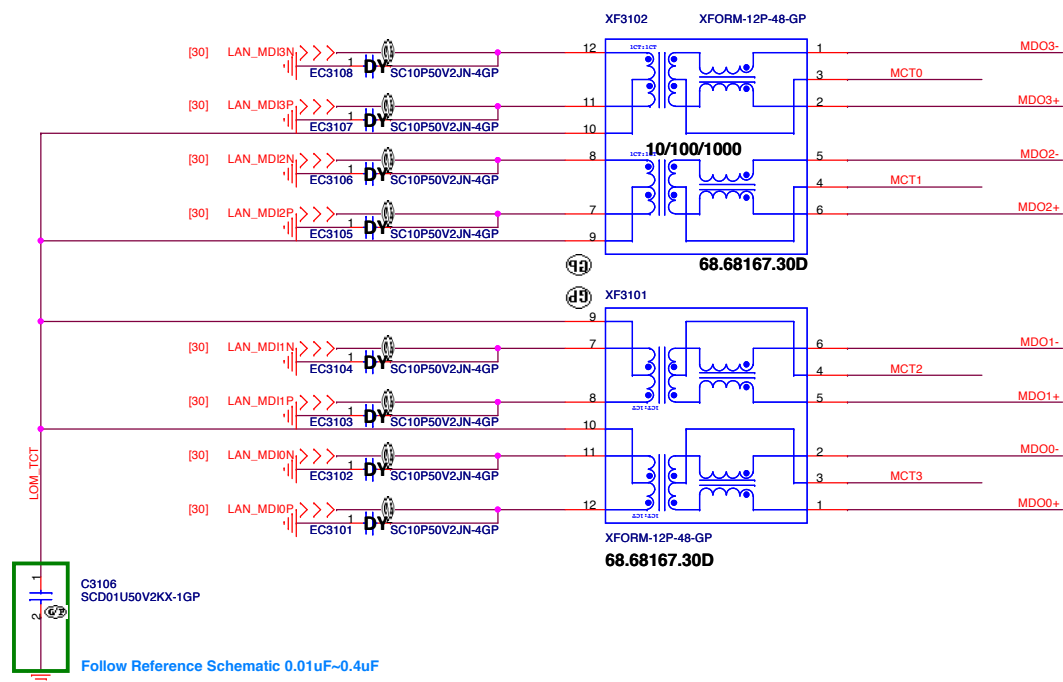
RTL8111GUS-CG	RTL8111G-ODT	RTL8106EUS-CG	RTL8106E-CG
71.08111.W03	71.08111.003	71.08106.003	071.08106.0003
SWR mode	LDO mode	SWR mode	LDO mode
10/100/1000M	10/100/1000M	10/100M	10/100M

[illegible]

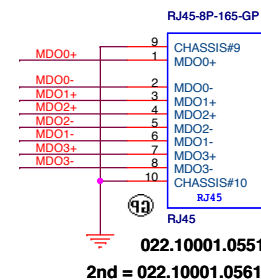
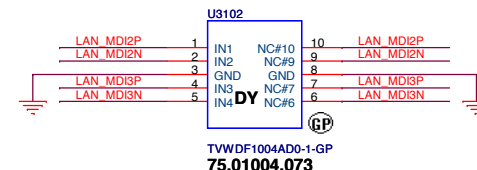
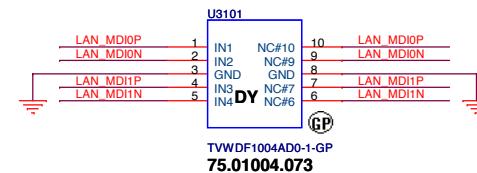
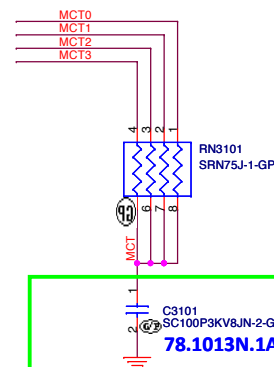
SSID = LOM

LAN TransFormer (10/100/1000M & 10/100M co-lay)

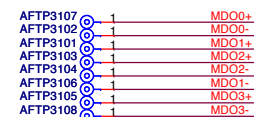
Layout note:
30 mil spacing between MDI differential pairs.



Follow Reference Schematic 0.01uF~0.4uF



Layout:
Place near RJ45



<Core Design>

DELL Wistron Corporation
21F, 88, Sec. 1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title			XFORM&RJ45	
Size	Document Number	Rev		
A3	Janus HSW 40/50/70	A00		
Date:	Monday, February 10, 2014	Sheet	31	of 104

5

4

3

2

1

D

D

C

C

B

B

A

A

(Blanking)

<Core Design>

			Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.		
Title					
(Reserved)Card Reader					
Size	Document Number				Rev
A4	Janus HSW 40/50/70				A00
Date: Friday, February 07, 2014			Sheet 32 of 104		

5

4

3

2

1

D

D

C

C

B

B

A

A

(Blanking)

<Core Design>

		Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title (Reserved)			
Size A4	Document Number Janus HSW 40/50/70		Rev A00
Date: Friday, February 07, 2014		Sheet 33 of	104

5

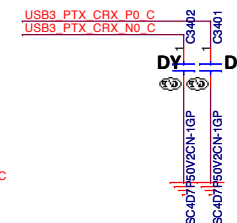
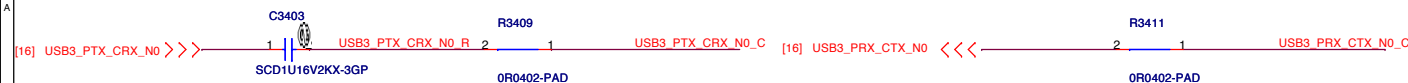
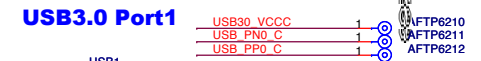
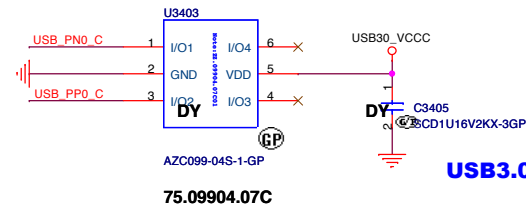
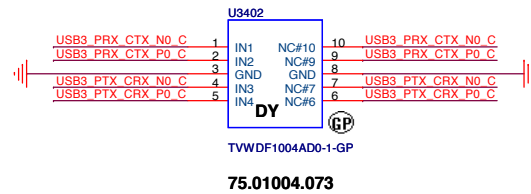
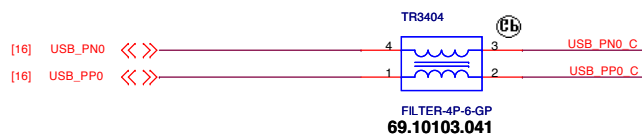
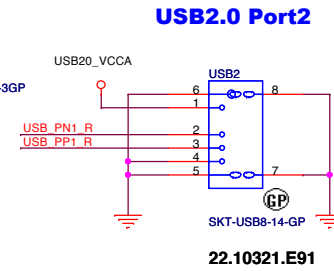
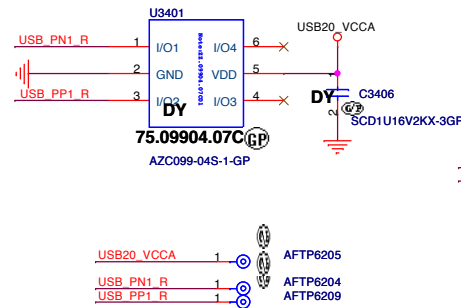
4

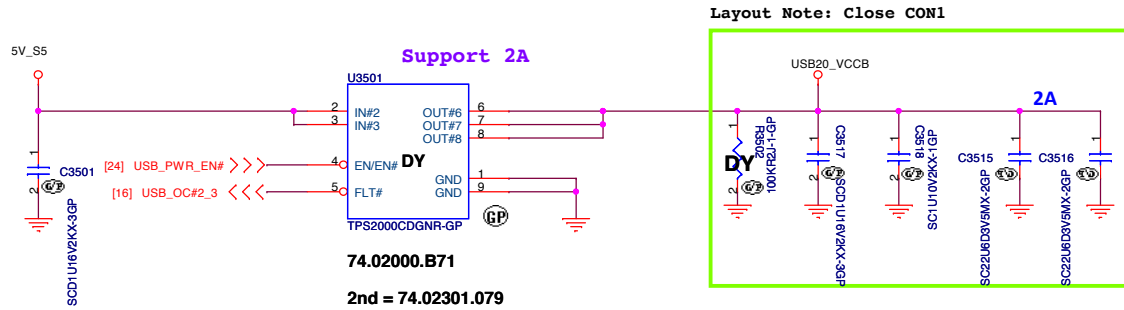
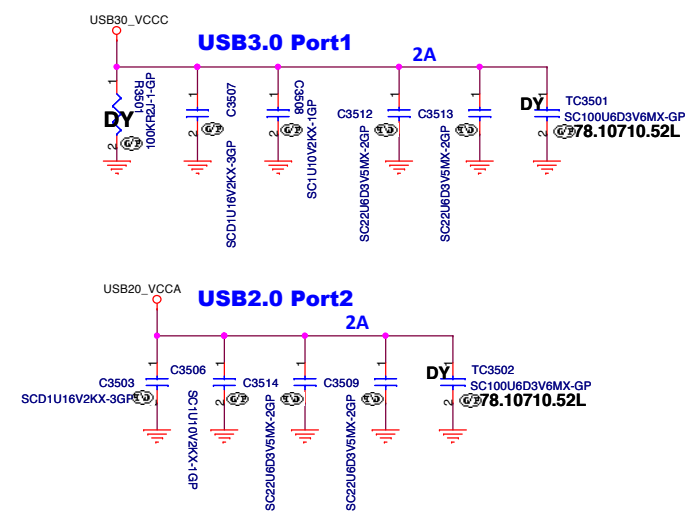
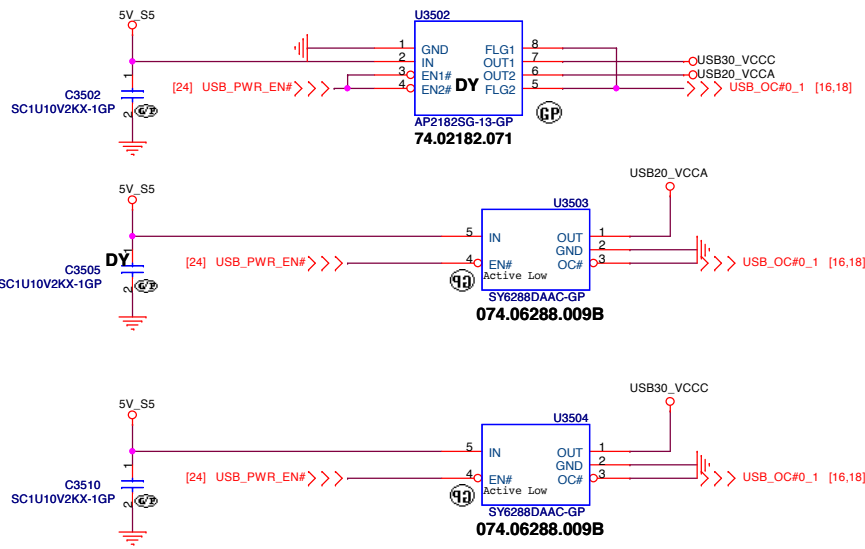
3

2

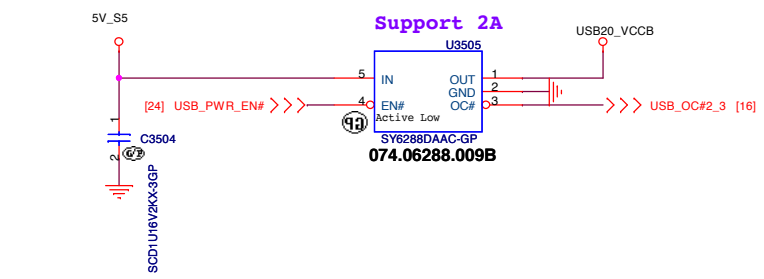
1

SSID = USB





USB2.0 Port3 (IO Board)



SODIMM1 M_VREF_CA_DIMMA 



Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

S3 Reduction Circuit

Janus HSW 40/50/70

Rev	A00
-----	-----

Sheet	37	of	104
-------	----	----	-----

(Blanking)

<Core Design>

		Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title (Reserved) 1D05_M			
Size A4	Document Number Janus HSW 40/50/70		Rev A00
Date: Friday, February 07, 2014		Sheet 39 of	104

(Blanking)

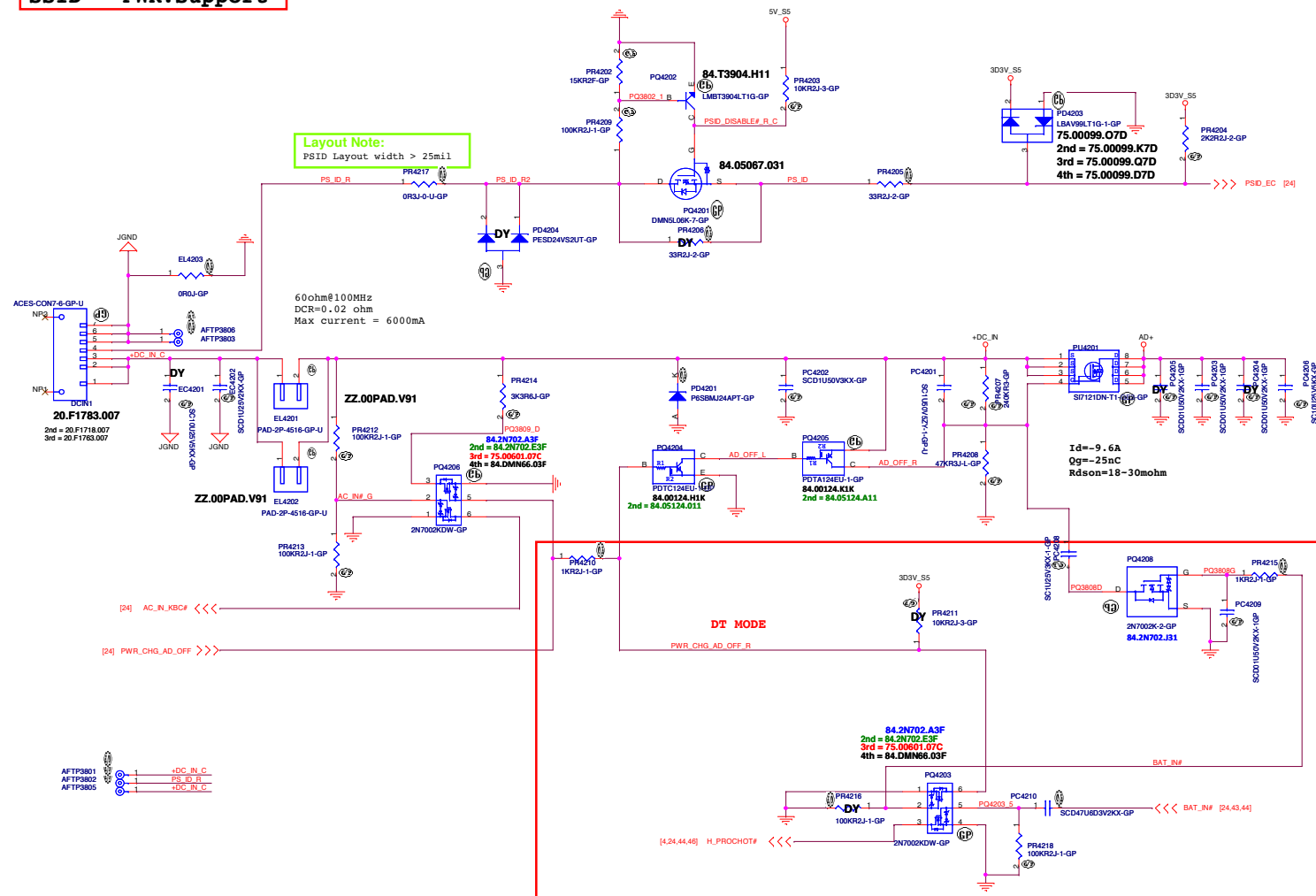
<Core Design>

		Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title			
<i>Reserved</i>			
Size A4	Document Number <i>Janus HSW 40/50/70</i>		Rev <i>A00</i>
Date: Friday, February 07, 2014		Sheet 40 of	104

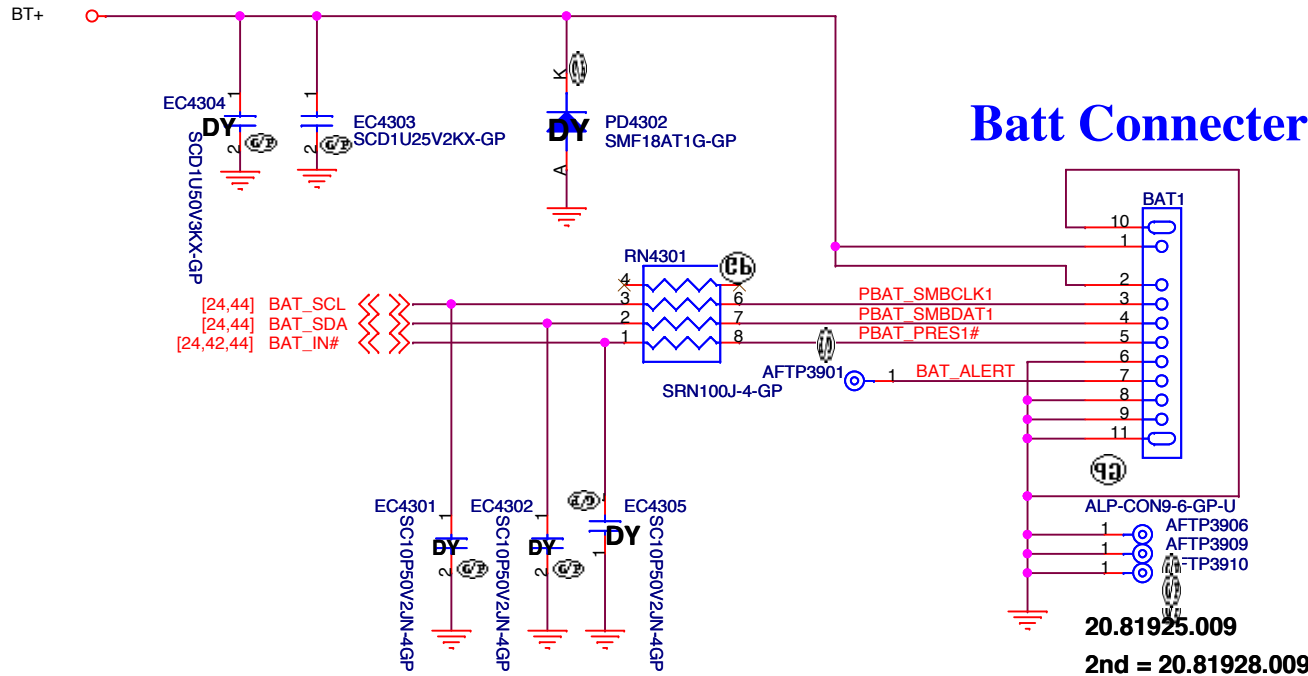
(Blanking)

<Core Design>

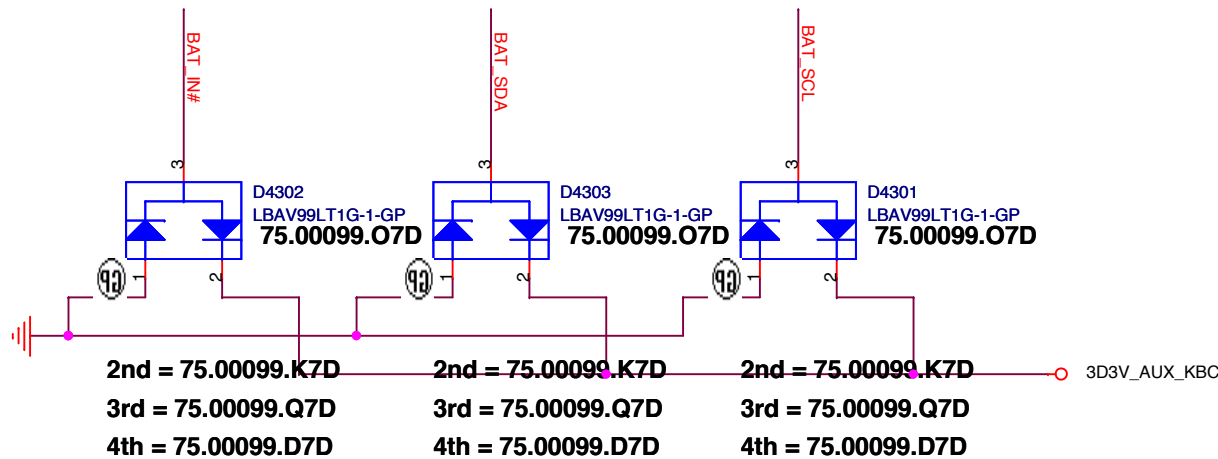
		Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title			
<i>Reserved</i>			
Size A4	Document Number <i>Janus HSW 40/50/70</i>		Rev <i>A00</i>
Date:	Friday, February 07, 2014	Sheet 41 of	104

SSID = PWR.Support

SSID = PWR.Support



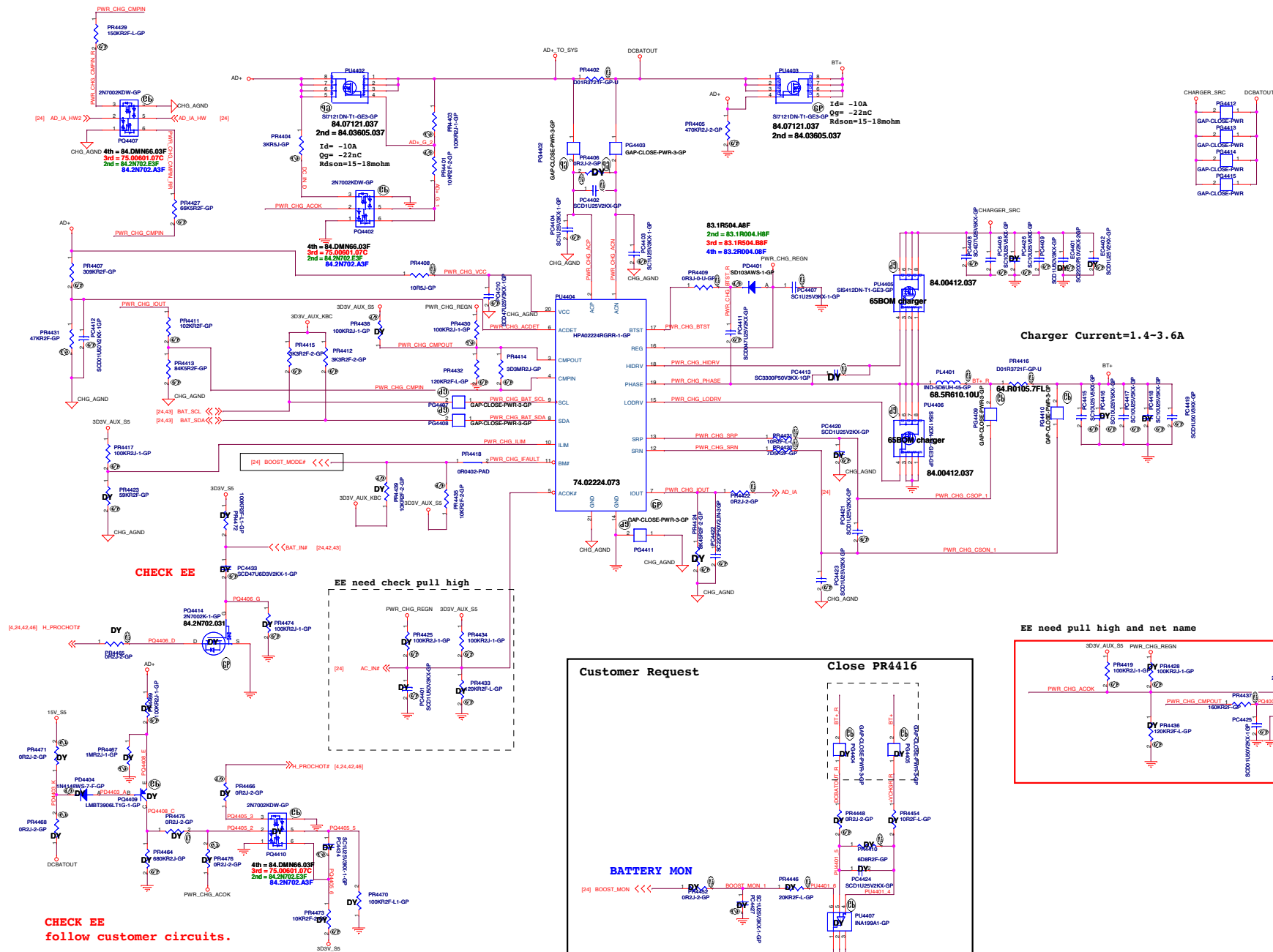
Placement: Close to Batt Connector



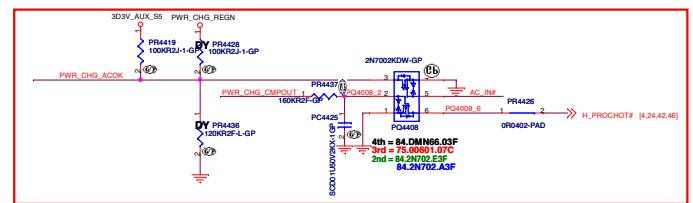
<Core Design>

DELL		Wistron Corporation	
		21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title			
BATT CONN			
Size A4	Document Number Janus HSW 40/50/70		Rev A00
Date: Friday, February 07, 2014	Sheet 43 of 104		

SSID = Charger



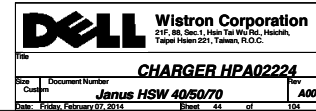
EE need pull high and net name

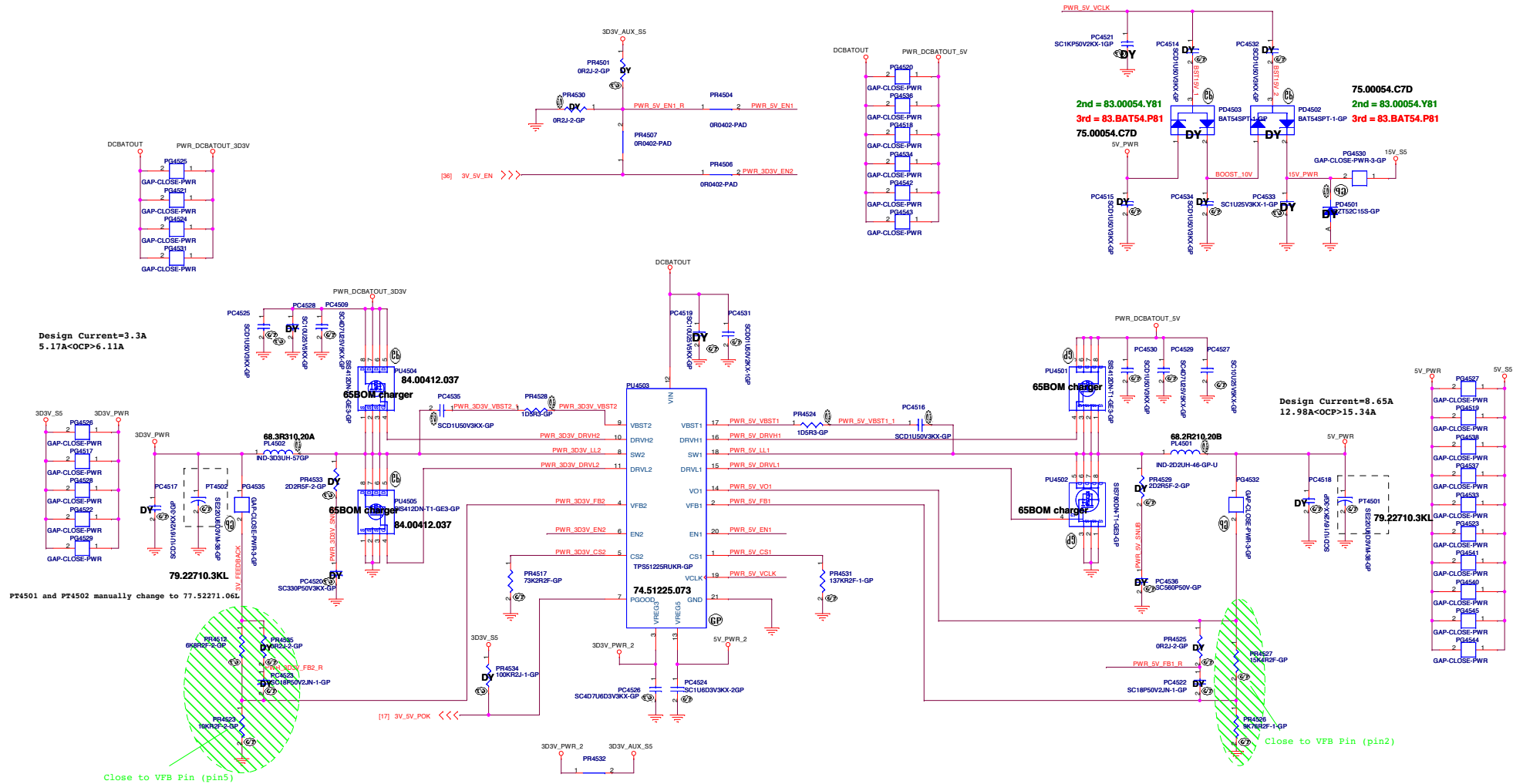


EC code only BQ24707

H_PROCHOT#	AD_IA_HW	AD_IA_HW2
45W	0	0
65W	1	0
90W	0	1

«Core Design»



SSID = PWR.Plane.Regulator_5v3p3v

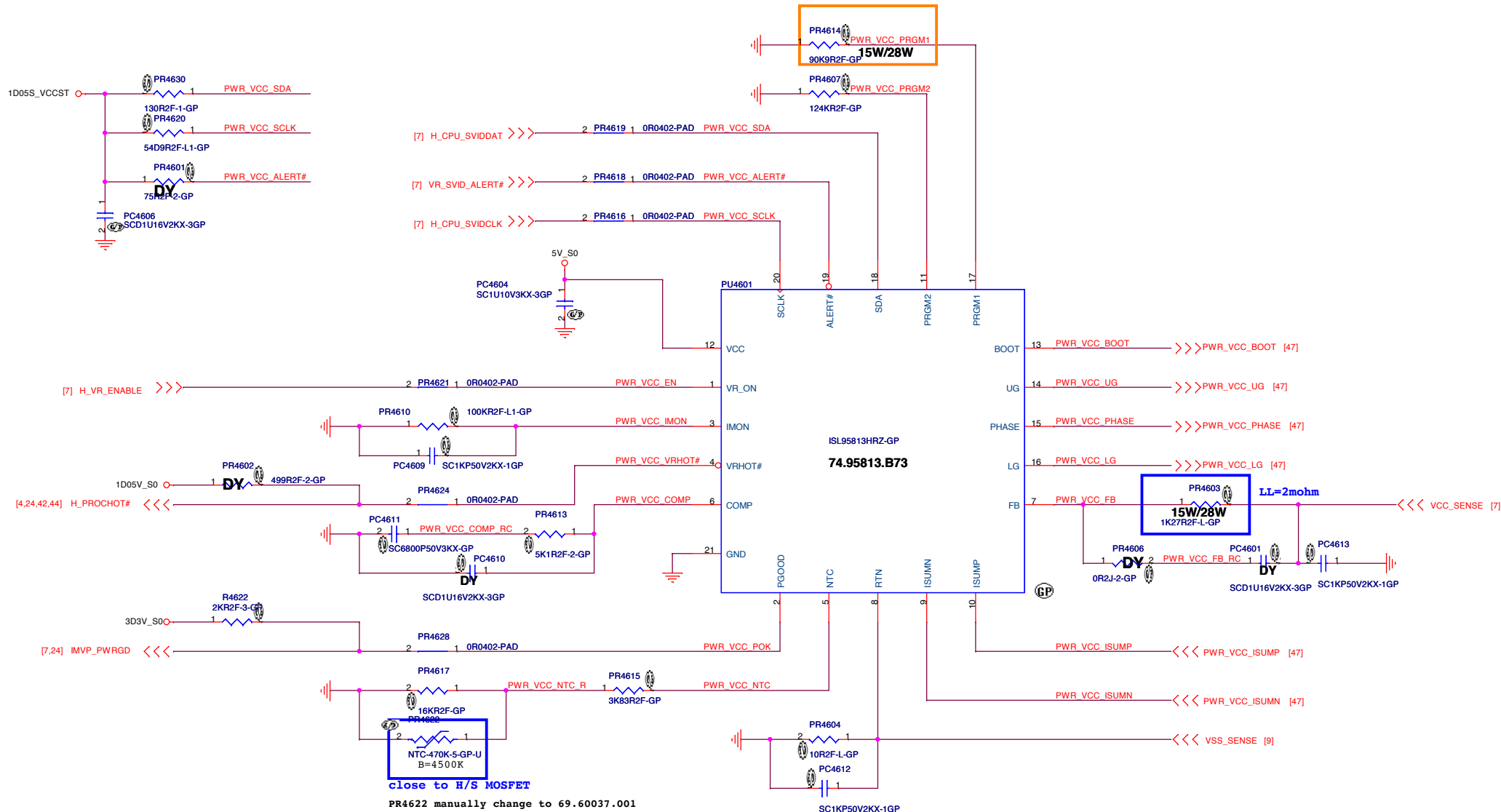
I/P cap: CHIP CAP C 10U 25V K0805 X5R/ 78.10622.51L
Inductor: CHIP IND 3.3UH P0603T-3R3NM Cyntec 28mohm/30mohm Isat =13.5Arms 68.3R310.20A
O/P cap:CHIP CAP POL 220U 6.3V M.6.3+4.5 /Matsuki/ 17mOhm / 77.52271.09L
H/S:SI5412 / 24mOhm/30mOhm@4.5Vgs / 84.00412.037
L/S:SI5780 / 14.5mOhm/17.5mOhm@4.5Vgs / 84.00780.037

TPS51225 & TPS51285 Co-lay

	TPS51225	TPS51285
PR4510	45.3KK	9.09K
PR4511	110K	22.1K

I/P cap: CHIP CAP C 10U 25V K0805 X5R/ 78.10622.51L
Inductor: CHIP CHOKO 2.2U PCMC063T-2R2MN 18mohm/20mohm Isat =14Arms 68.2R210.20B
O/P cap:CHIP CAP POL 220U 6.3V M 6.3*4.5/ Matsuki/ 17mOhm / 77.52271.09L
H/S:ISi412 / 24mOhm/30mOhm4.5Vgs / 84.00412.037
L/S:ISiS780 / 14.5mOhm/17.5mOhm4.5Vgs / 84.00780.037

SSID = CPU.Regulator



	PR4603	PR4614
15W	1.27K 64.12715.6DL	90.9K 64.90925.6DL
28W	1.58K 64.15815.6DL	113K 64.11335.6DL

<Core Design>

Wistron Corporation

21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.

Title

ISL95813_CPUCORE(1/2)

Size

A3

Document Number

Janus HSW 40/50/70

Rev

A00

Date:

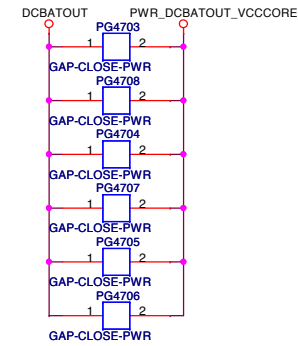
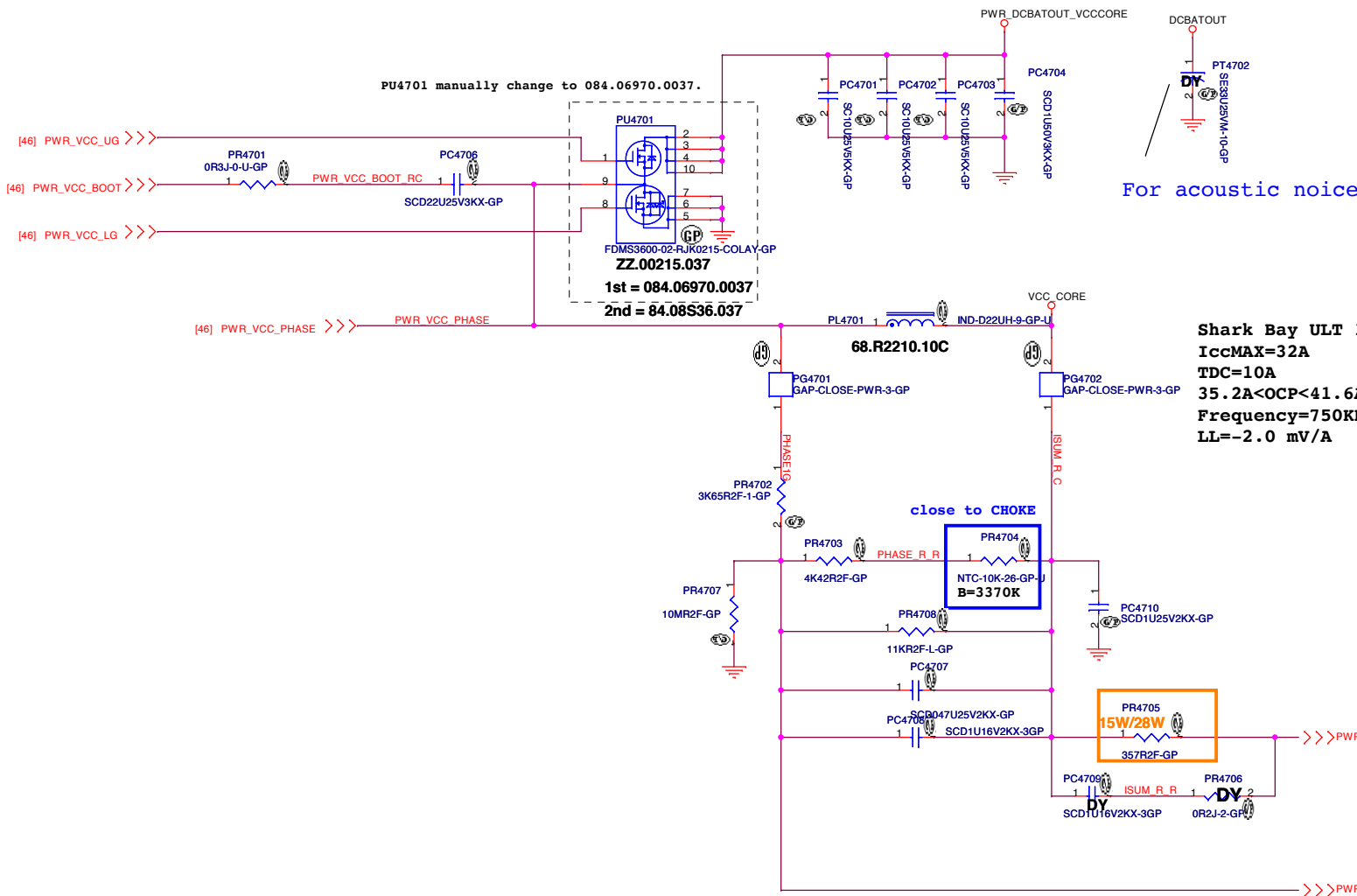
Friday, February 07, 2014

Sheet

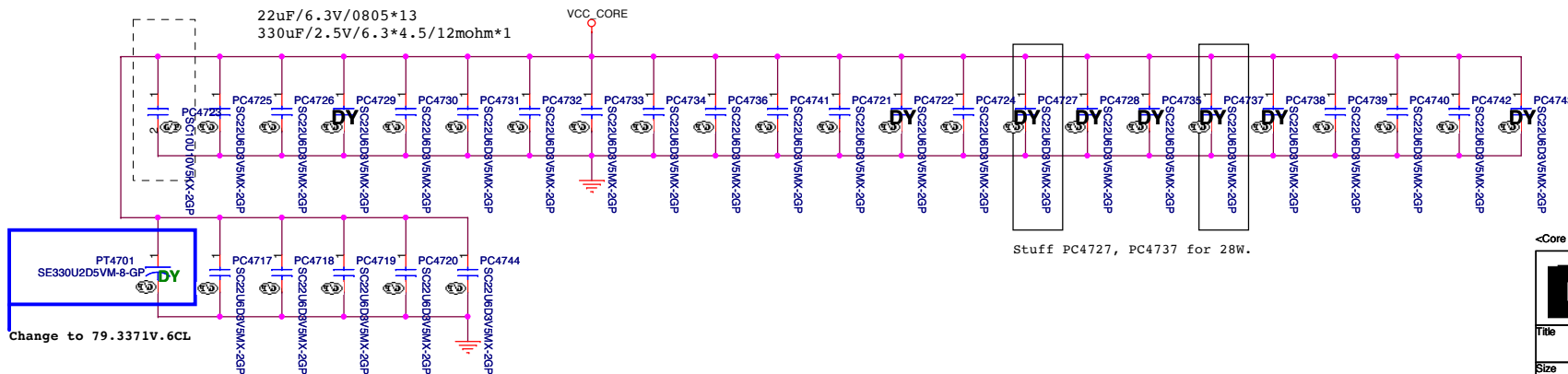
46

of

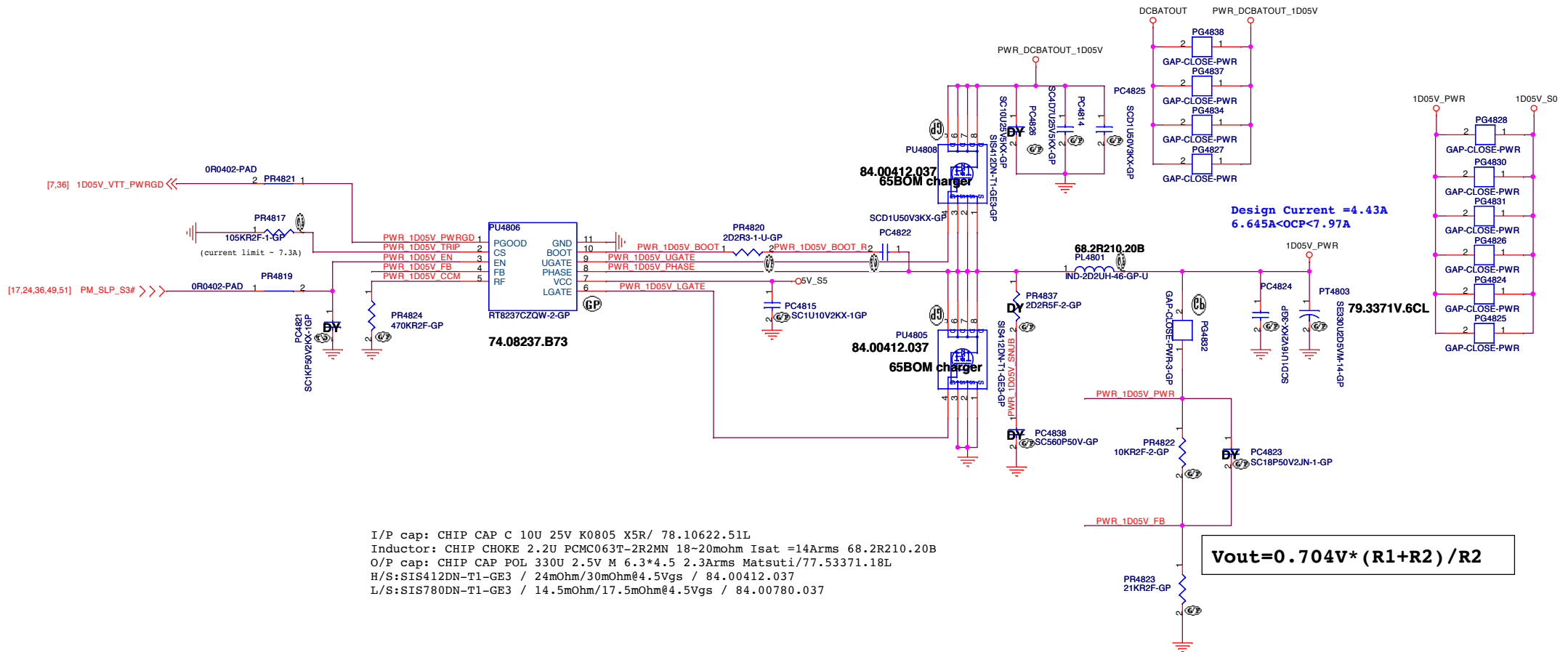
104



	PR4705 (Cyntec)	OCP
15W	357 ohm (64.35705.6DL)	38A
28W	412 ohm (64.41205.6DL)	48A
	PR4705 (Maglayers)	
15W	383 ohm (64.38305.6DL)	38A
28W	464 ohm (64.46405.6DL)	48A

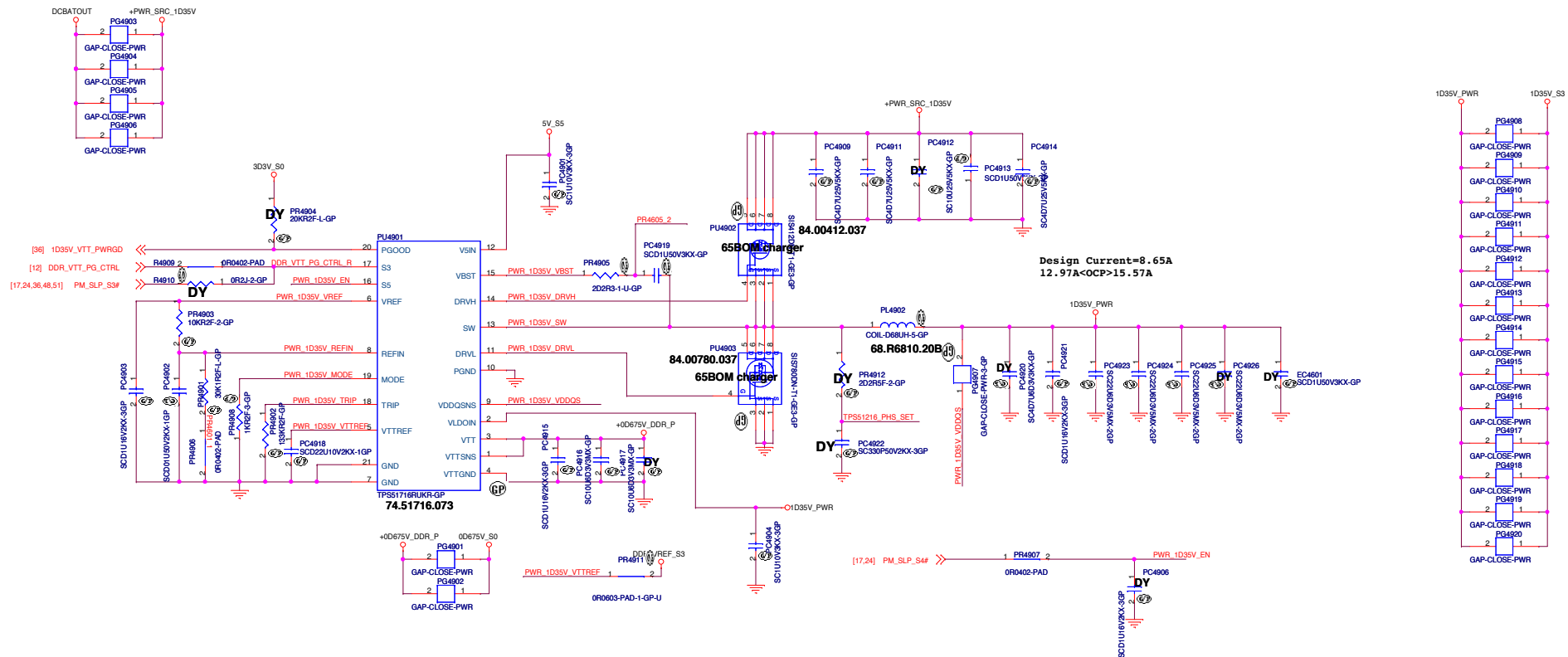


SSID = PWR.Plane.Regulator_1p05v



<Core Design>

SSID = PWR.Plane.Regulator_1p35v0p675v



State	S3	S5	VDDR	VTTREF	VTT
S0	Hi	Hi	On	On	On
S3	Lo	Hi	On	On	Off (Hi-Z)
S4/S5	Lo	Lo	Off	Off	Off

I/P cap: 10U 25V K0805 X5R/ 78.10622.51L
Inductor: CHIP IND 0.1UH M PCMC063T-R10MN 1.5~1.7mohm Isat =60Arms 68.R1010.10T
O/P cap: CHIP CAP POL 330U 2.5V M 6.3*4.5 2.3Arms Matsui/77.53371.18L
MOS: FET MOS FDMS3664S NC POWER56 / 84.03664.037 / Q1: 8.5~11mohm @Vgs=4.5V Q2: 2.6~3.2mohm @Vgs=4.5V

(Blanking)

<Core Design>



Wistron Corporation
21F, 88, Sec. 1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Reserved

Size
A3

Document Number
Janus HSW 40/50/70

Rev
A00

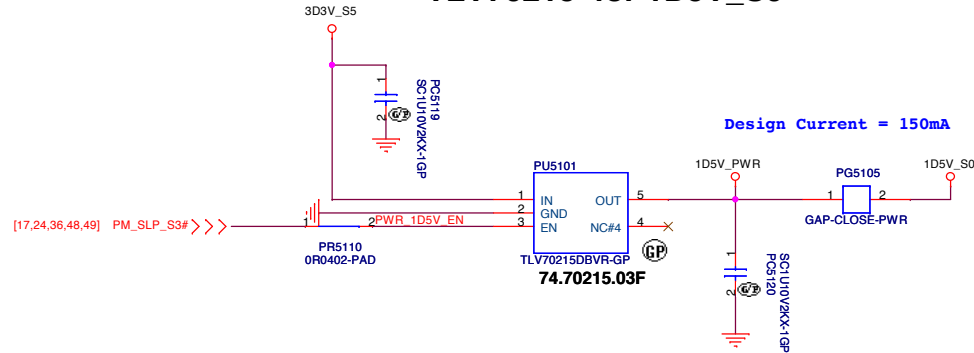
Date: Friday, February 07, 2014

Sheet 50 of 104

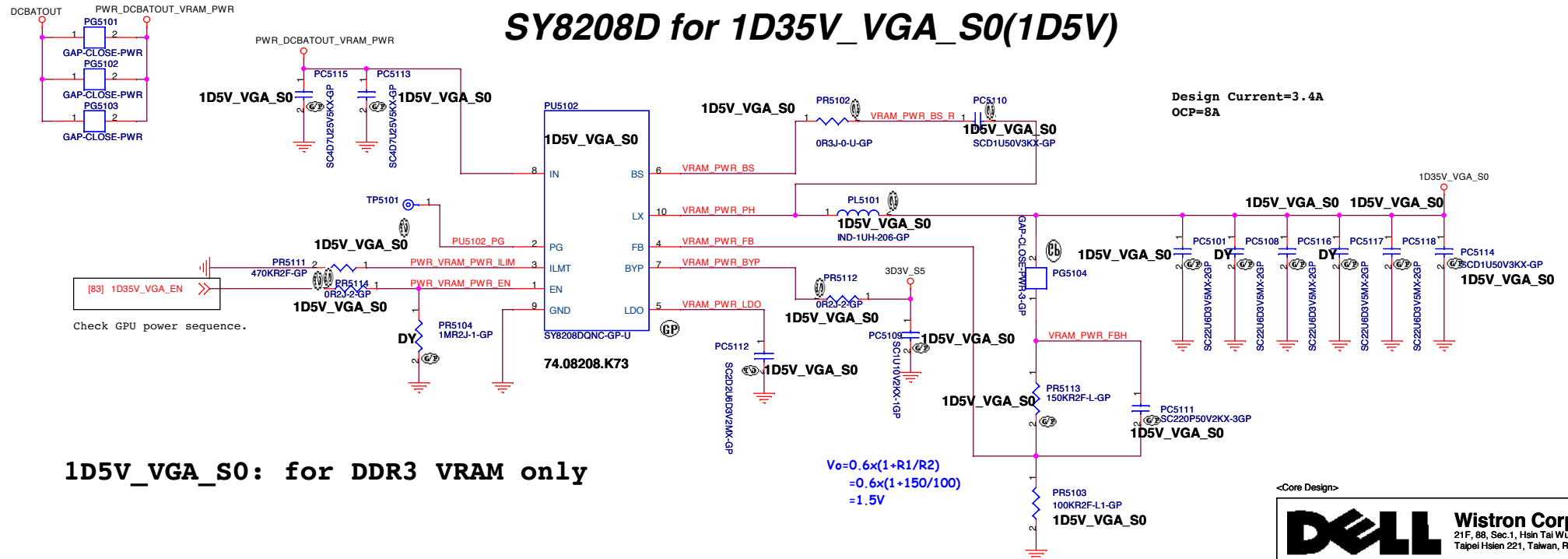
1

SSID = PWR.Plane.Regulator_1p5v

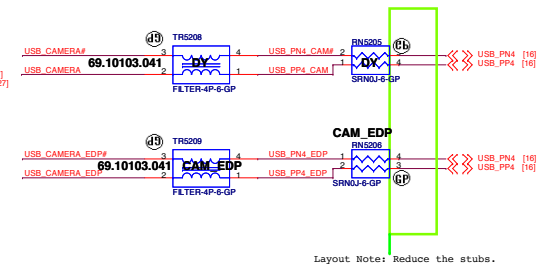
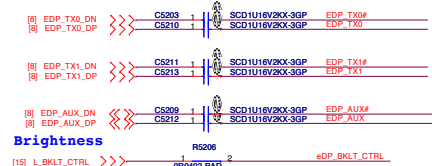
TLV70215 for 1D5V_S0



SY8208D for 1D35V_VGA_S0(1D5V)



1D5V_VGA_S0: for DDR3 VRAM only



(Blanking)

<Core Design>



Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

(Reserved)

Size	Document Number	Rev
A3	Janus HSW 40/50/70	A00

Date: Friday, February 07, 2014	Sheet 53 of 104
---------------------------------	-----------------

(Blanking)

<Core Design>



Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

HDMI Level Shifter/Connector

Size

A3

Document Number

Janus HSW 40/50/70

Date

Friday, February 07, 2014

Rev

X02

Sheet

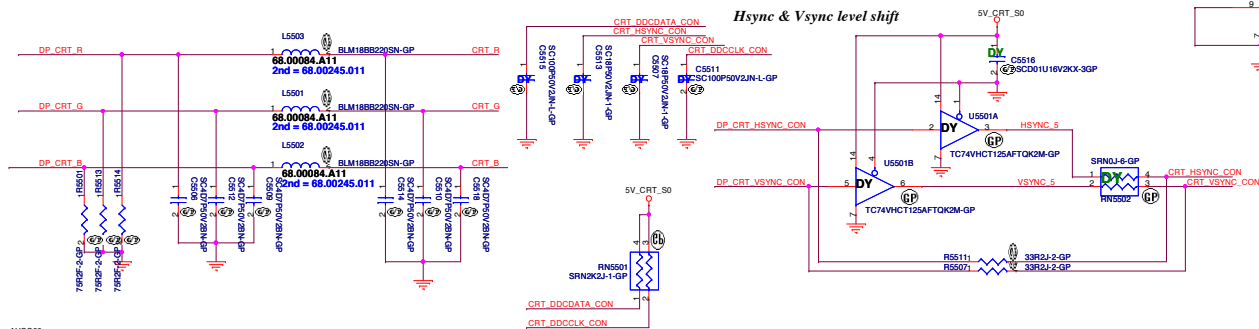
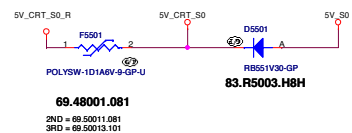
54

 of

104

The schematic shows the following connections:

- VCC_CRT_S0_R**: Connected to pin 9 of the SCDD01UH8V2KX-L1-GP.
- GND**: Connected to pins 10, 11, 12, 13, 14, 15, 16, 17, and 18 of the SCDD01UH8V2KX-L1-GP.
- CRT_DDCDATA_CON**: Connected to pin 12 of the SCDD01UH8V2KX-L1-GP.
- CRT_DDCCLK_ID3**: Connected to pin 13 of the SCDD01UH8V2KX-L1-GP.
- CRT_A**: Connected to pin 14 of the SCDD01UH8V2KX-L1-GP.
- CRT_B**: Connected to pin 15 of the SCDD01UH8V2KX-L1-GP.
- CRT_VSYNCON**: Connected to pin 16 of the SCDD01UH8V2KX-L1-GP.
- CRT_HSYNCON**: Connected to pin 17 of the SCDD01UH8V2KX-L1-GP.
- D-SUB-15-252-GP**: Connected to pins 18 through 22 of the SCDD01UH8V2KX-L1-GP.



The figure contains two schematic diagrams for the OR0603-PAD-1-GP-U package. The top diagram shows the AVCC33 connection. It features a 30V_S0 input connected to pin 1 of the package. Pin 2 is connected to a 10k resistor (R5504), which is then connected to the AVCC33 pin. A 0.1uF capacitor (C5523) is connected between AVCC33 and ground. The bottom diagram shows the VDD_DAC_33 connection. It features a 30V_S0 input connected to pin 1. Pin 2 is connected to a 10k resistor (R5503), which is then connected to the VDD_DAC_33 pin. A 0.1uF capacitor (C5521) is connected between VDD_DAC_33 and ground.

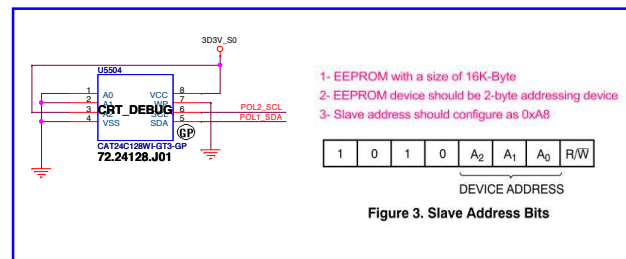
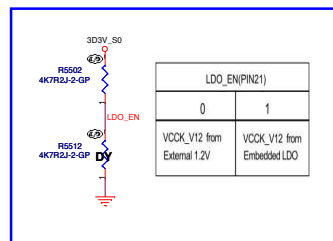
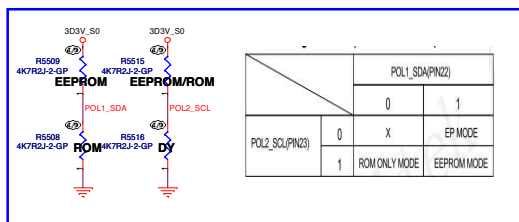
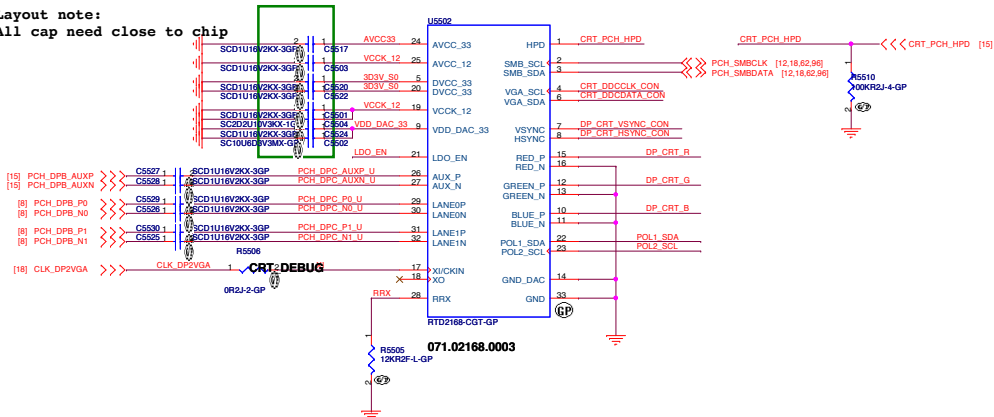
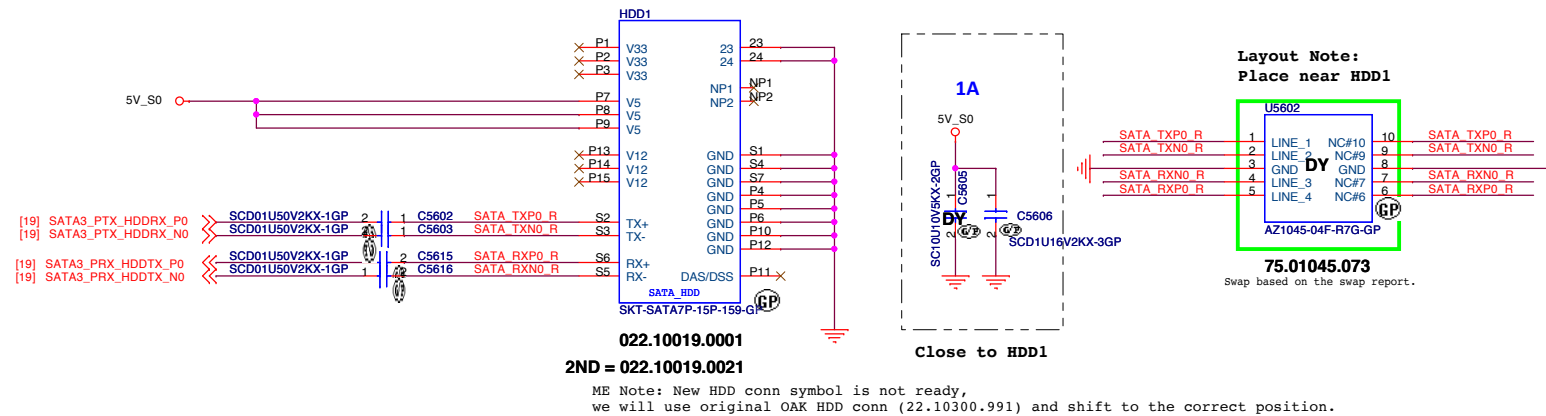
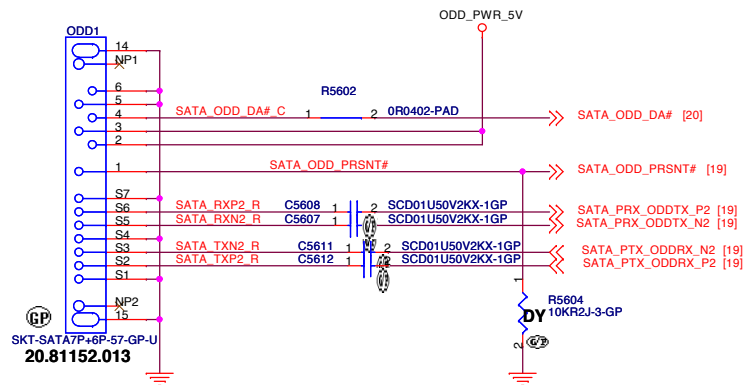


Figure 3. Slave Address Bits

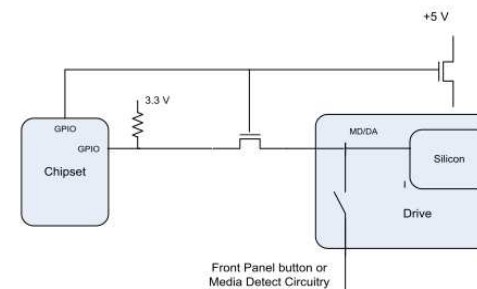
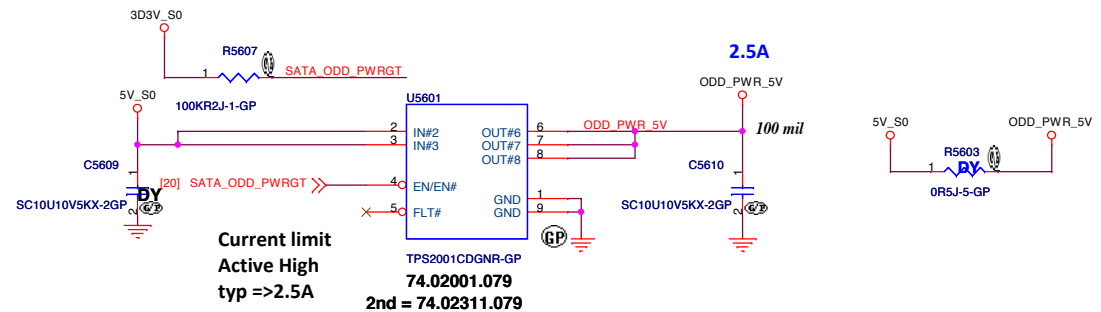
SATA HDD Connector



ODD Connector



SATA Zero Power ODD



<Core Design>



Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichin,
Taipei Hsien 221, Taiwan, R.O.C.

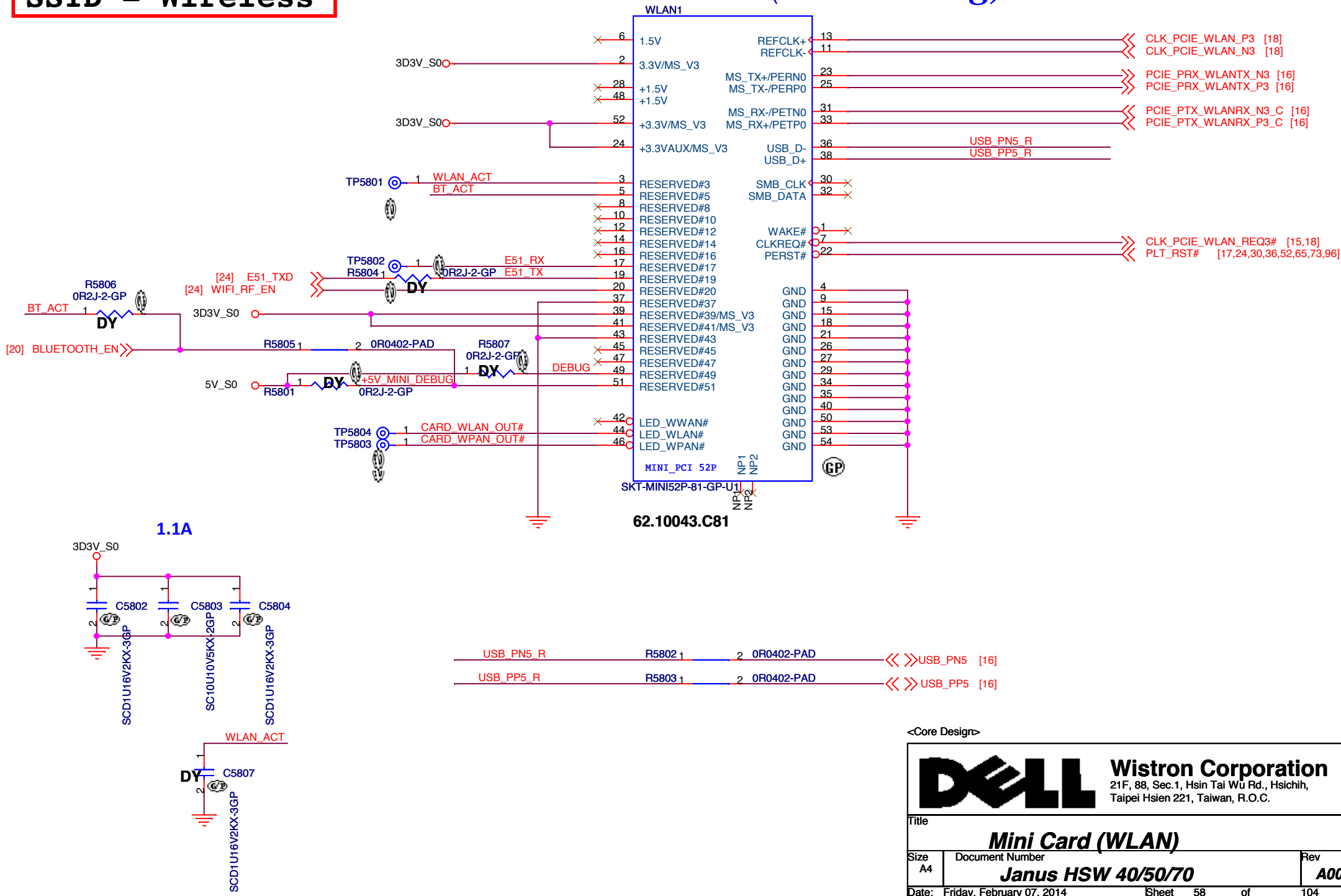
File			HDD/ODD	
Size	Document Number	Rev		
A3		Janus HSW 40/50/70		A00
Date:	Friday, February 07, 2014	Sheet	56	of 104

SSID = ESATA

(Blanking)

SSID = Wireless

Mini Card Connector(802.11a/b/g)



<Core Design>



Wistron Corporation

21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Mini Card (WLAN)

Size

A4

Document Number

Janus HSW 40/50/70

Rev

A00

Date: Friday, February 07, 2014

Sheet 58 of 104

(Blanking)

<Core Design>



Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Reserved

Size
A4

Document Number
Janus HSW 40/50/70

Rev
A00

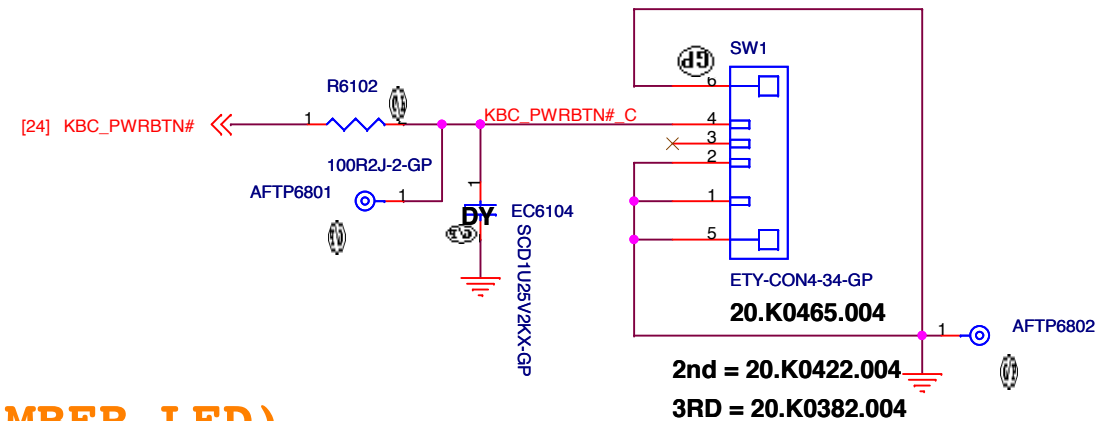
Date: Friday, February 07, 2014Sheet 59 of 104

(Blanking)

<Core Design>

			Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.		
Title (Reserved)					
Size A4	Document Number Janus HSW 40/50/70				Rev A00
Date: Friday, February 07, 2014			Sheet 60 of 104		

Power button



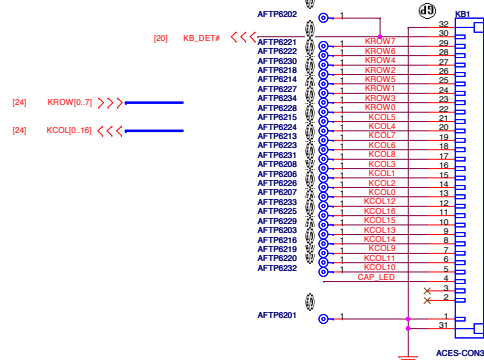
Battery LED2 (WHITE_LED)
Low actived from KBC GPIO



Sheet 61 of 104

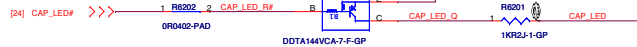
SSID = KBC

Internal Keyboard Connector (DVC40)



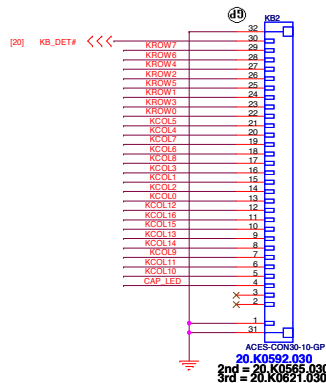
20.K0592.030
2nd = 20.K0565.030
3rd = 20.K0621.030

CAP LED Control
LOW actived from KBC GPIO

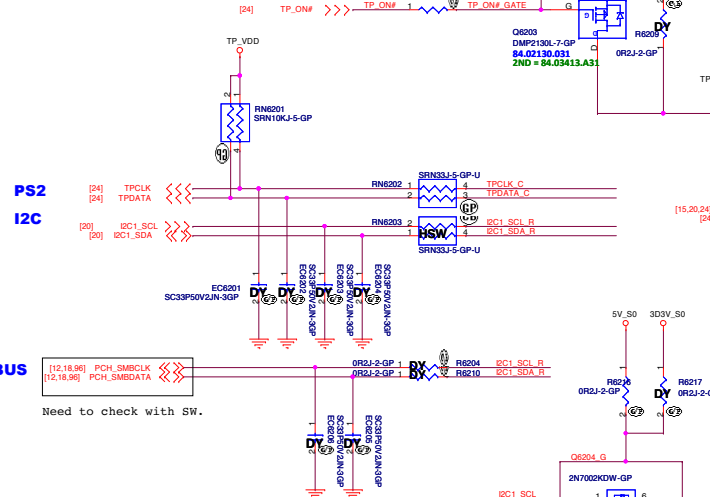


84.00144.N11

Internal Keyboard Connector (DVC50/DVC70)

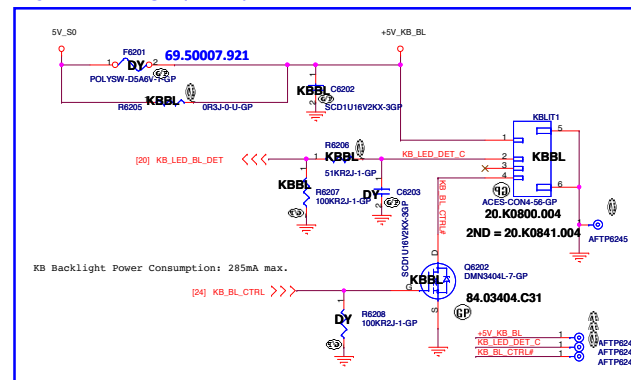


2nd = 20.K0565.030
3rd = 20.K0621.030

SSID = Touch.Pad

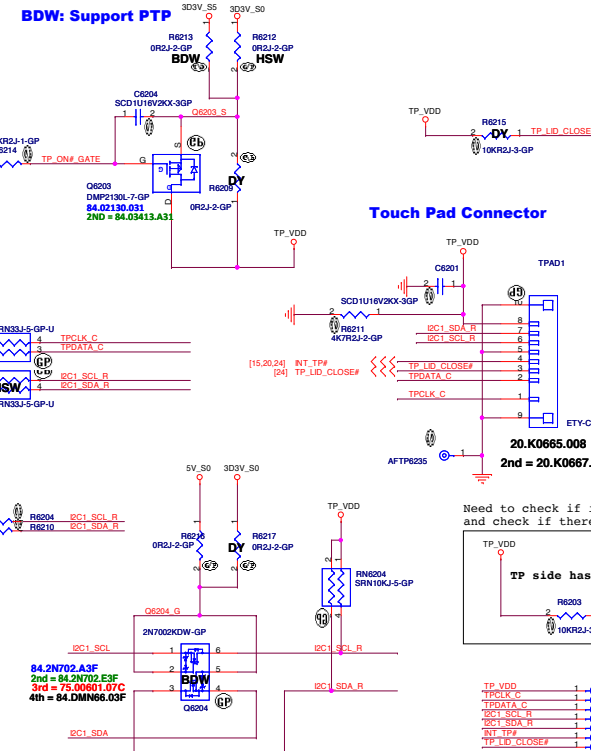
Need to check with SW.

Keyboard Backlight (DVC70)



KB Backlight Power Consumption: 285mA max.

BDW: Support PTP 3D3V_S5 3D3V_S0



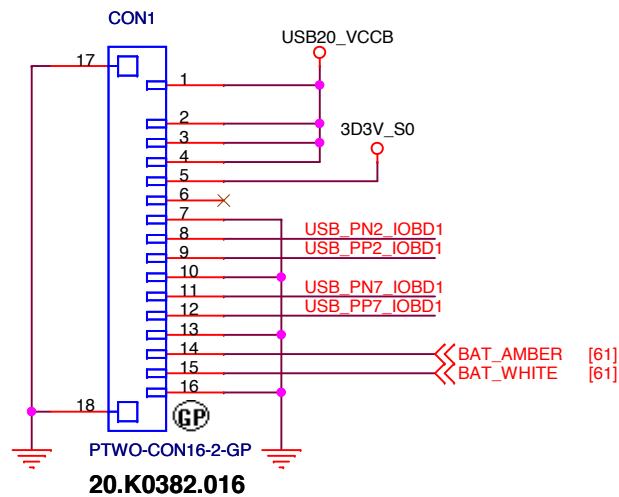
Touch Pad Connector

Pin number	Pin name
1	VDD
2	DAT(I2C)
3	CLK(I2C)
4	GND
5	ATTN
6	GPIO
7	DAT(PS2)
8	CLK(PS2)

Need to check if it is Active High or Active Low
and check if there is PH on TPAD side.

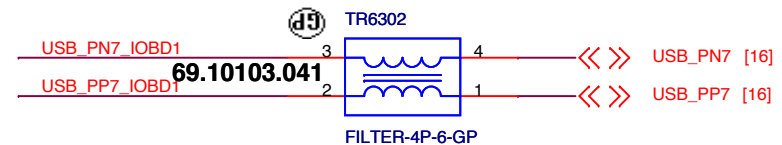
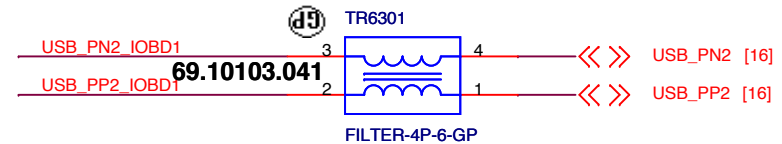
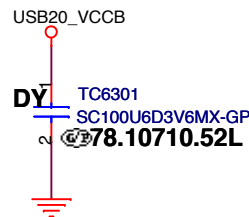
TP side has pull high

TP_VDD	1	1	AFTP623
TPCLK_C	1	1	AFTP623
TPDATA_C	1	1	AFTP623
I2C1_SCL_R	1	1	AFTP623
I2C1_SDA_R	1	1	AFTP623
INT_TP#	1	1	AFTP623
TP_LD_CLOSE#	1	1	AFTP623



USB2.0 Port3 Card Reader LED

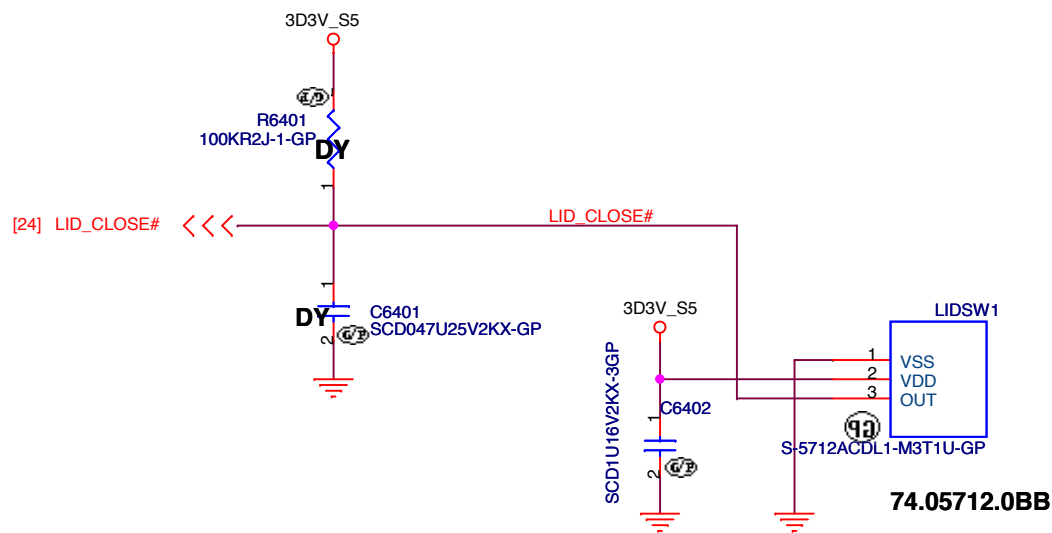
The maximum range of the PMOS output current in RTS5170 (Card Reader IC) is 400mA



<Core Design>

DELL		Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title IO Board Connector			
Size A4	Document Number Janus HSW 40/50/70		Rev A00
Date: Friday, February 07, 2014		Sheet 63 of 104	

SSID = User.Interface



<Core Design>



Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Hall Sensor

Size
A4

Document Number

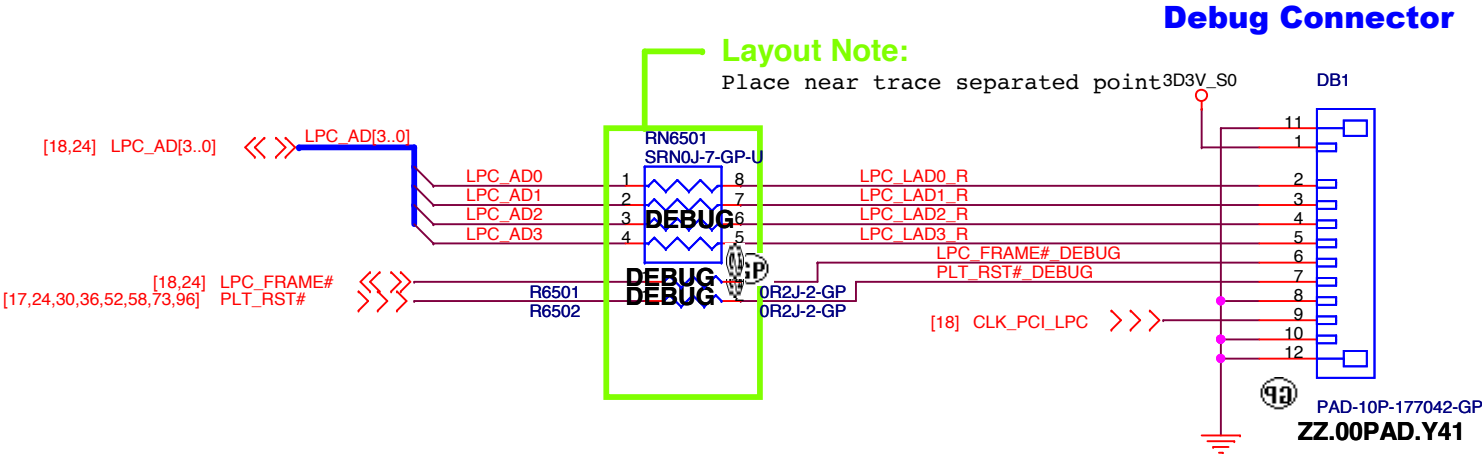
Janus HSW 40/50/70

Rev
A00

Date: Friday, February 07, 2014

Sheet 64 of 104

SSID = DEBUG PORT



20.D0075.110: Dummy Pad with solder mask is ZZ.00PAD.Y41
DB1 Optional: New one smaller LPC connector is 20.F1180.010.

<Core Design>

		Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title Dubug connector			
Size A4	Document Number Janus HSW 40/50/70		Rev A00
Date: Friday, February 07, 2014		Sheet 65 of 104	

(Blanking)

<Core Design>

		Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title			
<i>Reserved</i>			
Size A4	Document Number <i>Janus HSW 40/50/70</i>		Rev <i>A00</i>
Date: Friday, February 07, 2014		Sheet 66 of	104

(Blanking)

<Core Design>



Wistron Corporation
21F, 88, Sec. 1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Size
A3

Document Number
Janus HSW 40/50/70

Date: Friday, February 07, 2014

Rev
A00

Sheet 67 of 104

Reserved

(Blanking)

<Core Design>



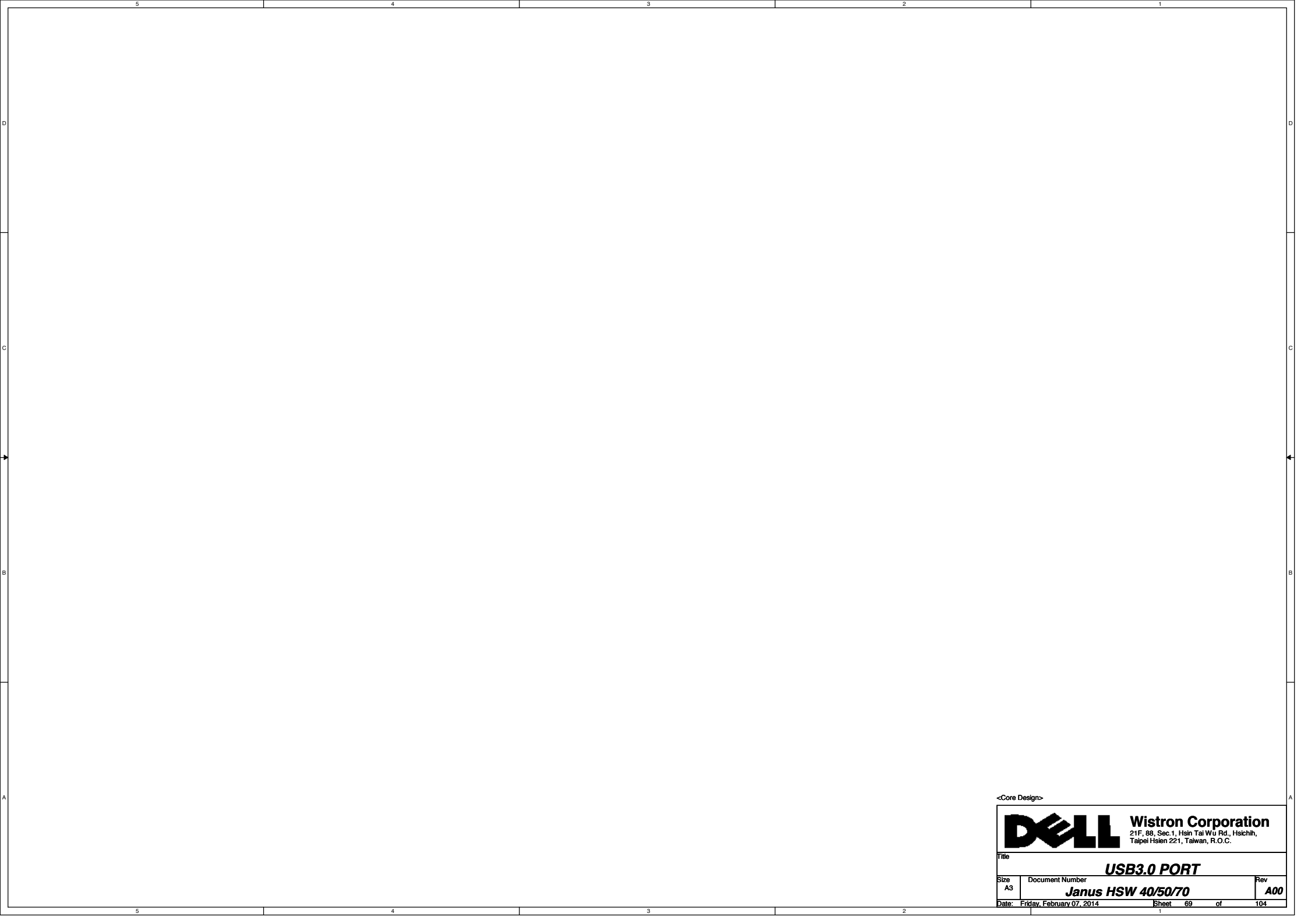
Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

RESERVED

Size A3	Document Number Janus HSW 40/50/70	Rev A00
------------	--	-------------------

Date: Friday, February 07, 2014	Sheet 68 of 104
---------------------------------	-----------------



<Core Design>

		Wistron Corporation 21F, 88, Sec. 1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title			
USB3.0 PORT			
Size	Document Number		Rev
A3	Janus HSW 40/50/70		A00
Date:	Friday, February 07, 2014	Sheet	69 of 104

(Blanking)

<Core Design>



Wistron Corporation
21F, 88, Sec. 1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Reserved

Size	Document Number	Rev
A3	Janus HSW 40/50/70	A00

Date:	Friday, February 07, 2014	Sheet	70	of	104
-------	---------------------------	-------	----	----	-----

(Blanking)

<Core Design>



Wistron Corporation
21F, 88, Sec. 1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Size

A3

Document Number

Janus HSW 40/50/70

Date: Friday, February 07, 2014

Sheet 71 of 104

Rev

A00

Reserved

(Blanking)

<Core Design>



Wistron Corporation
21F, 88, Sec. 1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Reserved

Size	Document Number	Rev
A3	Janus HSW 40/50/70	A00

Date:	Friday, February 07, 2014	Sheet	72	of	104
-------	---------------------------	-------	----	----	-----

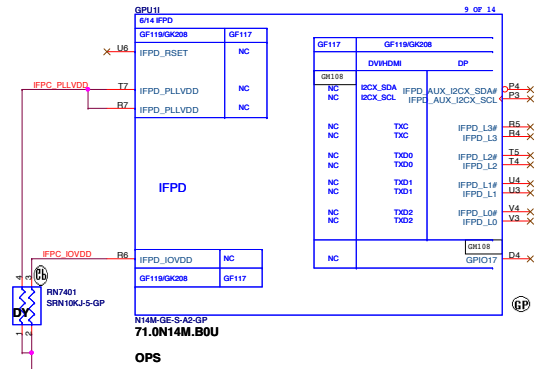
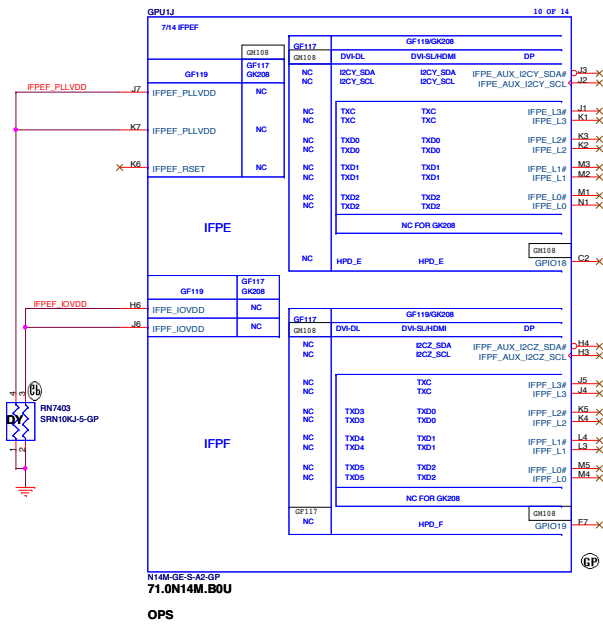
[illegible][illegible]

Table 3-33. SP_PLLVDD and VID_PLLVDD Power Rail Filtering Combined(RVL-06891-001)N15V- **GT** -S DDR3L Recommended MemoriesTable 15-2. Resistance Mapping to Hex Values

(DS-06814-001)

Table 9. N15V-GM Binary Strap Mode Mapping

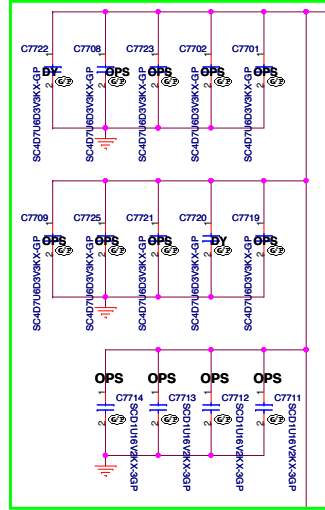
(RVL-06891-001)N15V- **GM** -S DDR3L Recommended Memories

(DS-06814-001)

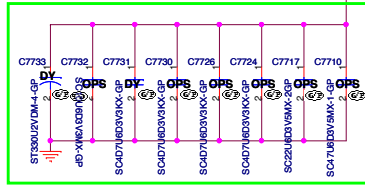
Table 10. Multi-Level Strap Differences

 Wistron Corporation 21F, 8B, Sec. 1, Hsin Tai Wu Rd., Hsuehli, Taipei Hsein 221, Taiwan, P.O.C.		
File		
GPU POWER(4/5)		
Size Custom	Document Number Janus HSW 40/50/70	New AC

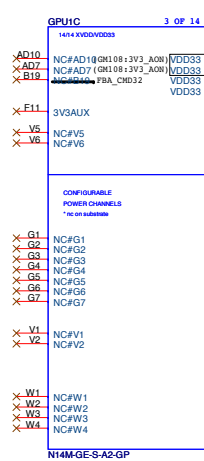
Under GPU



Near GPU



G10, G12:
If GC62.0 is implemented, connect to a 3V3 rail that will be on in GC6.
If GC62.0 is NOT implemented, connect to the same rail as VDD33.

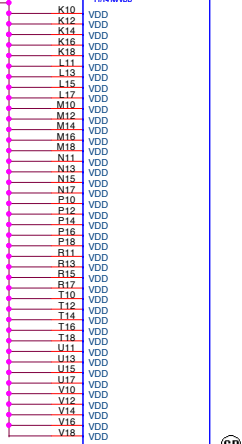


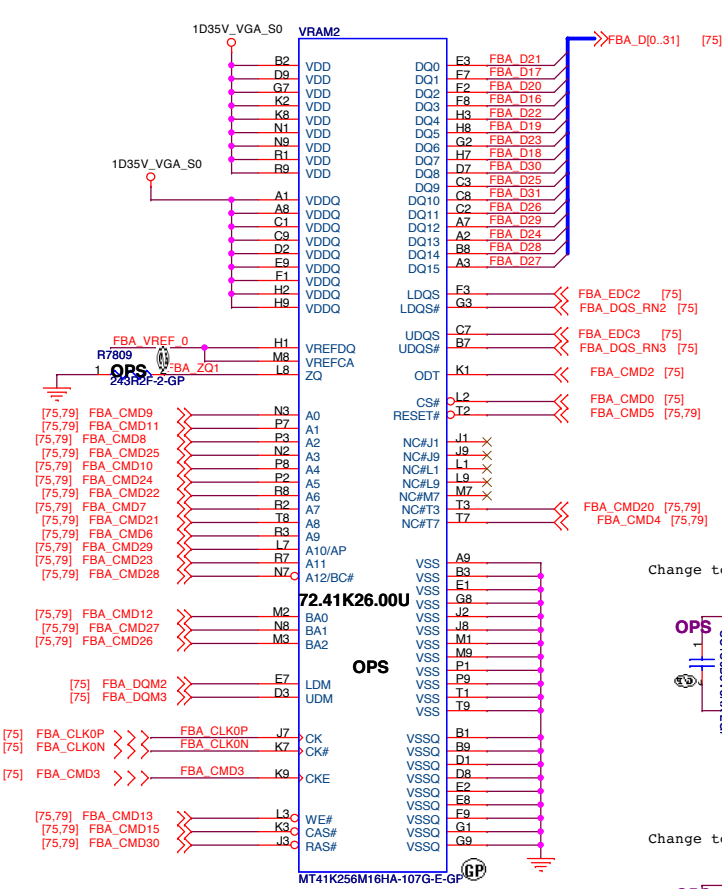
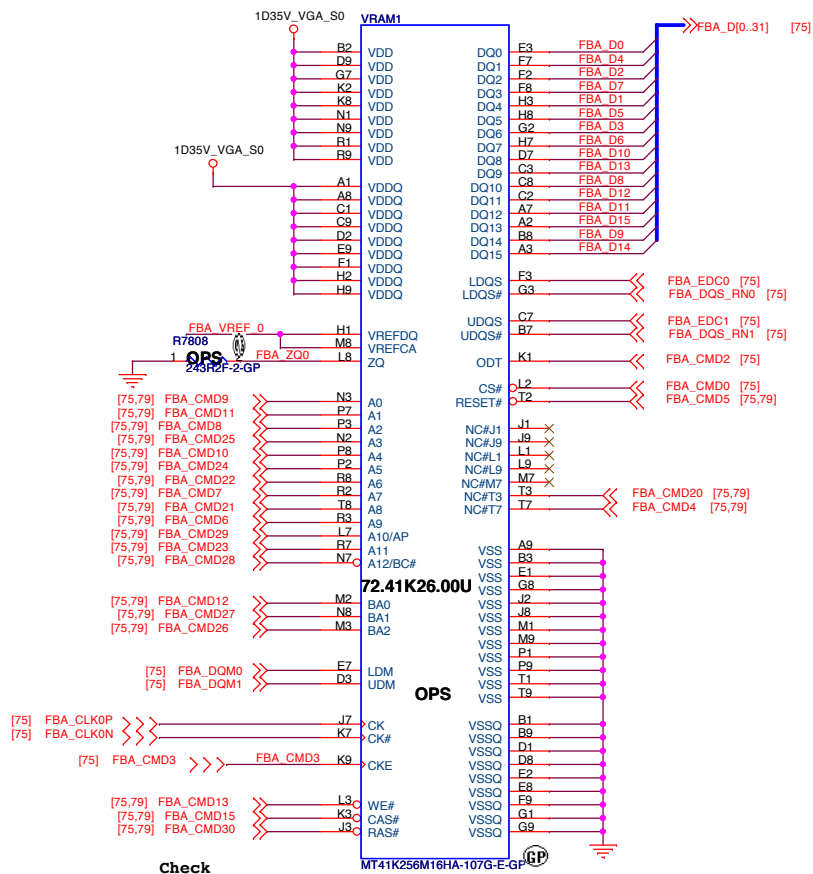
71.0N14M.B0U

OPS

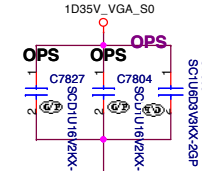
VGA_CORE

GPU1E 5 OP 14

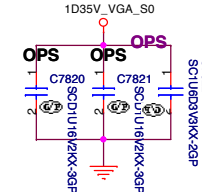




Place close VRAM1 VDD ball

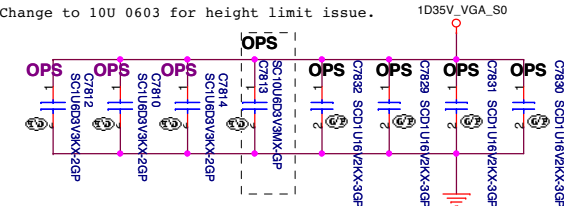


Place close VRAM2 VDD ball



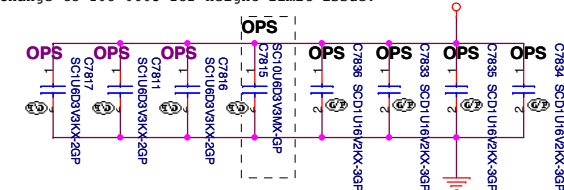
Place close VRAM1VDDQ ball

Change to 10U 0603 for height limit issue.

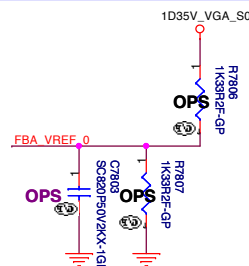


Place close VRAM1VDDQ ball

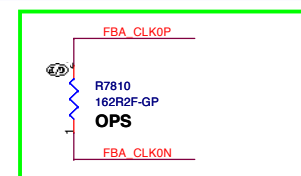
Change to 10U 0603 for height limit issue.



Frame Buffer Partition A-Lower Half



FBCLK Termination place on VRAM side



Layout Note: Place in the end.

FBVREF Termination

Type	FBVREF%	Voltage	GPU_GPIO10
Un-termination	50%	0.749V	High
Termination	70%	1.0617V	Low

20110613

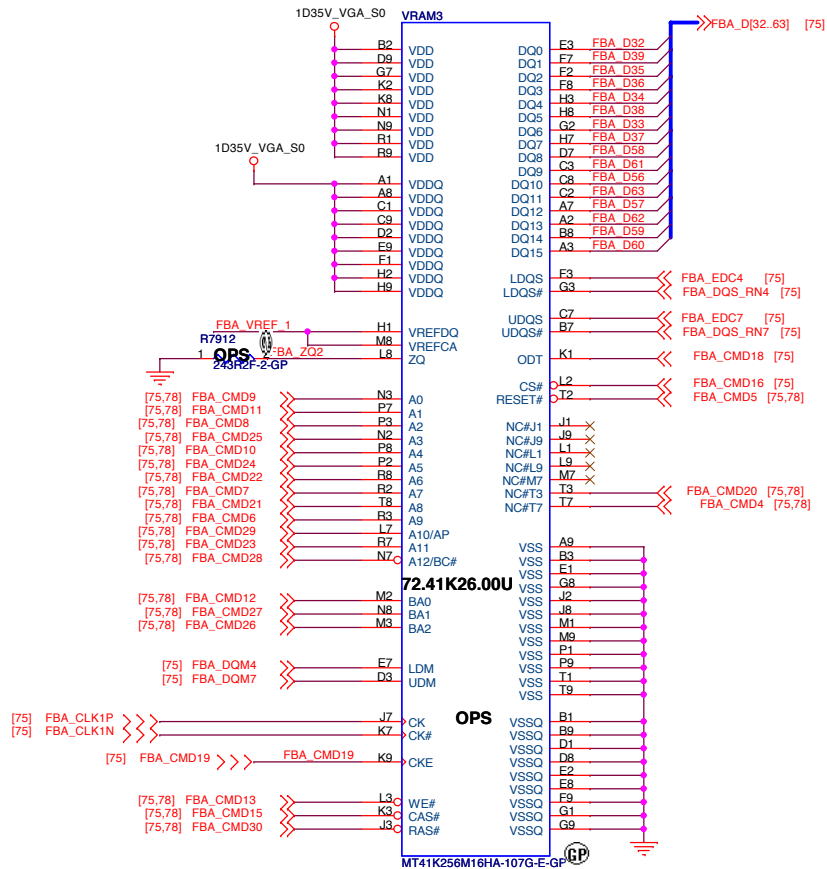
<Core Design>

Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

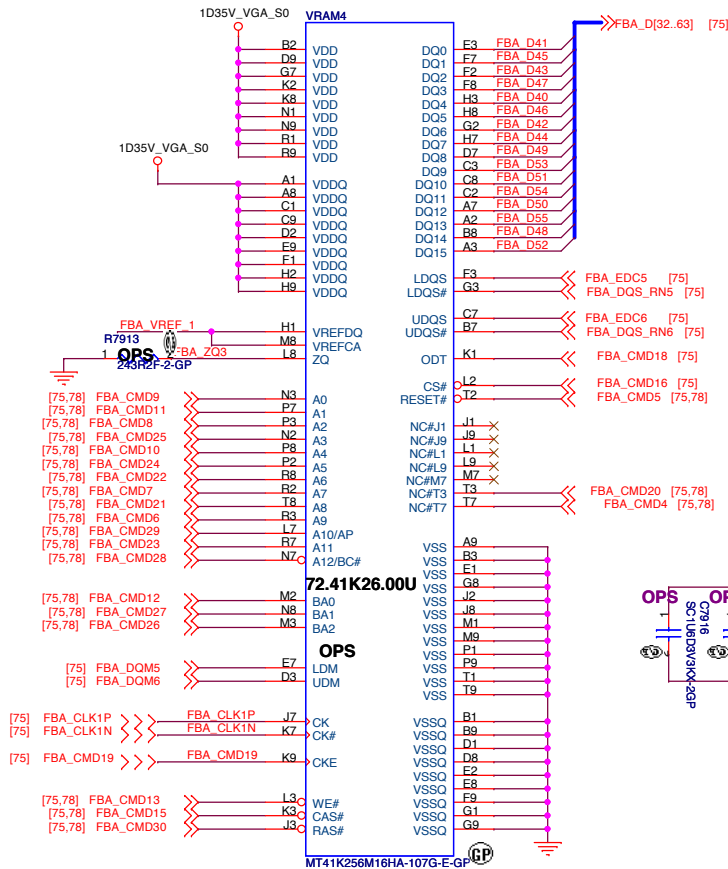
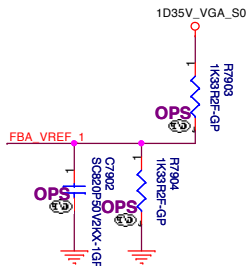
Title: **GPU-VRAM1,2 (1/4)**

Size A3 Document Number **Janus HSW 40/50/70** Rev **A00**

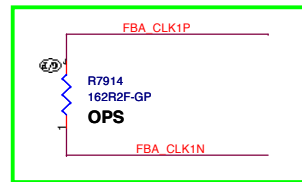
Date: Friday, February 07, 2014 Sheet 78 of 104



Frame Buffer Partition A-Lower Half

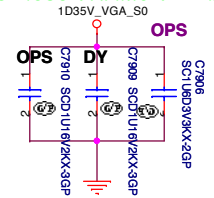


FBCLK Termination place on VRAM side

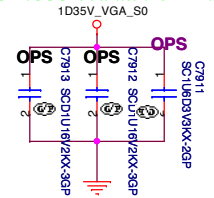


Layout Note: Place in the end.

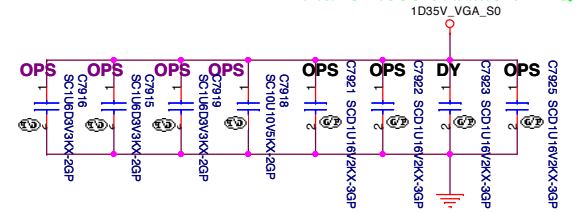
Place close VRAM3 VDD ball



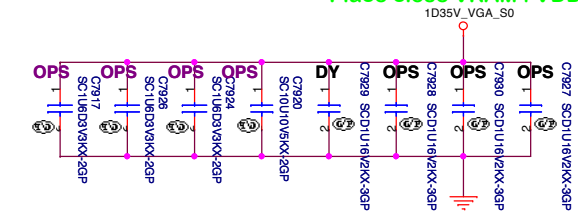
Place close VRAM4 VDD ball



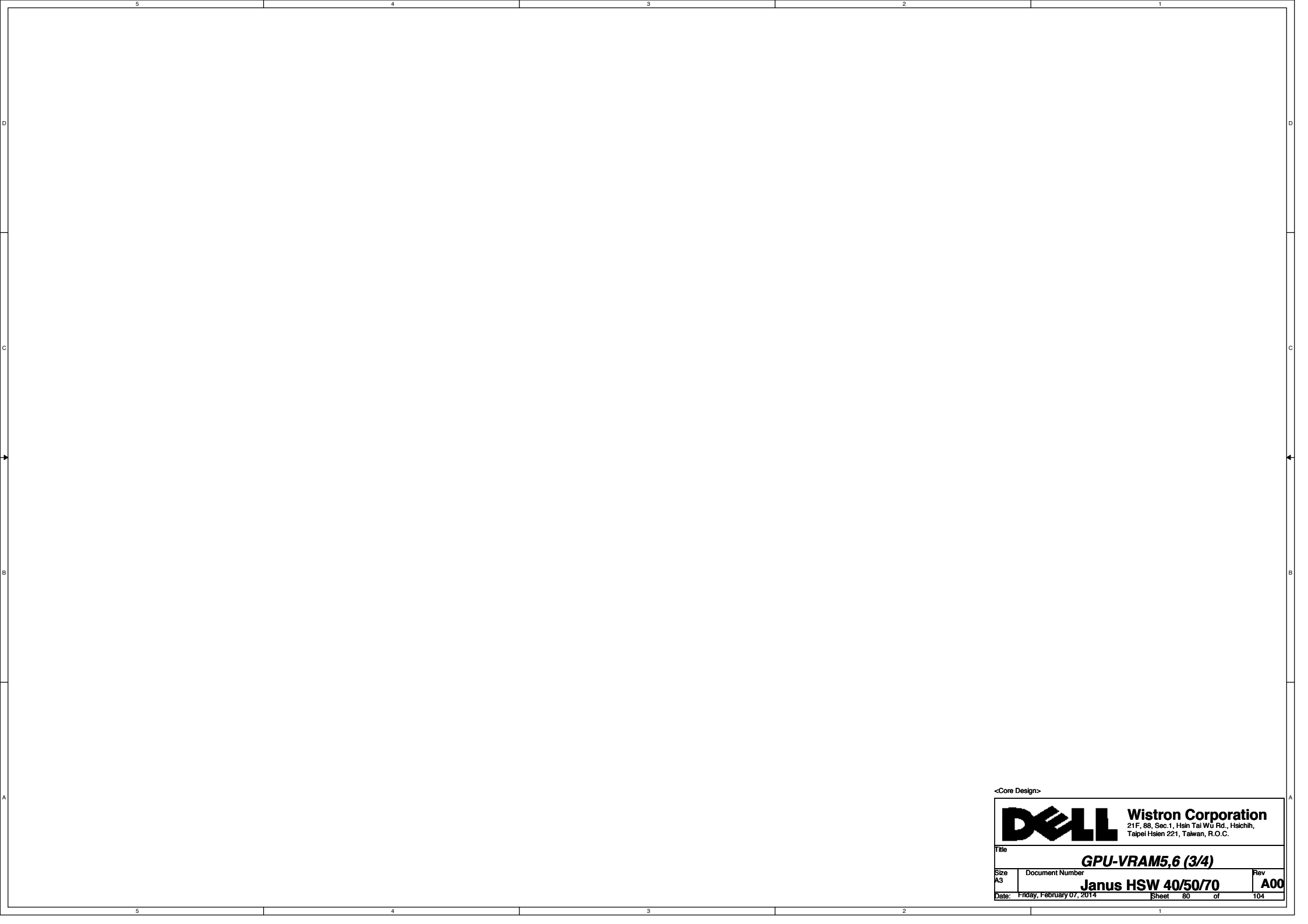
Place close VRAM3 VDDQ ball



Place close VRAM4 VDDQ ball

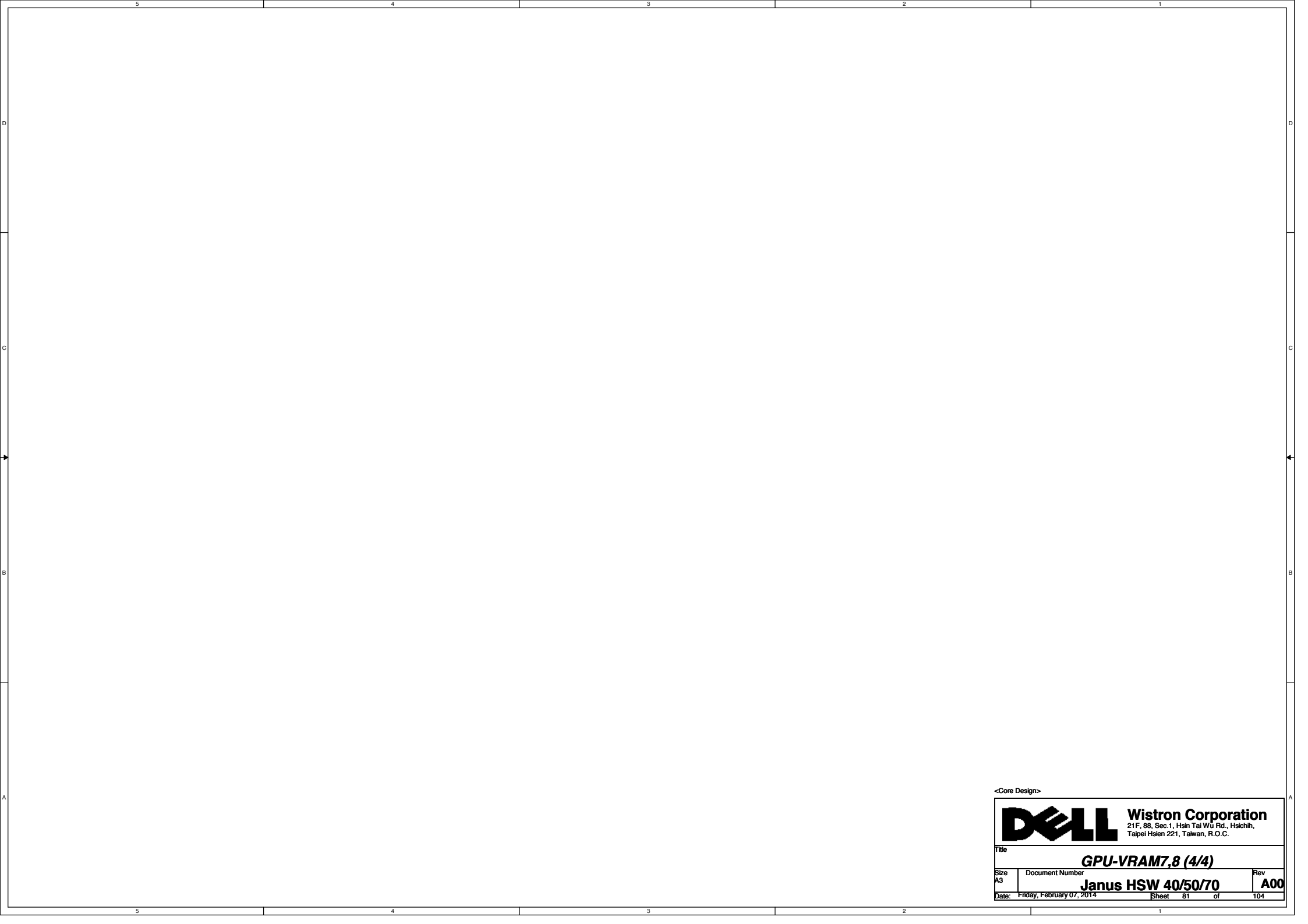


<Core Design>



<Core Design>

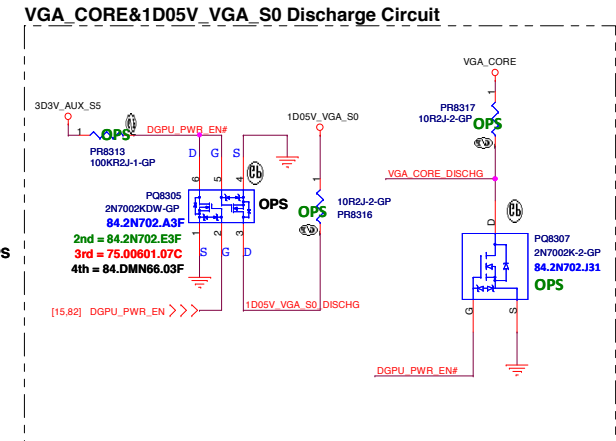
			Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.		
Title					
GPU-VRAM5,6 (3/4)					
Size	Document Number				Rev
A3	Janus HSW 40/50/70				A00
Date:	Friday, February 07, 2014		Sheet	80	of 104



<Core Design>

		Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title			
GPU-VRAM7,8 (4/4)			
Size	Document Number		Rev
A3	Janus HSW 40/50/70		A00
Date: Friday, February 07, 2014			
Sheet 81 of 104			


```
3D3V_VGA_S0 should ramp-up before VGA_Core
VGA_Core should ramp-up before 1D5V_VGA_S0
1D35V_VGA_S0 should ramp-up before 1D05V_VGA_S0
```

[illegible]

CTx (pF)	Rise Time (µs) 10% - 90%, COU _T = 0.1µF @ VIN; VOUT=0 ohm load							
	Typical values @ 25°C, 25V X7R 10% ceramic cap							
	5V	3.3V	1.8V	1.5V	1.2V	1.05V	1V	0.8V
0	107	72	46	41	36	34	33	29
220	425	276	146	122	103	91	88	74
270	489	316	172	139	121	107	104	84
470	774	487	272	224	181	159	154	123
680	1108	708	375	317	242	221	213	168
1000	1561	1007	546	441	364	314	299	234
2200	3600	2289	1240	1019	817	681	665	539
4700	7757	5092	2674	2203	1808	1592	1516	1177
10000	15700	10310	5601	4659	3674	3401	3197	2562

Table 1. Rise time vs. CTx value

<Core Design>



(Blanking)

<Core Design>



Wistron Corporation
21F, 88, Sec. 1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Reserved

Size	Document Number	Rev
A3	Janus HSW 40/50/70	A00

Date: Friday, February 07, 2014	Sheet 84 of 104
---------------------------------	-----------------

(Blanking)

<Core Design>



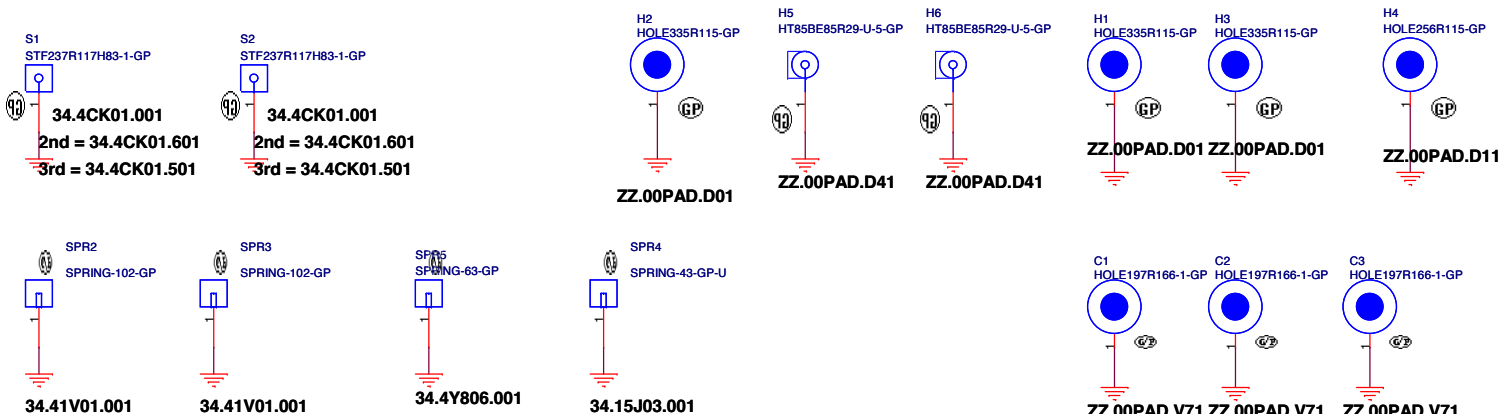
Wistron Corporation
21F, 88, Sec. 1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Reserved

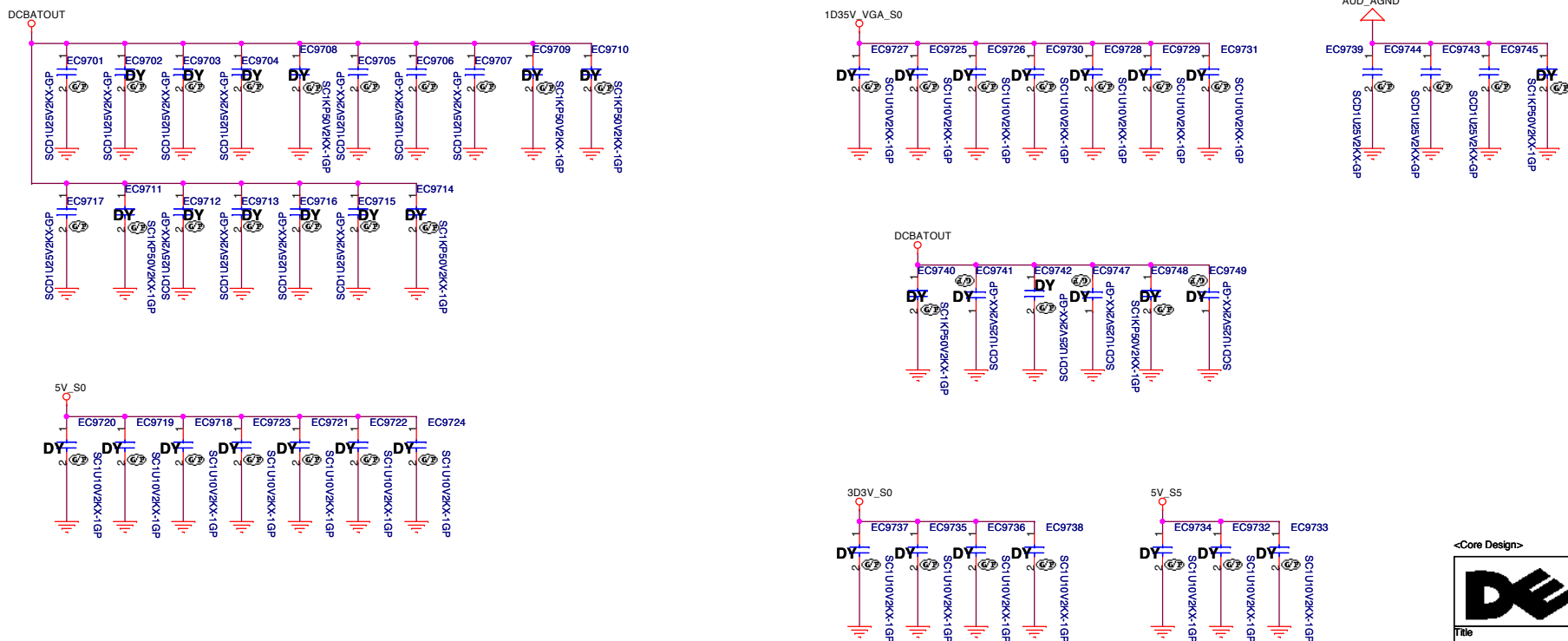
Size	Document Number	Rev
A3	Janus HSW 40/50/70	A00
Date:	Friday, February 07, 2014	Sheet 85 of 104

SSID = Mechanical



SSID = EMI

Mind the voltage rating of the caps.



(Blanking)

<Core Design>



Wistron Corporation
21F, 88, Sec. 1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Size
A3

Document Number
Janus HSW 40/50/70

Rev
A00

Date: Friday, February 07, 2014

Sheet 87 of 104

(Blanking)

<Core Design>



Wistron Corporation
21F, 88, Sec. 1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Size

A3

Document Number

Janus HSW 40/50/70

Rev

A00

Date: Friday, February 07, 2014

Sheet 88 of 104

(Blanking)

<Core Design>

			Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.		
Title (Reserved)Finger Print					
Size A4	Document Number Janus HSW 40/50/70				Rev A00
Date: Friday, February 07, 2014			Sheet 89 of 104		

(Blanking)

<Core Design>



Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Free Fall Sensor

Size A3	Document Number Janus HSW 40/50/70	Rev A00
------------	--	-------------------

Date: Friday, February 07, 2014	Sheet 90 of 104
---------------------------------	-----------------

(Blanking)

<Core Design>



Wistron Corporation
21F, 88, Sec. 1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Size

A3

Document Number

Janus HSW 40/50/70

Rev

A00

Date: Friday, February 07, 2014

Sheet 91 of 104

(Blanking)

<Core Design>

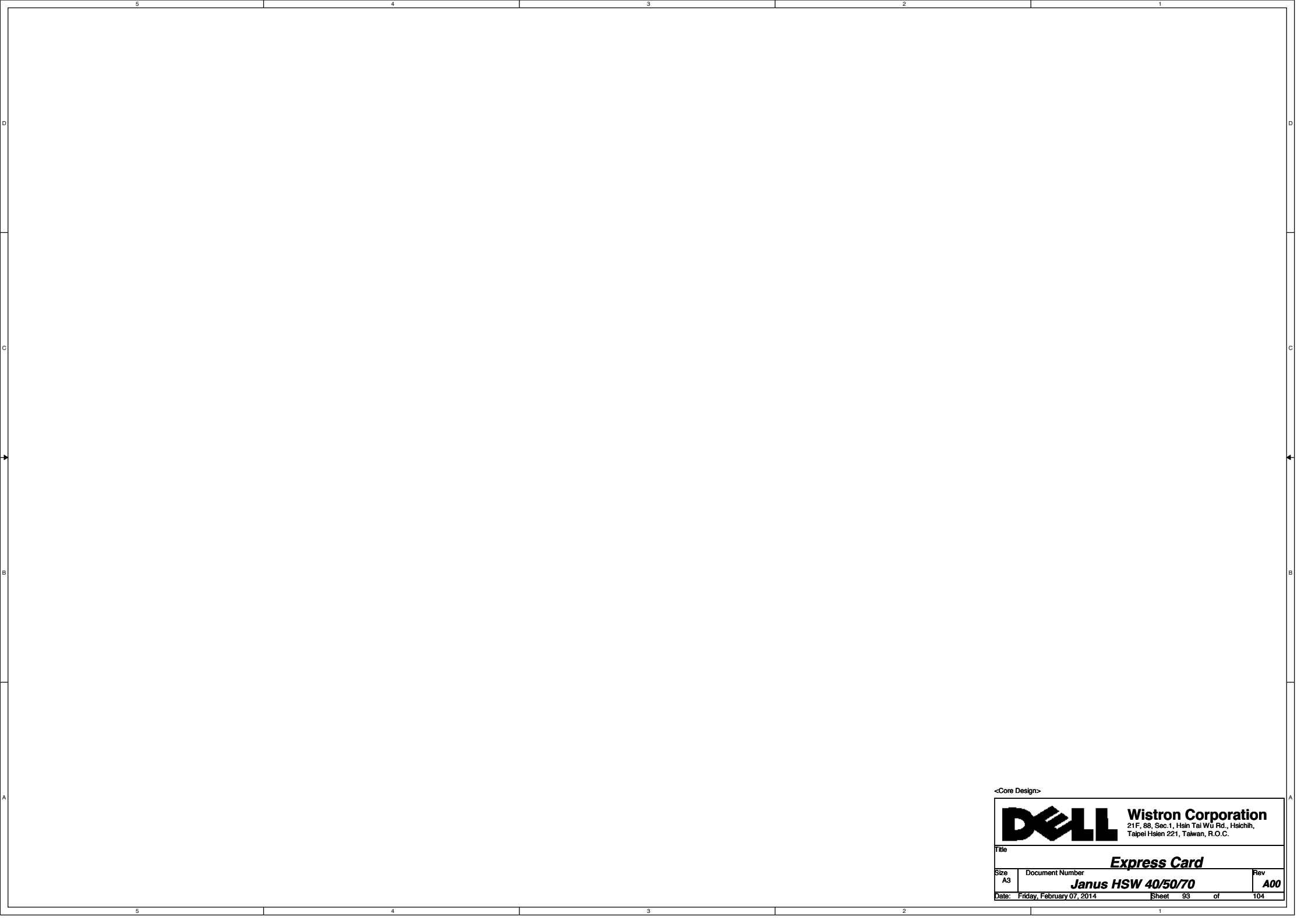


Wistron Corporation
21F, 88, Sec. 1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Reserved

Size	Document Number	Rev
A3	Janus HSW 40/50/70	A00
Date:	Friday, February 07, 2014	Sheet 92 of 104



<Core Design>

		Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title			
Express Card			
Size A3	Document Number Janus HSW 40/50/70		Rev A00
Date: Friday, February 07, 2014		Sheet 93 of	104

(Blanking)

<Core Design>



Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Size
A3

Document Number
Janus HSW 40/50/70

Date: Friday, February 07, 2014

Rev
A00

Sheet 94 of 104

LVDS Switch

(Blanking)

<Core Design>



Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

CRT Switch

Size
A3

Document Number
Janus HSW 40/50/70

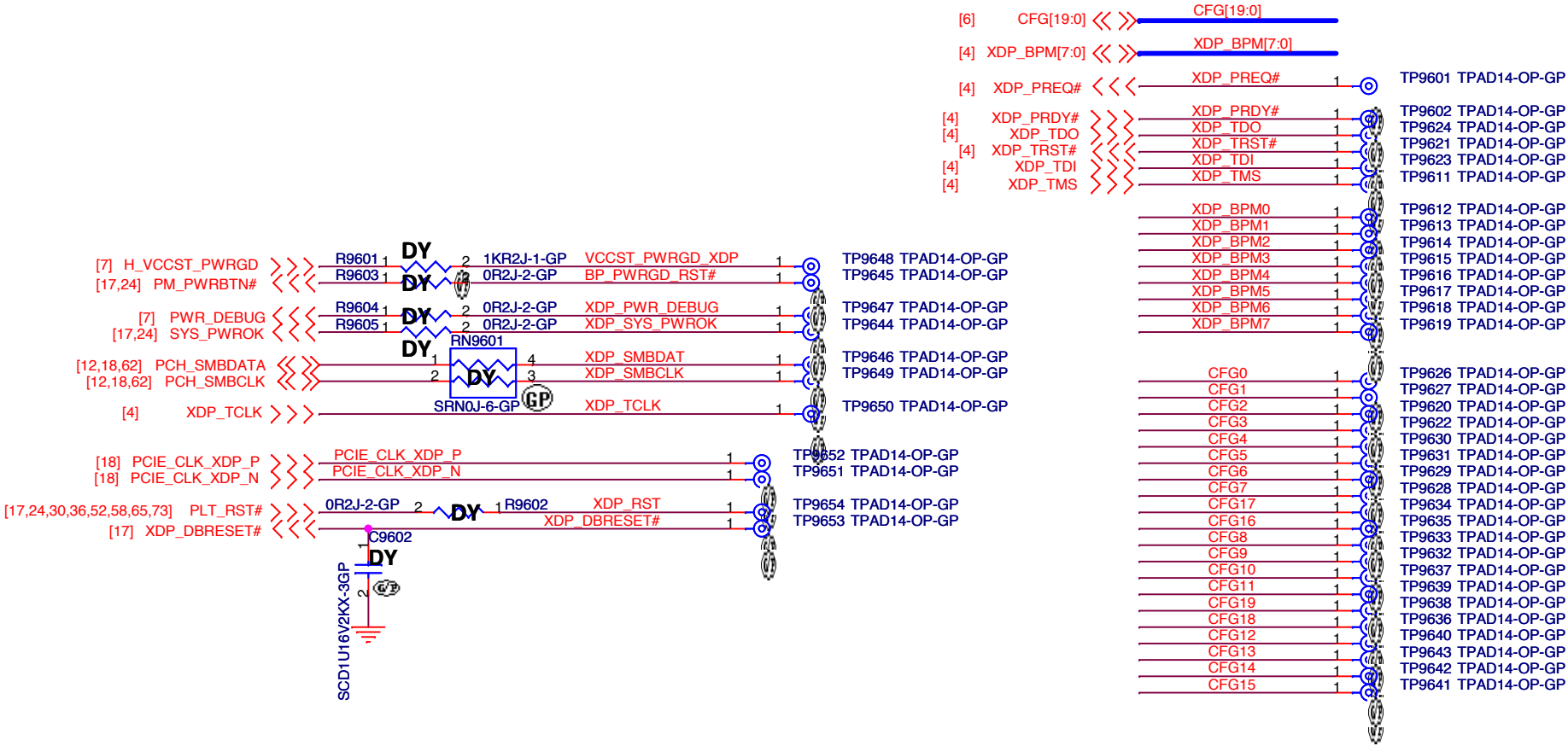
Date: Friday, February 07, 2014

Rev
A00

Sheet 95 of 104

SSID = XDP

CPU XDP



<Core Design>



Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

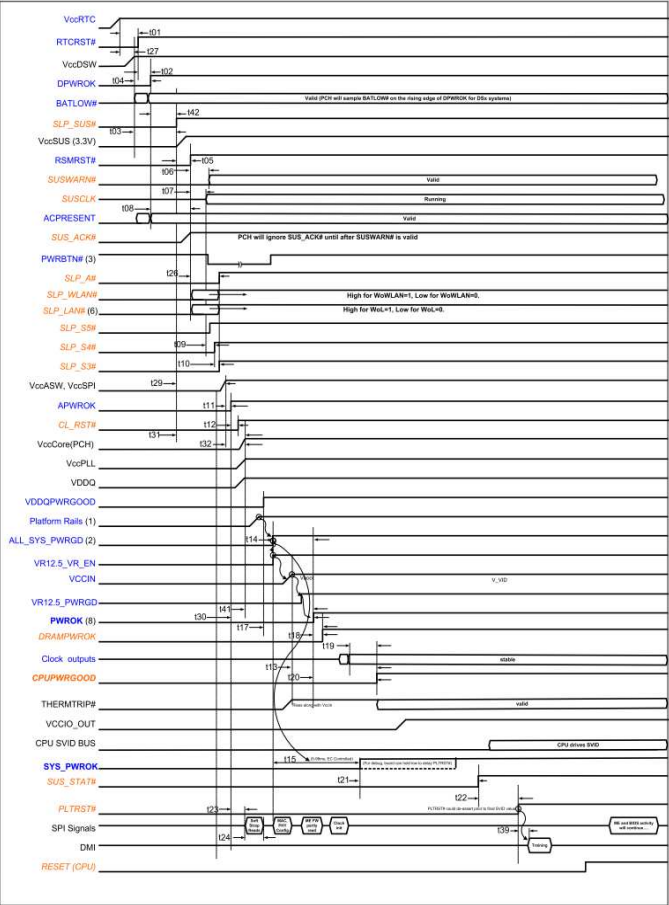
CPU/PCH XDP

Size A4	Document Number Janus HSW 40/50/70	Rev A00
------------	--	-------------------

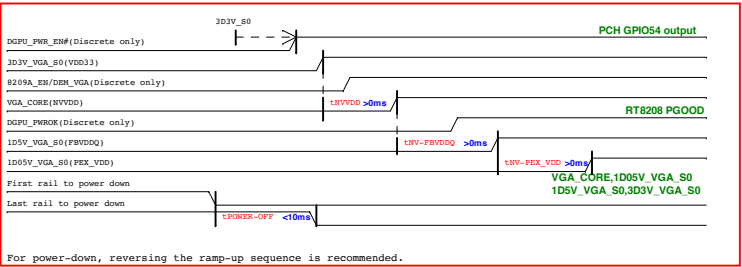
Date: Friday, February 07, 2014

Sheet 96 of 104

Shark Bay Platform Power Sequence

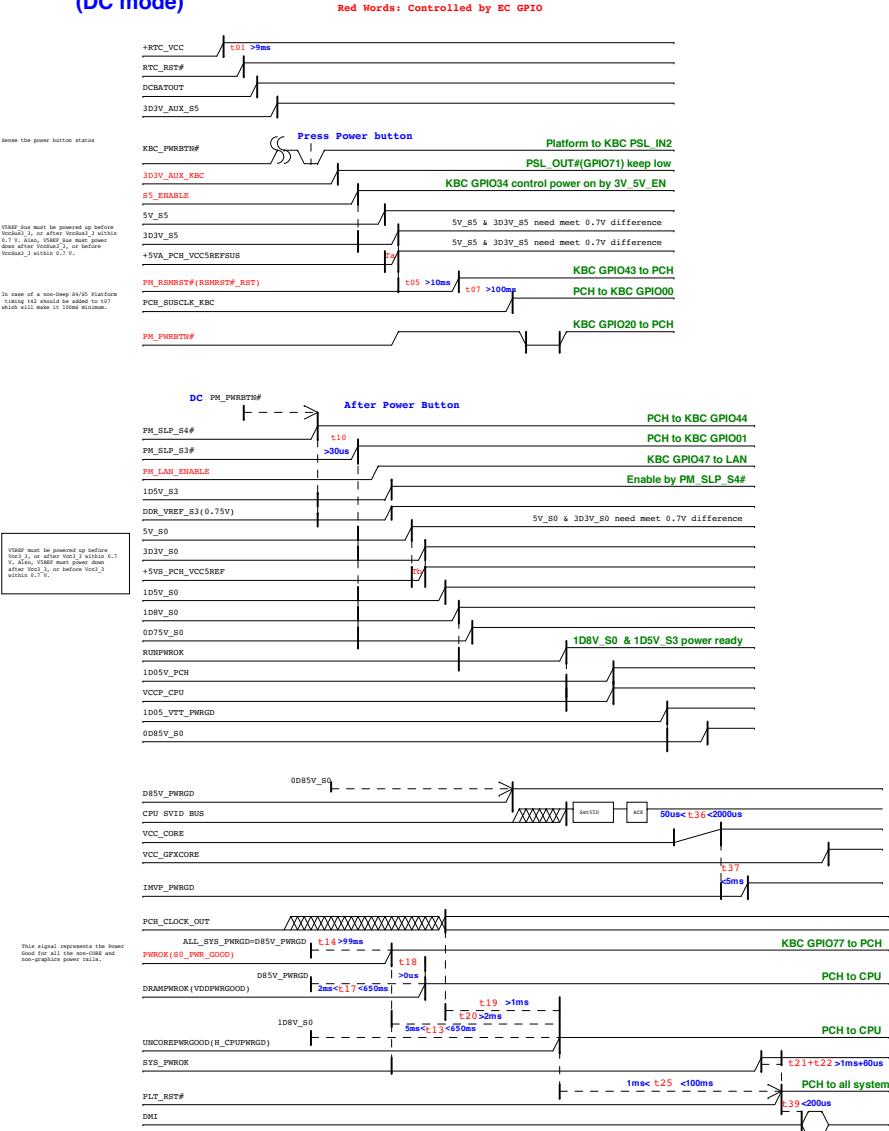


N14P-GT Power-Up/Down Sequence

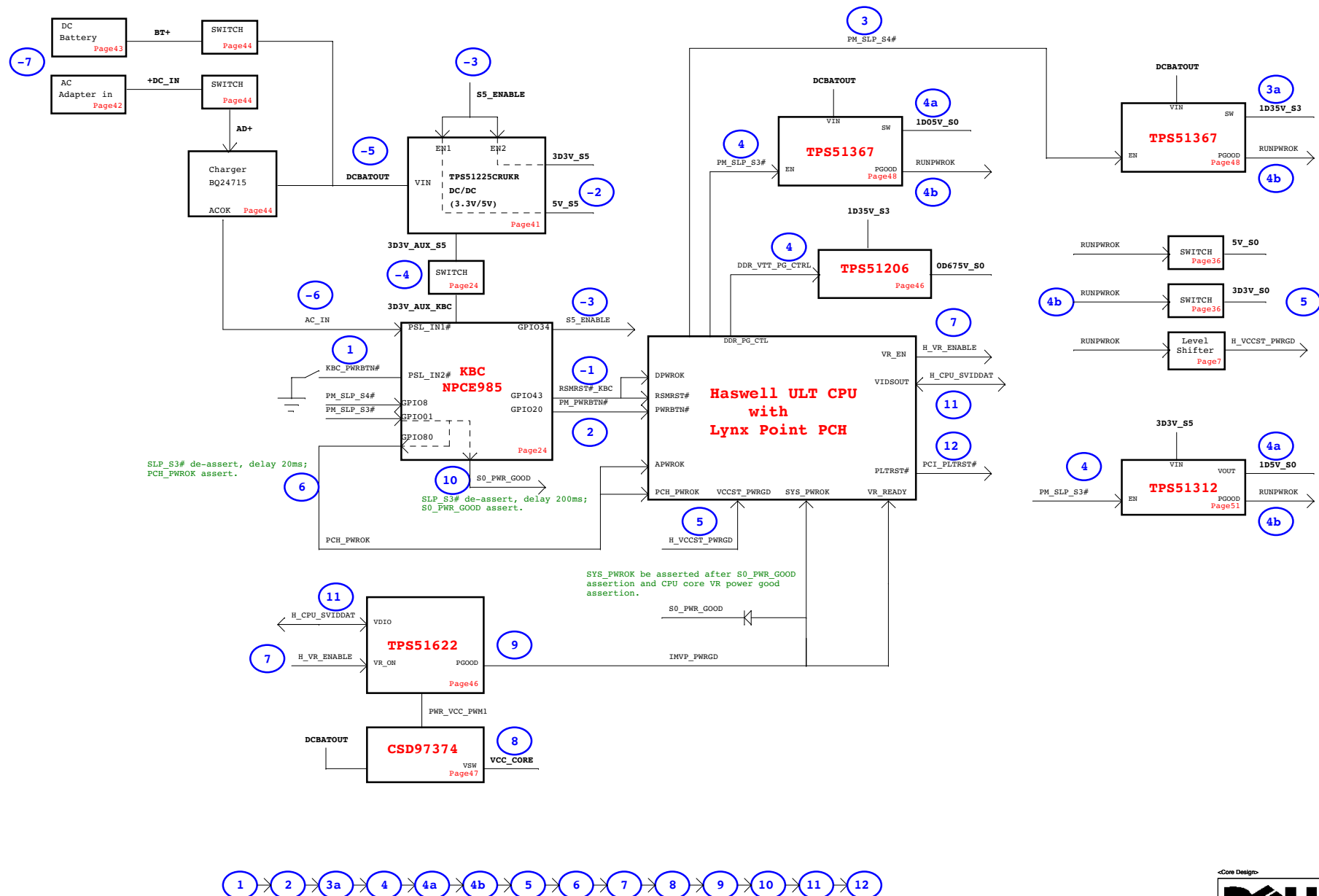


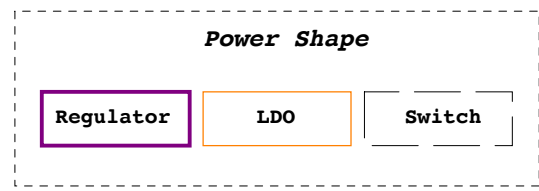
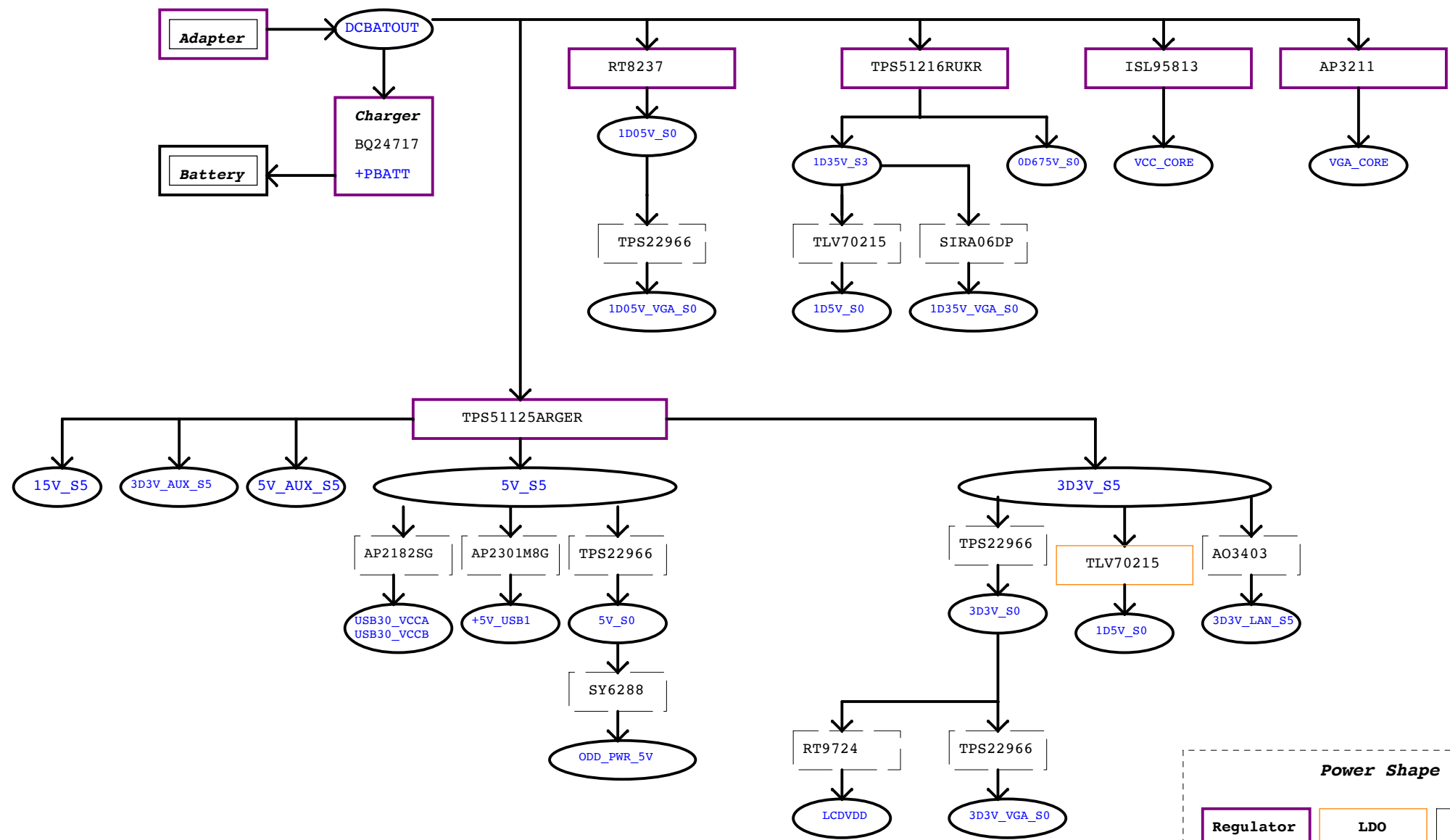
For power-down, reversing the ramp-up sequence is recommended.

(DC mode)

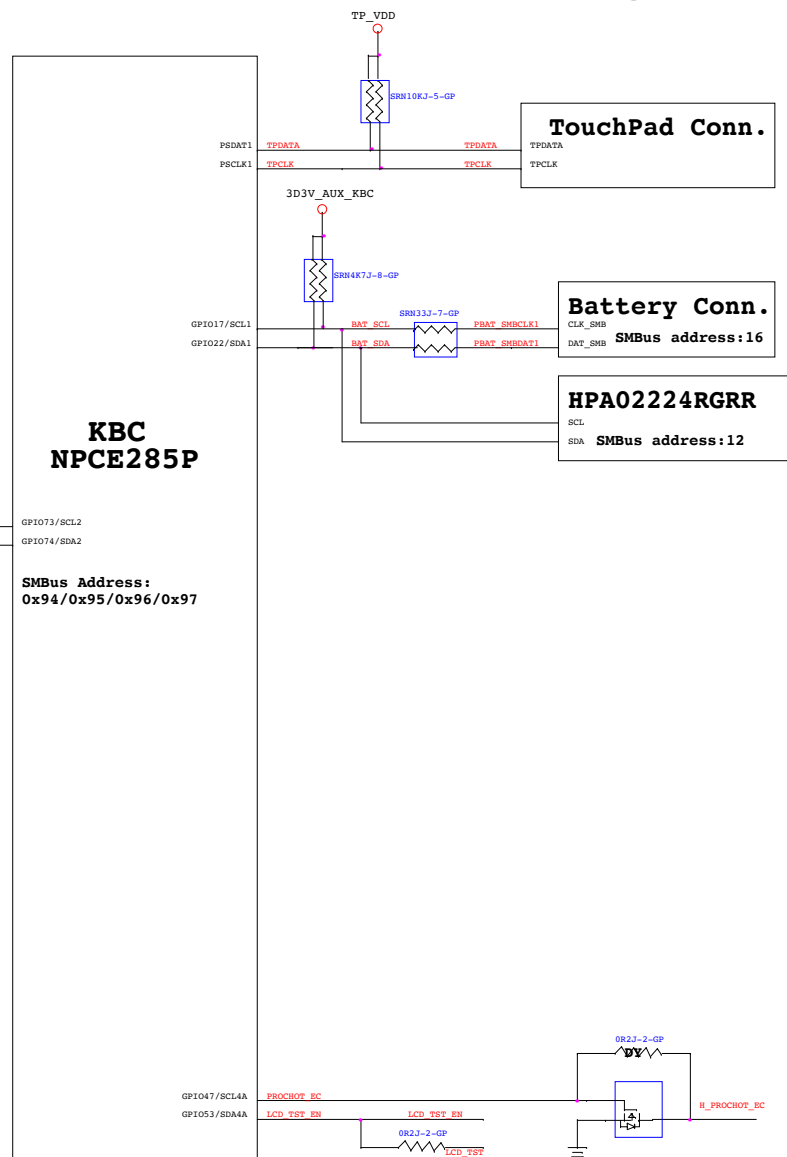
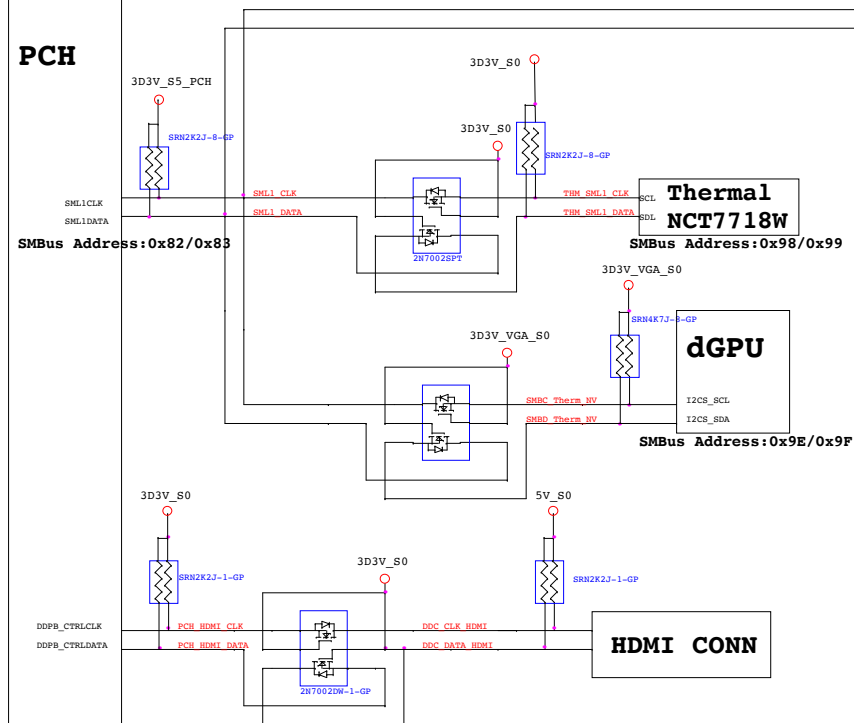


Wistron SHARK BAY POWER UP SEQUENCE DIAGRAM

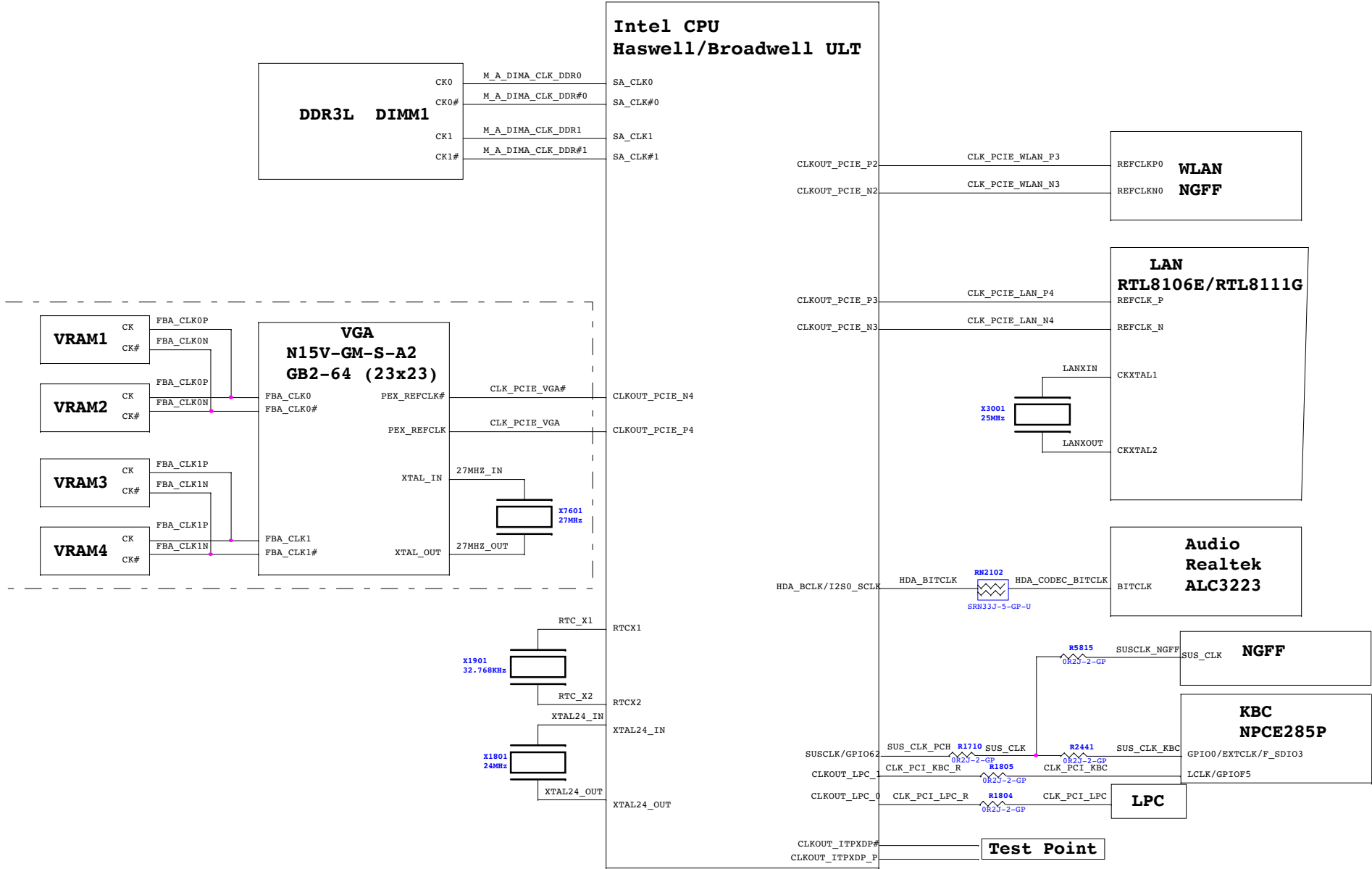




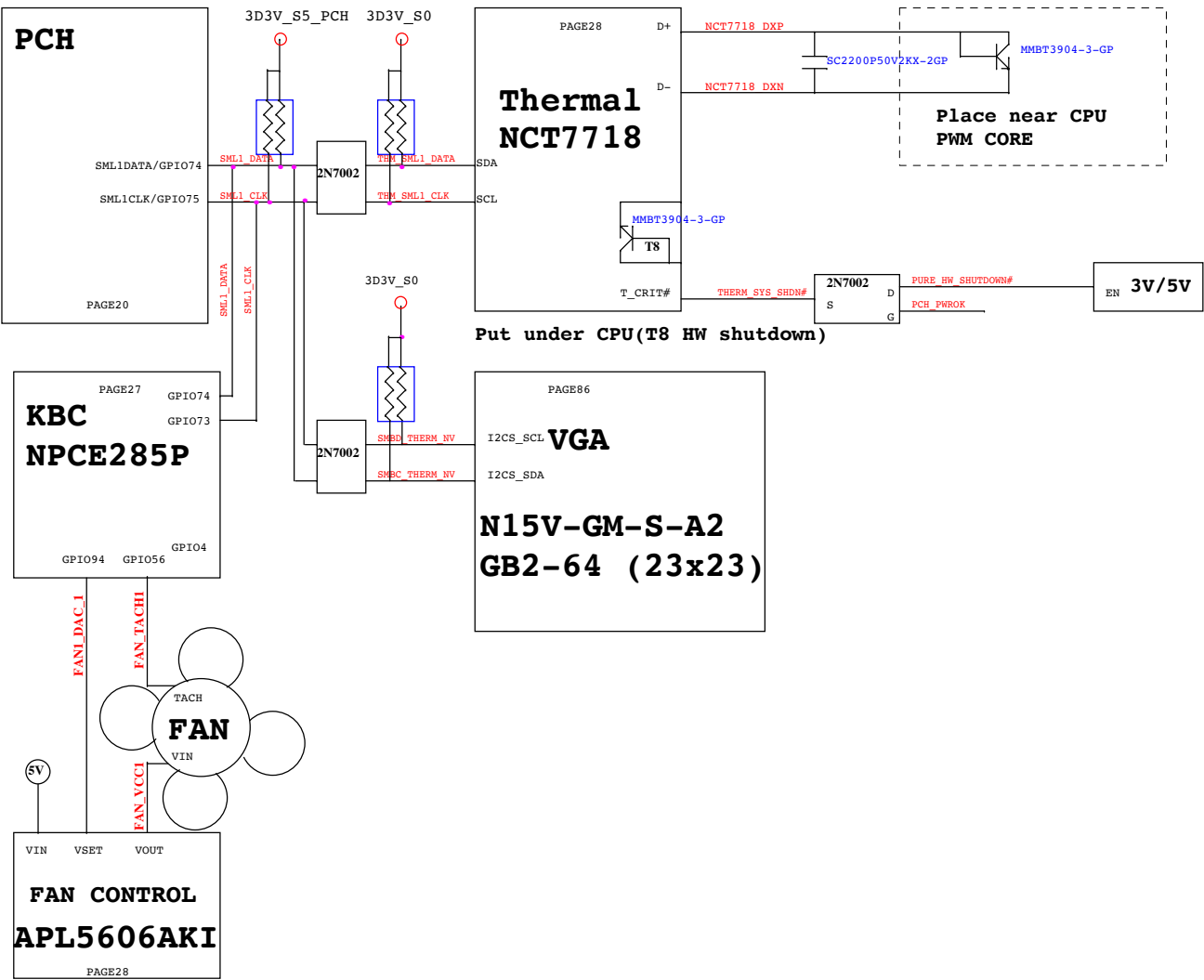
KBC SMBus Block Diagram



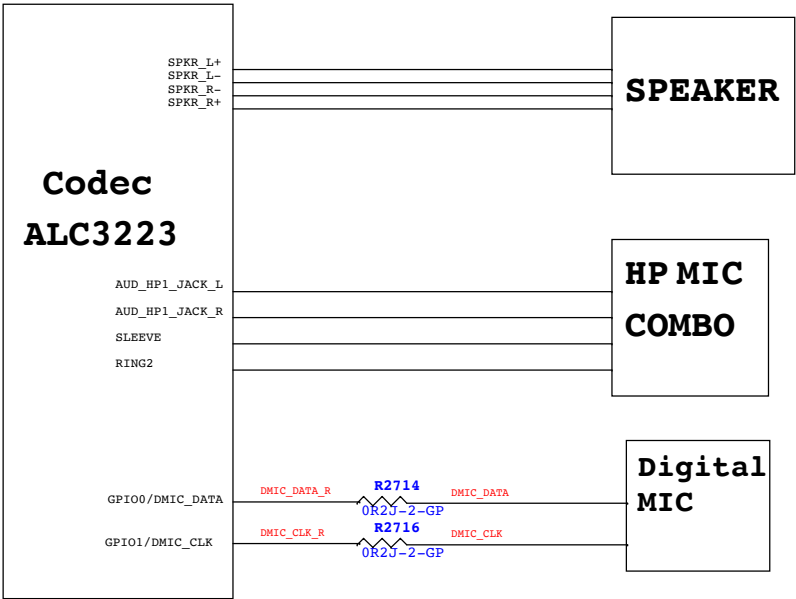
CLK Block Diagram



Thermal Block Diagram



Audio Block Diagram



[illegible][illegible]

5	4	3	2	1
VERSION	DATA	PAGE	Change Item	

<Core Design>



Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title

Change History

Size
A3

Document Number
Janus HSW 40/50/70

Date: Friday, February 07, 2014

Rev
A00

Sheet 104 of 104